

Algorithmic Addiction in Social Recommender Systems: Development of an Ethical AI Audit and Pro-Social Pivot Strategy under the EU Digital Fairness Act

CAPSTONE PROJECT (CONSULTING REPORT)

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Executive Summary

This executive summary concisely frames the client problem, methodological approach, principal findings, and actionable recommendations. Problem: major social platforms exhibit algorithmically mediated engagement dynamics that create elevated addiction risks for users and expose firms to intensifying regulatory threat, particularly from EU/DFA policy developments. Approach: we conducted a secondary qualitative synthesis of existing studies, constructed a comparative coding matrix to surface cross-platform mechanisms, and developed an Audit Toolkit to operationalize assessment of algorithmic harms for product, policy and legal teams. Key findings: (1) a systemic engagement-addiction risk is observable across TikTok, Instagram and LinkedIn-type platforms, driven by recommendation loops, short-form reinforcement and reward-timing; (2) these mechanisms are technologically diffuse but functionally similar, making platform-specific fixes insufficient; and (3) regulatory momentum is clear: the EU and the anticipated Digital Fairness Act signal a direction of travel toward stricter oversight of personalization and monetization practices. Headline recommendations: adopt the Ethical AI Audit Toolkit immediately to identify high-risk features and remediation pathways; execute a Pro-Social AI pivot that prioritizes attention stewardship and positive social outcomes in model objectives; and realign monetization with wellbeing by decoupling ad incentives from engagement-maximization, introducing alternative revenue models and embedding user wellbeing KPIs into executive compensation. These steps mitigate legal exposure and preserve long-term user trust and platform value.

Keywords

Keywords: algorithmic addiction, social recommender systems, dark patterns, Digital Fairness Act, ethical AI audit, pro-social design, EU platform regulation.

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List of Abbreviations

- AI – Artificial Intelligence
- AIAS-5 – Algorithmic Impact Assessment Scale
- CI/CD – Continuous Integration / Continuous Deployment
- CSR – Corporate Social Responsibility
- DFA – Digital Fairness Act
- DSA – Digital Services Act
- HITL – Human-in-the-Loop
- SDT – Self-Determination Theory
- UX – User Experience
- VLOP – Very Large Online Platform

1 Problem Description and Regulatory Environment

1.1 Motivation and Practical Relevance

Dark-pattern research has documented the anatomy of manipulative design in striking detail. Subscription flows are engineered to make cancelling harder than signing up. Illusory-urgency countdowns and misleading brand associations are built into standard UI templates (Sanikidze, 2026, p. 390). These are not accidental; behavioural evidence confirms that a "only two left" message shifts a neutral non-purchase into a perceived loss (Bansal, 2026, p. 135), precisely because scarcity cues activate loss-framing rather than deliberation. As AI-driven personalisation scales these techniques, the exploitation of predictable cognitive vulnerabilities becomes systematic rather than incidental, making a targeted audit methodology practically necessary.

European regulation is already moving to curb this trajectory. The DSA requires very large platforms to assess "systemic risks" to civic discourse and electoral processes, backed by transparency reports and non-profiled recommendation options (T et al., 2026, p. 12; Chukhray & Koval, 2026, p. 2). The AI Act goes further, banning subliminal manipulation and manipulative effects that significantly harm behaviour and placing such practices in the "unacceptable risk" tier (T et al., 2026, p. 12; Sanikidze, 2026, p. 390). A planned Digital Fairness Act would regulate addictive design and unfair personalisation as a distinct problem space bridging the DSA and consumer protection law. The regulatory direction is clear even before that Act is codified.

Global AI-governance proposals reinforce the same logic. Kostenko's AI Law Model requires systems to respect human dignity, autonomy and psycho-emotional comfort (Kostenko, 2025, pp. 61–62), prohibiting designs that manipulate behaviour or cause digital addiction. The fairness principle bans systems built on distorted data or that consolidate social inequality, with certification, suspension powers and public ethics-impact reports mandated for high-risk domains (Kostenko, 2025, pp. 73–74).

Digital-literacy gaps sharpen the urgency. Survey evidence on GDPR/DSA/AI Act awareness shows high self-assessed basic skills but persistent difficulty recognising hidden content and sophisticated manipulation (Chukhray & Koval, 2026, p. 2; T et al., 2026, p. 12). Users are sceptical that regulation alone guarantees safety, and individual awareness cannot carry the protective burden alone. OECD and UNESCO AI-ethics instruments articulate fairness and transparency norms (Bahangulu & Owusu-Berko, 2025, p. 9; T et al., 2026, p. 12; Catapang, 2026, p. 12), yet the "principles-to-practice gap" literature finds high-level declarations rarely specify operational procedures for data governance or post-deployment monitoring (Bahangulu & Owusu-Berko, 2025, p. 9; Catapang, 2026, p. 12). Accountability registries and human-in-the-loop oversight are becoming deployment prerequisites for high-risk sectors (Kostenko, 2025, p. 83; Grace Malachi et al., 2025, p. 11; Selvam & González Vallejo, 2025, p. 21; Kostenko, 2025, p. 108). Social recommender systems shaping attention at scale belong in the same category. An audit and pro-social pivot strategy that is both ethics-grounded and transparent is, on those grounds, academically necessary and regulatorily urgent.

1.2 Research Objectives and Central Questions

Interface tactics in digital retail are not isolated creative decisions. Research on urgency marketing and dark patterns shows they are systematically organised around psychological vulnerabilities and short-term revenue objectives (Bansal, 2026, p. 136). This project's first objective is a secondary-research

synthesis mapping such manipulation-oriented choices in recommender-driven environments through a coding matrix drawing on scarcity, psychological targeting and revenue optimisation constructs.

A second objective follows regulatory ambition. The AI Act bans subliminal manipulation and manipulative effects significantly harming behaviour; the DSA requires very large platforms to assess systemic risks to civic discourse and to offer at least one non-profiled recommendation option (T et al., 2026, p. 2248). A forthcoming Digital Fairness Act will complement this by targeting addictive design and unfair personalisation as its own problem space (Sanikidze, 2026, p. 390). Translating these obligations into operational audit questions is the practical task. The DFA functions here as policy trajectory, with the AI Act and DSA as the binding anchor frameworks (Sanikidze, 2026, p. 390; T et al., 2026, p. 2248).

A third objective is methodological. Qualitative and secondary designs examining AI systems and regulatory texts stress triangulating regulatory intent, platform practice and documented harm (Khare et al., 2026, p. 7; T et al., 2026, p. 6; Mariia, 2025, p. 3). This study commits to interpretive synthesis of secondary materials rather than proprietary telemetry. Black-box algorithms are assessed through documented UI behaviour and regulator findings, following Garr's convergent-parallel content-curation design and Chakraborty et al.'s mixed-methods education-ethics work as templates for evaluating algorithmic risk without internal model access (Mariia, 2025, p. 3; Chakraborty et al., 2025, p. 5). The methodology is calibrated to the opacity of the systems under review.

The central questions operate at the level of observable outputs and regulatory alignment. Do engagement-optimising recommendation architectures embed structural incentives similar to those identified in political recommender systems, where optimisation objectives and training dynamics contribute to polarisation (T et al., 2026, p. 6)? How far do existing and emerging EU instruments (AI Act, DSA, Digital Markets Act, projected DFA) function as a de facto governance regime for addictive design (Sanikidze, 2026, p. 390; T et al., 2026, p. 2248; Kustura, 2025, p. 2)?

A final objective draws on ethical AI marketing and corporate digital responsibility research. Asante and Adeoba (Asante & Adeoba, 2026, p. 11; Asante & Adeoba, 2026, p. 16) identify Communities of Practice and CSR as mechanisms turning internal ethical learning into policies and algorithm audits, with stakeholder trust as the core outcome. Catapang's Ethics-by-Design account maps human agency, fairness and transparency to specific lifecycle tasks rather than abstract principles (Catapang, 2026, p. 12). The working objective here is a practical audit toolkit and decision matrix, usable by management teams, linked to CSR governance and oriented toward sustaining engagement within EU risk-based regulation (Asante & Adeoba, 2026, p. 11; Catapang, 2026, p. 12; Kustura, 2025, p. 4). Whether that toolkit can close the principles-to-practice gap is the measure against which this study's recommendations will ultimately be judged.

1.3 Scope, Boundaries and Key Assumptions

This project draws on qualitative secondary research rather than primary empirical measurement or direct access to proprietary systems. Literature-based inquiry and reflexive thematic analysis synthesise academic, policy and industry documents (Asante & Adeoba, 2026, p. 8; T et al., 2026, p. 6), treating publications and regulatory texts as primary data and building understanding inductively. Claims about algorithmic addiction and manipulative interface patterns are mediated through documented analyses of recommendation architectures and regulatory regimes, not fresh behavioural experiments.

Black-box AI systems are examined through observable outputs and described objective functions, not

source-code inspection. Trade secrecy limits detailed technical analysis. This thesis adopts the same boundary as scholars studying democratic harms from recommendation architectures, framing audits around externally assessable properties: explainability obligations, independent audit requirements and risk classifications (T et al., 2026, p. 6; T et al., 2026, p. 16; Kostenko, 2025, p. 244), rather than unverified reconstructions of internal weights. All analytical conclusions about algorithmic risk derive from published regulatory findings and academic syntheses, not proprietary telemetry.

The regulatory frame draws on formally adopted EU instruments (DSA, AI Act) alongside extended AI law proposals specifying transparency, non-discrimination and audit obligations for high-risk systems, including algorithmic passports and national AI registers (Kostenko, 2025, p. 244; Kostenko, 2025, p. 243). The AI Law Model's digital-neutrality principle prohibits information-flow distortion and hidden censorship, and requires documented decision logic and independent audit with violations recorded in a public Register of Digital Neutrality Incidents (Kostenko, 2025, p. 109). Any forward-looking reference to a Digital Fairness Act (Sanikidze, 2026, p. 391) is treated as policy direction anchored in existing materials, not as codified law binding on current platform practice.

Organisational governance scope emphasises Communities of Practice, CSR mechanisms and AI ethics committees translating shared ethical learning into policies and accountability structures (Asante & Adeoba, 2026, p. 11; Asante & Adeoba, 2026, p. 16). Ethical AI marketing research (Asante & Adeoba, 2026, p. 11) restricts the governance focus to transparency, privacy, fairness and user control as the primary trust drivers, leaving technical compliance metrics outside scope.

Three key assumptions underpin the analysis. First, digital-literacy and regulatory-awareness surveys (Chukhray & Koval, 2026, p. 2) indicate persistent user vulnerability despite high self-rated competence. Second, human-in-the-loop oversight remains indispensable: regulatory texts call for supervision channels and appeal procedures, and educational HITL models show combining machine efficiency with human judgment sustains trust (Kostenko, 2025, p. 244; Selvam & González Vallejo, 2025, p. 21; Travieso-González, 2025, p. 3). Third, independent auditing and transparency are enforceable governance levers: AI law proposals establish mandatory audits and registration, and democratic recommendation governance work advocates annual third-party audits against content-policy and diversity benchmarks (T et al., 2026, p. 16; Kostenko, 2025, p. 244). The Toolkit this study develops therefore operates within published regulatory expectations and observable system behaviours; it makes no claims that require access to proprietary model weights or internal platform data.

1.4 Conceptual Clarification of Algorithmic Addiction

Short-video platforms offer the clearest empirical entry point. Neural-network recommendation engines continuously optimise content matching, using micro-duration stimuli and variable reward schedules to trigger dopamine release and induce an "auto-pilot mode" of persistent scrolling (Li, 2026, p. 2), already scrutinised under the DSA where infinite scrolling, autoplay and push notifications were investigated as systematic manipulative features impairing free and informed choice. College students illustrate the developmental-vulnerability dimension: their unstable self-identity and incompletely matured executive functions render them susceptible to excessive use, associated with diminished academic concentration, weakened social skills, and heightened anxiety, depression and behavioural addiction. Longitudinal data (Li, 2026, p. 2) specifically tracking recommendation-system effects on executive-function development in this cohort remain scarce, which constrains causal inference about platform-specific harm.

This report therefore defines algorithmic addiction as compulsive engagement emerging from the interaction of personalised recommendation logic, interface-level stimuli, and user vulnerabilities that limit reflective control. Social-media recommender research extends the concept beyond short video to broader feeds. Engagement-driven prioritisation (Hu, 2025, p. 19) amplifies misinformation and reinforces societal polarisation, creating a paradox: platforms both connect users and manipulate cognitive autonomy. Collaborative-filtering architectures systematically amplify high-arousal material; Instagram's prioritisation of body-negative imagery and the associated decline in self-esteem among adolescent women of color illustrates harm generated while users remain only partially aware of the underlying curation.

Neurocognitive indicators are stark. Users exposed to hyper-personalised feeds show higher depressive-symptom incidence than those on chronological content. Neuroimaging links algorithmically amplified "doomscrolling" (Hu, 2025, p. 3) to reduced prefrontal cortex activity mirroring substance-dependence patterns. Compulsive-checking evidence in dopamine-driven feedback loops confirms that recommender logics and interface affordances repeatedly cue high-arousal responses that alter emotional regulation and impulse control over time (Hu, 2025, p. 2; Li, 2026, p. 2). Hu's dataset recorded 23% higher depressive-symptom incidence among users on hyper-personalised feeds compared with chronological controls (Hu, 2025, p. 2; Li, 2026, p. 2).

Ethical and legal models supply normative boundaries. Kostenko's ethical-responsibility principle (Kostenko, 2025, pp. 61–62) prohibits systems that manipulate behaviour or cause digital addiction, framing psycho-emotional comfort and informed decision-making as mandatory design objectives. The digital-neutrality principle requires documented decision logic and independent audit, recognising proven manipulation as a threat to digital sovereignty (Kostenko, 2025, p. 109). Algorithmic addiction is thus treated not only as a behavioural pattern but as a violation of these duties: a state where optimisation for engagement and profit produces compulsive, autonomy-reducing usage (Kostenko, 2025, pp. 61–62; Kostenko, 2025, p. 109; Hu, 2025, p. 19; Li, 2026, p. 2).

Matrix factorisation-based recommenders frequently incorporate social ties to sharpen personalisation. Recent work demonstrates how both explicit friendships and inferred implicit connections can be fused into probabilistic matrix factorisation to improve predictive accuracy (Reafee et al., 2016, p. 2), a technique that can enhance relevance at the cost of reinforcing the same social-topology signals that drive engagement loops. Whether social-tie fusion raises compulsive-use rates in the same way as purely content-driven signals is not yet established in the published literature.

1.5 Positioning within Management and AI Ethics Scholarship

Management scholarship on AI treats ethical governance as an organisational learning and control problem, not a purely technical task. Communities of Practice (CoP) are conceptualised as social learning systems where professionals develop capabilities through participation and joint problem-solving (Asante & Adeoba, 2026, p. 3). Ethical AI in marketing is context-dependent and continually developing through inductive knowledge construction rather than fixed rulebooks. Reflexive thematic analysis is proposed to synthesise fragmented evidence on how organisational mechanisms support responsible AI implementation (Asante & Adeoba, 2026, p. 8). The implication for this thesis is direct: an audit framework must be embedded in governance routines, not delivered as a one-off checklist.

CSR is positioned as the governance bridge. It formalises CoP-generated insight into commitments, policies and accountability practices (Asante & Adeoba, 2026, p. 16; Asante & Adeoba, 2026, p. 11), con-

verting moral awareness into routines for AI design, data management and performance measurement. Algorithm audits, AI ethics committees and CSR metrics make ethical aspirations enforceable inside firms (Asante & Adeoba, 2026, p. 16). Stakeholder trust is the central outcome variable, resting on transparency, privacy, fairness and responsible personalisation rather than technical efficiency alone. Consumer acceptance of AI-driven advertising (Asante & Adeoba, 2026, p. 16; Asante & Adeoba, 2026, p. 11; Asante & Adeoba, 2026, p. 3) is conditioned on perceived control and clarity, implying any recommender audit framework must address these dimensions to be managerially relevant.

Parallel AI governance proposals supply an external normative frame. Kostenko's AI Law Model operationalises human dignity, autonomy and non-discrimination through mandatory procedures: notification of AI use, explainability, human control guarantees over critical decisions, and protection of vulnerable groups (Kostenko, 2025, p. 9), backed by national AI expert councils, accredited auditors and dynamic risk reclassification. This resonates with management calls for independent audits and external oversight committees (Kostenko, 2025, p. 9; Bahangulu & Owusu-Berko, 2025, p. 9). How well those councils and auditors perform under opaque black-box conditions is still tested mainly (Kostenko, 2025, p. 9) in financial and healthcare sectors; whether social media platforms would accept comparable oversight remains unresolved.

Digital consumer and financial-ethics research anchors this further. AI-embedded financial decision support raises explainability and automation-bias concerns (Raaj S. et al., 2026, p. 3; Cao, 2026, p. 2). Credit-decisioning evidence shows models trained on proxy variables can reproduce gender and intersectional discrimination even without protected attributes (Sebastian, 2025, p. 9; Deldjoo et al., 2023a, p. 7), meaning fairness audits focused only on overt attributes miss compounded subgroup disadvantage. Methodologically, ethical AI marketing and dark-pattern regulation studies converge on secondary qualitative designs synthesising regulatory texts and survey evidence where proprietary data access is constrained (Khare et al., 2026, p. 6; Asante & Adeoba, 2026, p. 8), echoed by constructivist grounded theory treating platforms as moral environments shaping identity and intention-behaviour gaps (Raaj S. et al., 2026, p. 3; Raaj S. et al., 2026, p. 5). These approaches validate this thesis's reliance on literature-based inquiry for theory-driven audit tools under black-box opacity. Personalisation-based dark-pattern research shows interface design induces decision distortion via loss aversion and anchoring (Zulfikar, 2026, p. 4), implying audit protocols must test manipulative elements and their cognitive mediators directly. Expert-AI-interaction studies show interfaces that surface uncertainty and let users reweight evaluative criteria preserve professional agency (Tseng & Lee, 2026, p. 6), a practical check that CSR-framed audits should adopt rather than leave to individual discretion.

2 Literature Review and Theoretical Foundations

2.1 Architecture of Social Recommender Systems

2.1.1 Core Components of Recommendation Pipelines

Three interlocking layers determine what a recommendation pipeline surfaces: data ingestion, model computation and governance instrumentation. In social environments, data capture is anchored on implicit feedback: clicks, viewing time and browsing activity that serve as proxy indicators of interest without directly encoding satisfaction (Polatidis et al., 2026, p. 21). Interaction data typically enters via distributed messaging into cloud data lakes. Financial decision-support systems ingest transaction streams through

Apache Kafka with row-level security and cryptographic tenant isolation. Yet historical bias enters at ingestion: neutral variables like geolocation reconstruct protected attributes despite formal omission of sensitive fields (S, 2026, p. 4), and social recommender pipelines relying on similar behavioural telemetry face the same structural risk. Bias can enter a pipeline at ingestion, before any model stage has had the opportunity to detect or correct it.

Pre-processing transforms raw logs into model-ready vectors. Credit-scoring bias-mitigation engines operate in three phases: reweighting at pre-processing, adversarial debiasing during training, and reject-option classification at post-processing (S, 2026, p. 4), aligning outputs with fairness constraints while preserving predictive performance. Core recommender models then operate on these signals. Content-based techniques retrieve items similar to prior interactions, limiting discovery. Collaborative filtering better supports serendipity but can surface niche items unevenly, with algorithms like Bayesian Personalised Ranking biased toward already-popular items.

Evaluation layers translate model behaviour into measurable outcomes. Beyond-accuracy metrics (diversity, novelty, catalogue coverage) assess whether a system broadens exposure. Bandit-based recommenders operationalise exploration to counteract "rich get richer" feedback loops, though the correlation between these metrics and actual user perception remains unclear (Jannach & Jugovac, 2019, p. 15). Industrial deployments often report only high-level business metrics. Dataset curation shapes outcomes further. Benchmarks like MovieLens, Amazon and MIND each emphasise different properties: explicit ratings, implicit scale, or temporal and trust relationships. Public availability enables reproducible evaluation (Polatidis et al., 2026, p. 21) but introduces popularity and demographic biases that constrain generalisability to real-world social settings, a gap that field-deployment audits are better placed to close than benchmark comparisons alone.

Governance components (explainability, audit logging and bias mitigation) are increasingly embedded alongside these stages. Financial AI architectures run asynchronous SHAP/LIME computations in parallel with inference, hashing explanation vectors onto immutable ledgers for regulatory audit (S, 2026, p. 4). This demonstrates that pipelines can be instrumented for real-time transparency rather than treating ethics as an external layer added after deployment.

2.1.2 Feedback Loops between User Behavior and Ranking

Feedback dynamics arise whenever user interactions are recycled as training signals determining subsequent rankings. Short-video ecosystems illustrate this directly. Recommendation engines form a closed loop: "personalized → need satisfaction → behavioral data accumulation → algorithm optimization → re push." More precise targeting increases dependence while "information cocoons" narrow exposure, and engagement metrics function simultaneously as optimisation targets and as data sources entrenching the patterns they measure. Engagement metrics serve simultaneously as optimisation targets and as data sources that entrench the patterns they measure.

Neuropsychological mechanisms accentuate this. Dopamine release tied to anticipation rather than completed reward produces "pursuit → transient satisfaction → further pursuit" cycles. Prolonged stimulation induces Δ FosB expression (Li, 2026, p. 3), altering reward-circuit gene expression and weakening self-regulatory capacity especially among adolescents. Ranking policies that weight recent watch time do more than sequence content; they shape the neurocognitive conditions under which the next click occurs.

Broader social-media evidence connects such loops to documented outcomes. Engagement-oriented pri-

oritisatation (Hu, 2025, p. 19) amplifies misinformation and polarisation while users remain largely unaware of the mechanism. TikTok's system reinforces filter bubbles while improving loyalty. YouTube's recommender, responsible for over 70% of views, prioritises watch time often at diversity's expense. Instagram and Facebook amplification of partisan material in election contexts intensified polarisation (Mariia, 2025, p. 2; Mariia, 2025, p. 23; Mariia, 2025, p. 29; Hu, 2025, p. 19). Infinite scrolling and variable reward schedules correlate with 22% higher compulsive-checking rates (Hu, 2025, p. 2), and neuroimaging links prolonged engagement to reduced prefrontal activity resembling substance dependence, indicating that the pattern is neurological, not merely habitual.

Algorithmic bias compounds this. Systems trained on skewed interaction data reproduce cultural stereotypes. Regulatory analysis under the DSA/AI Act finds engagement-driven biases in core algorithms remain largely unaddressed despite transparency reporting (Mariia, 2025, p. 5; Mariia, 2025, p. 8; Mariia, 2025, p. 9), so feedback loops privileging sensationalist content persist. Proposed responses include fairness metrics and explanatory AI tools exposing why items appear in feeds.

The bounded-rationality research adds an interpretive layer: users satisfice and rely on availability and anchoring shortcuts (Mahay, 2025, p. 4), so click signals may misrepresent stable preferences and instead reflect transient cognitive constraints. Heuristically biased traces are exactly what ranking models learn from. Click-based training signals carry systematic distortions on that basis, distortions fairness audits should treat separately from downstream algorithmic output.

2.1.3 Data Flows, Logging and Profiling Practices

Consider AI-mediated educational systems as an illustrative case. IoT-enabled learning platforms aggregate biometric logs (voice, keystroke dynamics, facial recognition), behavioural analytics and network metadata into unified datasets distinguishing genuine activity from fraud (Jayanthi, 2025, p. 5). Pre-processing pipelines impute and normalise these into model-ready profiles. Financial AI architectures show a parallel high-volume, multi-tenant pattern: transaction streams ingested via Kafka into row-level-secured storage still admit historical bias at ingestion, since neutral variables like geolocation act as precise demographic proxies within deep neural networks (S, 2026, p. 4). A telemetry-parsing layer exposes incoming vectors to a domain-specific bias-mitigation engine so ethical evaluation is integrated into the live flow, not batched retrospectively. Social recommender systems with similar ingestion architectures inherit the same bias risk.

Logging practices are a regulatory requirement, not an ancillary concern. Financial systems run asynchronous SHAP/LIME computations in parallel with inference, hashing explanation vectors onto immutable ledgers for tamper-evident audit (S, 2026, p. 4). HITL grading deployments time-stamp every AI and instructor decision with reviewer identifiers (Selvam & González Vallejo, 2025, p. 16; Selvam & González Vallejo, 2025, p. 17), making overrides traceable at the level of individual rubric dimensions. Ethically-grounded education frameworks extend logging toward explicit privacy guarantees: anonymisation, encrypted audit-log storage, opt-out options, and clear "AI generated" vs. "instructor reviewed" attribution aligned with FERPA and GDPR (Selvam & González Vallejo, 2025, p. 20; Jayanthi, 2025, p. 5).

Regulatory AI law models generalise these into structured duties for high-risk systems. The TSAI regime requires owners to preserve event logs and comply with data-protection and transparency requirements, operationalised through an algorithmic passport specifying data sources, legal grounds, log storage periods and audit information (Kostenko, 2025, p. 242; Kostenko, 2025, p. 244). For AI in public communications, auditable interaction logging distinguishes permitted dual-use tools from prohibited manipulative

systems; logging intensity scales from increased to critical risk tiers, and platforms amplifying manipulative content without such safeguards bear public responsibility, with independent audit mandated to verify watermarking, labelling and log-based evidence collection (Kostenko, 2025, p. 95; Kostenko, 2025, p. 97). Privacy impact assessments and routine independent audits are practical complements to these cryptographic safeguards (Badhan et al., 2024, p. 6). Together, these requirements build logging into platform governance as an operational layer running alongside inference, not a compliance record assembled after the fact.

2.2 Engagement-Optimized Interface Design in Social Platforms

2.2.1 Infinite Scroll and Continuous Content Streams

Remove the stopping cue, and the user keeps scrolling. Infinite scroll achieves this by coupling a single low-effort gesture to a personalised next item rather than a chronological queue (Li, 2026, p. 2; Li, 2026, p. 3). Each scroll both delivers content and refines the model. The closed cycle, "personalized → user need satisfaction → behavioural data accumulation → algorithm optimization → re push," depends on this frictionlessness to remain active. Removing friction is an explicit design objective, not an incidental property of the interface.

Neuroscience explains why frictionlessness is not neutral. Dopamine surges at the moment of expectancy before the next item, producing a "pursuit → transient satisfaction → further pursuit" cycle. Prolonged stimulation induces Δ FosB expression (Li, 2026, p. 3; Li, 2026, p. 2) in reward circuitry, undermining self-regulatory capacity, particularly among adolescents whose executive functions are still maturing. Variable reward schedules, identified (Hu, 2025, p. 2) as core elements of "dopamine driven feedback loops," yield 22% higher compulsive-checking rates; neuroimaging links prolonged use to reduced grey-matter density in impulse-control regions, the same structural change observed in substance-dependence populations.

Regulatory analysis treats continuous streams as potentially manipulative design. The EU DSA investigated TikTok's infinite scrolling, autoplay and push notifications as systematic manipulative features impairing free decision-making. Article 25 DSA prohibits manipulative interfaces outright. Chinese algorithmic-recommendation rules separately require providers to offer non-personalised options and functions to disable recommendation entirely (Li, 2026, p. 2; Li, 2026, p. 7). TikTok's "For You Page" is described by DSA regulators (T et al., 2026, p. 16) as an exceptionally effective, opaque attention-capture system, notable precisely because users cannot identify the signals driving its curation.

Proposed mitigations target frictionlessness directly. Randomised content or customisable filters can disrupt information cocoons. Active exit nodes, usage-duration reminders and timed pauses reintroduce deliberation (Li, 2026, p. 7). "Why this content" explanations move from black-box manipulation toward visible, DSA-aligned assistance (Li, 2026, p. 7; Mariia, 2025, p. 8). This thesis treats continuous streams as ethically problematic specifically where infinite scroll combines engagement maximisation, ranking opacity and absent exit cues for vulnerable groups (Li, 2026, p. 2; Li, 2026, p. 3; Hu, 2025, p. 2; T et al., 2026, p. 16). The absence of an exit cue is itself a design choice that regulators are now beginning to classify as manipulative.

2.2.2 Autoplay, Short-Form Clips and Sound Design

Every completed clip immediately triggers another. That is the design intention of autoplay: continued consumption is the path of least resistance, not deliberate stopping (Li, 2026, p. 2; Li, 2026, p. 3). Dominant

formats are 15–60 second vertical clips chained through infinite swipe navigation. The extremely low cost of swiping enables effortless session extension. Douyin usage runs near 60 minutes across 21+ daily sessions, with 41.3% of users opening the app multiple times daily and 56.9% exceeding 60 minutes (Li, 2026, p. 3). These are not edge-case patterns; they describe the median user's daily experience.

For college students and adolescents with immature self-control, this pattern is associated with heightened addiction susceptibility, diminished academic concentration, and increased anxiety and depression (Li, 2026, p. 2; Li, 2026, p. 6). Sound design amplifies the effect: dynamic music paired with vertical clips produces a "sunflower seed effect" (Li, 2026, p. 3) that disrupts default-mode-network functioning and compromises self-regulatory capacity. Neurobiologically, the content-switching mechanism resembles a slot machine. Each swipe carries unpredictable payoff. Dopamine surges during anticipation rather than consumption. Chronic stimulation downregulates dopamine receptors and elevates reward thresholds (Li, 2026, p. 4), reducing the hedonic value of natural rewards in a self-reinforcing cycle.

Broader social media evidence corroborates this. Infinite scrolling and variable reward schedules correlate with 22% higher compulsive-checking rates. Neuroimaging links prolonged use to reduced prefrontal activity and grey-matter density in impulse-control regions (Hu, 2025, p. 2), mirroring substance dependence. The pattern mirrors substance dependence rather than pre-existing preference: autoplay and high-arousal soundscapes function as a stimulus package tuned to neural addiction pathways.

Regulatory sources treat this as potential manipulative design. The DSA investigated TikTok's infinite scrolling, autoplay and push notifications on precisely these grounds (Li, 2026, p. 2). Kostenko's AI Law Model prohibits gamification of addiction and exploitation of age-related vulnerabilities (Kostenko, 2025, p. 223), mandating for minors secure defaults, bans on endless scrolling and autoplay, and visible "no personalization" switches. Reliability provisions require systematic incident logging and independent audits for such systems (Kostenko, 2025, p. 57; Kostenko, 2025, p. 79). Autoplay, short clips and sound design sit at the intersection of dopamine-driven feedback loops and explicit regulatory concern, especially where adolescents form the core user group (Li, 2026, p. 2; Li, 2026, p. 3; Hu, 2025, p. 2; Kostenko, 2025, p. 223). Whether mandatory age-verification for these features would reduce the documented harm among adolescents remains an open empirical question.

2.2.3 Notification Systems and Re-Engagement Triggers

A notification can call a user back into the personalised cycle before they have decided to return. Push notifications and streak reminders operate alongside infinite scrolling and autoplay as core elements of an "algorithmic addiction loop" (Li, 2026, p. 2; Li, 2026, p. 3), acting as re-entry cues that simultaneously refresh the data used for further targeting. Each notification both refreshes engagement data and provides new targeting material for subsequent pushes.

Neurocognitively, the harm begins before the app opens. Notifications signalling potentially rewarding content trigger the "pursuit → transient satisfaction → further pursuit" cycle through anticipatory dopamine release. High-frequency stimulation induces Δ FosB expression (Li, 2026, p. 3; Li, 2026, p. 2) that supports addiction-like states, a particular concern for adolescents whose executive functions are still maturing. Compulsive-checking behaviour documented in broader "dopamine driven feedback loop" research (Hu, 2025, p. 2) necessarily interacts with notification policies that repeatedly invite re-entry.

Regulatory responses address this directly. Kostenko's AI Law Model prohibits gamification of addiction and exploitation of age-related vulnerabilities; for minors it bans endless scrolling, autoplay, streaks and

forced quests outright, restricting notifications to minimal, neutral informational messages with visible "no personalization" switches (Kostenko, 2025, p. 223). The confidentiality principle further prohibits using personal data for profiling aimed at influencing users without explicit consent (Kostenko, 2025, p. 44), directly constraining notification pipelines built on fine-grained behavioural histories. Dark-pattern research confirms that nagging notifications wear down resistance and generate decision fatigue, exploiting System 1 heuristics over deliberative System 2 processing (Khare et al., 2026, p. 2; Khare et al., 2026, p. 8).

These findings converge on a working taxonomy. High-frequency, personalised, difficult-to-disable prompts are high-risk for vulnerable users. Low-frequency, informational, consent-aligned notifications are lower risk (Li, 2026, p. 2; Li, 2026, p. 3; Hu, 2025, p. 2; Kostenko, 2025, p. 223). Clinical research on adolescents adds a caution: many conceal severe distress and seek coping strategies privately rather than disclose to caregivers (Azasu et al., 2026, p. 9), so notification-based re-engagement aimed at minors risks aggravating hidden crises and needs explicit safeguards. Since the underlying architectures rely on offline clustering and sequence mining to surface highly contextual re-engagement moments (Noorian Avval & Harounabadi, 2023, p. 2), limiting sequential, context-aware personalisation in notification pipelines may matter as much as restricting message content itself.

2.3 Behavioral Economics and Choice Architecture

2.3.1 Default Effects and Friction Costs in Digital Journeys

India's IRDAI banned pre-checked add-on insurance defaults in 2022 (Khare et al., 2026, p. 8) precisely because preselected options reduce cognitive effort enough to produce uninformed consent at population scale. Social platforms apply the same mechanism to engagement-maximising settings, embedding them into the recommender substrate so that status-quo bias keeps users there without requiring an affirmative choice (Hermansyah, 2025, p. 152; Hermansyah, 2025, p. 153). Friction costs work through the complementary mechanism. Roach-motel and subscription-trap patterns push exit costs up, and 47% of users report cancellation difficulty (Liu & Keshavarz, 2026, p. 4; Khare et al., 2026, p. 8); "nagging" notifications then create decision fatigue that steers them back toward platform-preferred options. Dual-process cognition explains the synergy: defaults hold users in fast System 1 processing, while sludge suppresses the reflective System 2 evaluation that meaningful cancellation or consent decisions require (Khare et al., 2026, p. 4). Choice screens and consent banners provide only partial relief. Counter-nudges, misleading button placement, and obfuscatory language can restore platform-aligned defaults without technically overriding regulatory requirements (Hermansyah, 2025, p. 9; Khare et al., 2026, p. 12), which is why DMA research ties friction reduction to the intensity of enforcement rather than the existence of rules alone (Hermansyah, 2025, p. 152; Hermansyah, 2025, p. 160). Default bias introduced at the consent layer propagates downstream into content selection, since recommender-driven engagement accounts for a large share of platform value (Guan, 2024, p. 3). Survey evidence reinforces the underlying awareness-trust gap: users widely recognise that algorithms shape what they see, yet this recognition does not translate into confidence in AI-curated content (Zhang & Liu, 2026, p. 20). Whether that low trust makes users more or less susceptible to platform-framed defaults is not established by this survey and is treated here as an open question, not a documented finding. Default settings and friction costs are treated throughout this analysis as central manipulation-risk indicators, in keeping with scholarly calls (Khare et al., 2026, p. 12) for bright-line prohibitions and mandatory UX audits rather than case-by-case adjudication.

2.3.2 Salience, Scarcity and Loss Aversion in Interface Design

Indian dark-pattern audits find "false urgency/scarcity" (misleading countdowns) and "interface interference" (visual hierarchy manipulation) on approximately 70% of audited platforms, layered over nagging and subscription traps (Khare et al., 2026, p. 8). The cognitive mechanism is prospect theory's loss asymmetry: losses loom larger than equivalent gains, so a countdown timer converts a routine purchase into a fabricated emergency by drawing System 1 attention and suppressing reflective appraisal (Bansal, 2026, p. 135; Bansal, 2026, p. 136). Executive interviews document the targeting logic: budget-conscious consumers receive "pay more later" narratives while higher-income segments receive exclusivity framing (Bansal, 2026, p. 138), a deliberate vulnerability-targeting distinction unrelated to product relevance. Executives describe deploying scarcity cues as routine competitive practice (Bansal, 2026, p. 135). The aggregate architecture is a three-layer system: a cognitive-trigger layer using scarcity and social proof, an algorithmic-personalisation layer delivering behaviourally-tailored timing, and a temporal-friction-reduction layer compressing the decision window (Bansal, 2026, p. 139). DSA Recital 69 flags vulnerability-exploiting targeted advertising as producing severe adverse effects; Article 26 mandates ad-targeting disclosure; and sensitive-data profiling is categorically banned (Quinelato, 2024, p. 18). These provisions establish the normative boundary between ordinary persuasion and the manipulation this taxonomy documents (Khare et al., 2026, p. 12; Sanikidze, 2026, p. 390). Because recommender personalisation shapes cultural-content distribution at scale (Rafie et al., 2025, p. 3), optimisation choices carry inherent normative weight and can reinforce the cognitive vulnerabilities urgency cues are designed to exploit.

2.3.3 Temporal Discounting and Instant Gratification Online

At approximately 60 minutes per day across more than 21 daily sessions (Li, 2026, p. 3), average Douyin usage is the direct product of micro-duration clips and infinite swipe navigation making "just one more" nearly effortless. Micro-duration formats align structurally with temporal discounting: immediate low-cost rewards outcompete distant, effortful alternatives, producing measurable correlations between usage patterns and diminished academic concentration among adolescents (Li, 2026, p. 2; Li, 2026, p. 3). The neuropsychology accelerates the effect. Dopamine surges occur at the moment of anticipation, not consumption, mirroring slot-machine variable rewards; chronic stimulation elevates reward thresholds and reduces the hedonic value of slower, learning-oriented activities (Li, 2026, p. 4). Hu's "algorithm driven engagement worship" documents the cumulative impact: 22% higher compulsive-checking rates and 23% higher depressive-symptom incidence on hyper-personalised feeds (Hu, 2025, p. 2; Hu, 2025, p. 20). Neuroimaging links doomscrolling to reduced prefrontal activity resembling substance-dependence profiles. A 2,239-participant I-PACE study found that past and present temporal focus positively correlated with addiction while future orientation was protective (Li, 2026, p. 4). The mediation pathway split by self-control type: inhibitory self-control triggered a "resistance backfire effect," while initiatory self-control supported substitute-behaviour initiation. Re-ranking approaches can moderate popularity and exposure biases while retaining acceptable recommendation accuracy (Sánchez et al., 2023, p. 3), confirming that maximising instant engagement is a design choice, not a technical constraint. DSA Article 25 bans interfaces that "substantially impair" free decisions (Quinelato, 2024, p. 20; Li, 2026, p. 2), and the DSA's active TikTok investigation has scrutinised infinite scroll and autoplay specifically. Literacy surveys show users remain susceptible to instant-gratification design despite high self-reported confidence in everyday digital tasks (Chukhray & Koval, 2026, p. 31). Individual countermeasures alone are unlikely to close that gap; platform-side structural change (Kostenko, 2025, p. 99) sits closer to the source of the problem.

2.4 Nudging, Dark Patterns and Ethical Boundaries

2.4.1 From Helpful Nudges to Manipulative Dark Patterns

Ethical choice architecture and dark-pattern manipulation share a structural surface but differ in what they do to user comprehension and reversibility (Liu & Keshavarz, 2026, pp. 32–33). A recommender that hides cheaper alternatives or makes refusal deliberately difficult does not clarify a choice; it defeats one. Drip pricing and confirm-shaming trigger fast System 1 processing; obstruction and forced action constitute "sludge" (Khare et al., 2026, p. 2; Khare et al., 2026, p. 4) that actively suppresses the reflective System 2 evaluation protective decisions require. Large-scale marketing crawls document the scale: thousands of dark-pattern instances including asymmetric cookie consent and hidden fees increase immediate compliance while eroding perceived fairness and long-term customer equity (Liu & Keshavarz, 2026, p. 33). Indian empirical evidence illustrates the power asymmetry directly: 87% of major platforms deploy at least one dark pattern, yet only 39% of users recognise the term while 68% report having experienced deception (Khare et al., 2026, p. 2). India's Advertising Standards Council has now defined thirteen dark-pattern categories (Khare et al., 2026, p. 3); judicial rulings against concealed BookMyShow fees confirm that interface-level deception falls under consumer law; and IRDAI has banned pre-checked insurance defaults outright. The gap between regulatory text and actual interface behaviour is substantial. Regulatory choice screens provide only partial protection, because platforms can neutralise them with misleading button placement and obfuscatory language that preserves nominal autonomy while undermining its practical exercise (Hermansyah, 2025, p. 159). For the Ethical AI Audit Toolkit in Section 5, this motivates a clear evaluative boundary: interface elements qualify as acceptable nudges when they simplify decisions without degrading comprehension or reversibility, and fall within this report's dark-pattern categories when they rely on information distortion, interface asymmetry, social or temporal pressure, choice obstruction, or forced action (Liu & Keshavarz, 2026, pp. 32–33; Khare et al., 2026, p. 2; Khare et al., 2026, p. 4).

2.4.2 Common Dark Interface Strategies in Social Media

Interface interference and obstruction appear on approximately 70% and 63% of major Indian platforms respectively; drip pricing, bait-and-switch, and privacy zuckering affect 75%, 48%, and 44% of consumers in the same content analysis (Khare et al., 2026, p. 7; Khare et al., 2026, p. 9). Social platforms adapt these patterns from e-commerce and embed them in feeds, consent flows, and notifications tightly coupled to recommender engines (Khare et al., 2026, p. 7; Liu & Keshavarz, 2026, p. 3), with the social-media equivalents being ambiguous consent dialogs and auto-enrolment into personalised feeds. Even where regulators introduce choice screens or interoperability mandates, dominant platforms respond with friction and defaults that neutralise the intervention (Hermansyah, 2025, p. 3; Hermansyah, 2025, p. 10). AI-enabled environments add escalation patterns. "Adaptive UI and friction shifting" uses reinforcement learning to dynamically adjust interface steps, trapping users in roach-motel cancellation journeys; "chat-bot coercion" deploys conversational agents with sycophantic tone and fabricated consensus to frame cancellation as socially deviant (Sanikidze, 2026, p. 11). Large language models compound this further by mass-producing indistinguishable fake testimonials, while confirm-shaming personalises guilt-inducing opt-out language using demographic history (Sanikidze, 2026, p. 11); fabricated "friends liked this" indicators manufacture belonging-based pressure at scale. Non-compliance data confirm the systemic character: 87% of major Indian platforms deploy at least one dark pattern, 81% self-declared "dark pattern free" during CCPA audits, and complaint resolution reached only 28% (Khare et al., 2026, p. 7; Khare et al., 2026, p. 12). Each pattern in this cluster, interface interference, obstruction, AI-driven friction shifting, and

conversational coercion, directly impairs autonomous exit and is classified as a central item in this report's working taxonomy (Liu & Keshavarz, 2026, p. 3; Khare et al., 2026, p. 7; Sanikidze, 2026, p. 11).

2.4.3 Moral and Legal Contours of Manipulative Design

Kostenko's ethical-responsibility principle formulates the moral dimension explicitly: AI systems must counter manipulative design through mandatory notification, explainability, and human control over critical decisions (Kostenko, 2025, p. 8; Kostenko, 2025, pp. 61–62), and addiction-causing systems are categorically prohibited. The TSAI regime provides the legal counterpart: access to digital infrastructure is conditioned on registration, risk classification, and regular audits via an algorithmic passport (Kostenko, 2025, p. 243; Kostenko, 2025, p. 244), with violations triggering fines or identifier suspension. Indian e-commerce evidence documents how the underlying patterns manifest: interface interference, obstruction, and drip pricing are systemic features exploiting dual-process cognition, prompting calls for bright-line prohibitions and mandatory UX audits (Khare et al., 2026, p. 2; Khare et al., 2026, p. 4; Khare et al., 2026, p. 13). European instruments converge on the same objectives. The DSA bans interfaces that "substantially impair" free decisions; the DMA bans gatekeeper interface pressure; and the anticipated Digital Fairness Act targets addictive product design directly (Sanikidze, 2026, p. 389; Sanikidze, 2026, p. 390), shifting EU policy from reactive enforcement toward ex-ante prevention. Georgia's EU-aligned consumer law demonstrates that existing consumer protection concepts already reach manipulative interfaces: the "Maxbosch"/BOSCH case sanctioned false countdown timers and deceptive brand imagery without AI-specific statutes (Sanikidze, 2026, p. 390; Sanikidze, 2026, p. 391), and Georgia's E-commerce Guideline requires disclosure of recommender parameters. For this report's audit framework, the working distinction follows: ethical design is documentable through algorithmic passports and audit trails; manipulative design, defined as intentional exploitation of cognitive vulnerability with opaque targeting and addiction-inducing engagement, falls within practices that AI law models seek to prohibit outright or subject to registration and sanction (Kostenko, 2025, p. 8; Kostenko, 2025, pp. 61–62; Sanikidze, 2026, p. 389; Khare et al., 2026, p. 13).

2.5 Motivational Psychology and Self-Determination

2.5.1 Autonomy, Competence and Relatedness in Digital Contexts

Kostenko's ethical-responsibility principle establishes that meaningful autonomy requires more than an off-switch: design must preserve users' capacity to understand and contest algorithmic influence (Kostenko, 2025, pp. 61–62), and systems causing addiction-inducing manipulation are prohibited outright. The accompanying right to a "psycho-emotionally safe digital environment" includes audit tools for one's own activity (Kostenko, 2025, p. 119; Li, 2026, p. 2), placing infinite scroll and high-frequency notifications within the scope of autonomy harm when they undermine critical thinking. Neural-network recommendation combined with variable reward schedules pushes users into "auto pilot" scrolling, already flagged by the DSA as potential manipulative design, with disproportionate impact on college students whose executive function is still maturing (Li, 2026, p. 2). Neuroimaging evidence confirms autonomy is constrained at the neurocognitive level, not only at the consent-screen level (Hu, 2025, p. 2). Competence faces a parallel gap. Literacy surveys report high self-assessed confidence in routine digital tasks but a mean score of only 2.31 out of 5 for recognising manipulation (Chukhray & Koval, 2026, p. 31). Kostenko's digital-hygiene rights, covering activity monitoring and profiling termination, address this "second-level divide" directly (Kostenko, 2025, p. 119). The Algorithmic Impact Navigator reinforces the audit implication:

broad stakeholder engagement surfaces harms that standard risk assessments miss (Leon, 2026, p. 2), and recommender audits should involve users and civil society alongside technical teams (Adanyin, 2024, p. 25). Relatedness manipulation is the least visible vector. Inflated engagement metrics via bots on Douyin and Xiaohongshu, combined with undisclosed influencer sponsorship, convert social validation into a tool for manufacturing artificial consensus (Wang, 2025, p. 13). Algorithms that preferentially amplify popular content reinforce herd behaviour and FOMO by embedding false relatedness signals directly into the recommendation architecture. Annual external audits and accessible complaint mechanisms, backed by revocable-consent requirements, are the operational remedies the literature proposes (Kostenko, 2025, p. 119; Kostenko, 2025, p. 81; Silva & Valença, 2026, p. 7). User-awareness campaigns can reduce susceptibility at the margins, but the structural conditions that degrade autonomy, competence, and relatedness remain intact until the design architecture changes (Kostenko, 2025, pp. 61–62; Kostenko, 2025, p. 119; Chukhray & Koval, 2026, p. 31; Wang, 2025, p. 13).

2.5.2 Impact of Algorithmic Environments on Intrinsic Motivation

Blooket's token system illustrates the design choice concisely: tokens function as a "currency for creative agency" when embedded in goal-oriented narratives, satisfying autonomy and competence; poorly calibrated extrinsic rewards produce the opposite (Anisa & Maisa, 2025, p. 7), crowding out internal satisfaction through overjustification. Short-video recommendation ecosystems represent the corrosive configuration at scale. Dopamine surges at the moment of scrolling anticipation drive a "pursuit → transient satisfaction → further pursuit" cycle concentrated among adolescents with lower self-control (Li, 2026, p. 2; Li, 2026, p. 3). Hu documents the downstream effect: 22% higher compulsive checking and reduced prefrontal activity in impulse-control regions (Hu, 2025, p. 2), making initiation of demanding future-oriented tasks harder even when underlying values are intact. Kostenko's ethical-responsibility principle bans addiction-causing systems (Kostenko, 2025, pp. 61–62); the accompanying confidentiality principle prohibits profiling-driven influence without explicit consent (Kostenko, 2025, p. 44). The anticipated Digital Fairness Act's addictive-design focus (Sanikidze, 2026, p. 390) and dual-process dark-pattern research on System 1/System 2 exploitation (Khare et al., 2026, p. 6) converge on the same motivational harm: architectures that substitute platform-defined reward schedules for users' own goals. The substitution is rarely disclosed. Counter-evidence matters for audit design. Gradually fading extrinsic-reward prominence internalises habits rather than locking users into point dependency (Anisa & Maisa, 2025, p. 7). An AI-augmented mental-health intervention in Nigeria found users valued AI as "good for first help, but not the only help," with human intermediaries essential to preserve self-directed coping (Grace Malachi et al., 2025, p. 9). Motivational impact in this report is operationalised as an externally assessable condition: whether design and data practices leave room for users' own long-term goals, evaluated against ethical-responsibility duties (Kostenko, 2025, p. 62; Li, 2026, p. 2; Hu, 2025, p. 2; Khare et al., 2026, p. 6).

2.5.3 Experiences of Control, Flow and Compulsion in App Use

1.074 billion Chinese short-video users, representing 95.4% of netizens, show excessive use linked to diminished academic concentration and behavioural addiction, concentrated among adolescents with lower self-control (Li, 2026, p. 2). Short-video platforms produce this outcome by coupling micro-duration clips with infinite swipe navigation (Li, 2026, p. 2; Li, 2026, p. 3), creating a closed personalised-delivery cycle that draws users into "auto pilot" scrolling. Dopaminergic dynamics blur flow and compulsion: surges occur at anticipation rather than consumption, mirroring slot-machine variable rewards. Chronic stimulation induces Δ FosB expression and elevates reward thresholds (Li, 2026, p. 3; Li, 2026, p. 4), reinforcing

neurobiological conditions that weaken self-governed choice even during sessions that feel like effortless immersion. Individual variation follows the I-PACE model. A 2,239-participant study found past and present temporal focus positively correlated with addiction while future orientation was protective (Li, 2026, p. 4), mediated by a "resistance backfire effect" under inhibitory self-control versus substitute-behaviour initiation under initiatory self-control. Broader social-media evidence records 22% higher compulsive-checking rates and reduced grey-matter density in impulse-control regions (Hu, 2025, p. 2; Hu, 2025, p. 20). Kostenko's model prohibits algorithms causing psychological pressure or forming addiction (Kostenko, 2025, pp. 61–62; Kostenko, 2025, p. 119), grounding the prohibition in an inalienable right to a digital environment free from exhaustion and manipulative influence. Recommendation architectures that systematically shift users from goal-directed engagement into dopamine-driven compulsive checking satisfy Kostenko's conditions for categorical prohibition. Audit practice should therefore classify subjective reports of "getting lost" in feeds and post-use exhaustion as misalignment signals requiring investigation.

2.6 European Digital Governance Frameworks

2.6.1 Digital Services Regulation of Online Platforms

The Digital Services Act (DSA) reframes recommender systems as objects of systemic accountability rather than neutral technical components. Platforms exceeding 45 million EU users ("very large online platforms," VLOPs) must conduct systemic risk assessments, submit to annual independent audits, and offer users at least one non-profiling recommendation option (Kustura, 2025, p. 7; T et al., 2026, p. 12). Three audit tiers emerge: internal risk assessments by providers, independent compliance audits under Chapter III, and vetted-researcher data access under Article 40(4) for regulators (Meßmer & Degeling, 2023, p. 18; Meßmer & Degeling, 2023, p. 19). Article 27 additionally obliges platforms to disclose the main parameters driving recommendations.

Whether these obligations have teeth is contested. Audits risk becoming procedural formalities where regulators (Bahangulu & Owusu-Berko, 2025, p. 9; Kustura, 2025, p. 9) lack the technical expertise to interrogate ranking algorithms, a fragmentation already visible in uneven GDPR enforcement across member states. Analysts nonetheless anticipate a "Brussels Effect," with DSA transparency and audit norms establishing a de facto global benchmark (Kustura, 2025, p. 8; Sikalo, 2026, p. 7). The systemic risk assessment requirement, non-profiling option, and audit-tier structure form the compliance floor this report's Ethical AI Audit Toolkit must internalise (T et al., 2026, p. 12; Kustura, 2025, p. 2).

2.6.2 Artificial Intelligence Risk Governance in the EU

Four risk tiers define the EU AI Act's architecture. Unacceptable risk (social scoring, real-time biometric surveillance, exploitation of psychological vulnerabilities) is banned outright. High-risk systems in education, employment and law enforcement require conformity assessment, technical documentation and human oversight. Limited risk carries transparency duties; minimal risk faces no mandatory obligations (Kostenko, 2025, p. 10; T et al., 2026, p. 12). Kostenko characterises this EU regime as the most systemic globally, contrasting it with the United States' sectoral soft-law approach and China's centralised, state-centric model. ISO/IEC JTC 1/SC 42 standards, for example ISO/IEC 24027 on bias and 24028 on trustworthiness, converge conceptually with the Act's risk categories but remain voluntary (Sankaran, 2025, p. 4; Sankaran, 2025, p. 5).

Institutional capacity remains the critical bottleneck. Even within the EU, auditing opaque, black-box models remains weak (Bahangulu & Owusu-Berko, 2025, p. 9; Kustura, 2025, p. 2), echoing uneven GDPR

Table 1: Selected AI risk governance features by region

Feature	EU AI Act	US federal	China
Legal form	Binding regulation	Soft guidance	Administrative regs
Risk tiers	4 level scheme	None unified	Domain specific
High risk duties	Conformity, oversight	Sectoral	State centric
Audit capacity	Mandated, still weak	Fragmented	Centralised

enforcement across member states. The AI Act interlocks with the DSA on systemic risk duties for VLOPs and with the DMA on gatekeeper competition duties. Together, these instruments push algorithmic accountability from individual rights toward infrastructural questions of market power. Whether national competent authorities can acquire the algorithmic-systems expertise to operationalise these duties within the EU AI Act's compliance deadlines is still an open question.

Concrete harm evidence is visible in adjacent domains. Algorithms trained on skewed school data have been shown (Luo, 2025, p. 2) to systematically underrate lower-income students, with exposed labels measurably reducing teacher engagement (a "digital self-fulfilling prophecy") the Act's dataset and audit requirements are designed to prevent. The Act's dataset and conformity assessment obligations are, in this light, a direct response to documented feedback loops between training-data composition and real-world outcome disparity.

2.6.3 Policy Trajectory toward a Digital Fairness Regulation

A gap sits at the centre of current EU digital regulation: neither the GDPR, the DSA, nor the AI Act currently classifies social recommender algorithms as high risk in their own right (T et al., 2026, p. 2248; Kustura, 2025, p. 8). Sanikidze describes a planned Digital Fairness Act (DFA), proposed for late 2026, that would regulate addictive product designs and unfair personalisation "across the board" (Sanikidze, 2026, p. 390), filling the gap between the DSA and the Unfair Commercial Practices Directive. If enacted at its proposed scope, the DFA would be the first instrument to address interface-level addiction risk as a primary regulatory target rather than a secondary side-effect of privacy or competition enforcement.

The existing regulatory stack covers overlapping but incomplete terrain: GDPR governs data privacy, the DSA governs platform accountability and non-profiled recommendation options, the DMA governs gatekeeper competition, and the AI Act governs risk-tiered system deployment. Enforcement practice already previews what a fairness-focused regime might catch. Georgia's Competition and Consumer Agency has sanctioned dark patterns (false countdown timers, fabricated scarcity, misleading brand associations) under general unfair-practices law, though those cases have not yet reached AI-driven dark patterns specifically (Sanikidze, 2026, p. 390). Recent DSA and UCPD enforcement against platform X and Temu, including fines for deceptive verification flows, signals regulatory impatience with interface pressure of exactly this kind.

Provider-commissioned reviews vary widely in independence (Dantas et al., 2025, p. 13), meaning credible compliance will require evidence of substantive, replicable testing rather than declarative documentation alone. For this thesis, which treats the DFA as a policy trajectory rather than settled law (T et al., 2026, p. 2248; Sanikidze, 2026, p. 390), the operational implication is to build audit dimensions for addictive design and unfair personalisation ahead of the DFA's formal text. Doing so ahead of formal enactment carries the acknowledged risk that the DFA's final scope differs from current proposals.

2.7 Transparency, Data Access and Audit Obligations

2.7.1 Platform Transparency Reporting Requirements

Transparency reporting under the DSA functions as the primary control surface for recommender accountability. VLOPs must publish recurring reports on disputes, suspensions and moderator staffing under Article 24, disclose the main parameters driving recommendations under Article 27, maintain an advertising repository under Article 39, and grant data access to regulators under Article 40 (Meßmer & Degeling, 2023, p. 19; Kustura, 2025, p. 7).

Table 2: Key DSA transparency reporting elements and audit implications

Regulatory hook	Required disclosure	Relevance for Ethical AI Audit
Art. 24 reports	Disputes, suspensions	Indicator of downstream harms and appeal pressure linked to recommender outputs
Art. 27 parameters	Main recommendation signals	Entry point for interrogating engagement optimisation and profiling dependence
Art. 39 ad repository	Ad content and targeting info	Enables cross checking of content vs ad recommendation consistency
Art. 40 data access	Data sharing to regulators	Conditions for external, scenario based audits of systemic risks

Meßmer and Degeling read these clauses as the backbone of a multi-actor audit regime, but warn that without detailed implementing acts, platforms could perform partial self-assessment while remaining formally compliant (Meßmer & Degeling, 2023, p. 6; Meßmer & Degeling, 2023, p. 19). Sebastian anticipates algorithmic impact assessments for high-risk domains such as credit scoring that resemble environmental impact reviews, with continuous rather than one-off disclosure because model drift can introduce new discrimination over time (Sebastian, 2025, p. 15). Kostenko’s AI Law Model generalises this into a transparency principle (Kostenko, 2025, p. 70) requiring disclosure of data sources, architectures, error probabilities and uncertainty scenarios, with covert or manipulative AI use treated as categorically unacceptable. The convergence across these three accounts points toward a common floor: disclosure must be specific enough to enable re-testing, not merely symbolic enough to satisfy a procedural checklist.

DSA reporting is a floor, not a ceiling. An internal audit dashboard should track recommendation exposure by content category, repeat exposure to flagged content, and non-profiled-option uptake. Translating Article 27 parameters and Kostenko’s disclosure requirements (T et al., 2026, p. 12; Kostenko, 2025, p. 70) into plain-language external statements lets regulators reconstruct how engagement objectives interact with fairness obligations. Without that translation layer, formal transparency duties remain inaccessible to non-specialist scrutiny.

2.7.2 Data Access for Regulators and Independent Researchers

Without data access, systemic auditing of recommenders is impossible. Article 40 of the DSA obliges VLOPs to grant access to Digital Services Coordinators, the Commission, and vetted researchers studying systemic risks such as civic-discourse effects (Meßmer & Degeling, 2023, p. 19; T et al., 2026, p. 12). Meßmer and Degeling operationalise this as a scenario-based model, mapping specific risk narratives

(eating-disorder promotion, for instance) to tangible Article 40 data requests rather than generic access (Meßmer & Degeling, 2023, p. 15). Kostenko's AI Law Model pushes the principle beyond EU platform law into cross-border "extraterritorial digital jurisdiction" (Kostenko, 2025, p. 249): any AI system affecting a State's territory or market falls under that State's incident-reporting, audit-cooperation and evidence-preservation duties, transmitted through supply chains via flowdown clauses. In practice, this means a platform operating solely from outside the EU could still be audited on the basis of users it reaches within a member state's jurisdiction.

Table 3: Data access regimes for AI and platform oversight

Dimension		DSA / EU platforms	AI Law Model (Kostenko)
Primary addressee		VLOPs with 45+ million users	All AI systems with impact on State territory
Legal hook		Art. 40 data access, vetted researchers	Extraterritorial digital jurisdiction, cross border liability
Data objects		Platform data for systemic risk research; ads repository (Art. 39)	Technical materials, logs, audit results, algorithmic passports
Research access		Vetted researchers for systemic risk studies	API access to registry attributes for scientific research
Logging requirement		Implied through audit and risk assessment duties	Explicit event logging, evidentiary preservation, log based arbitration

Formal access rights do not execute themselves. Regulators still struggle with black-box models that resist interpretation (Bahangulu & Owusu-Berko, 2025, p. 9), and Meßmer and Degeling anticipate a fragmented mix of mandatory and voluntary audits across platforms rather than a uniform regime (Meßmer & Degeling, 2023, p. 19). Work on AI-enhanced auditing in finance points toward human-AI collaboration: AI handles large-scale analytics while human experts retain contextual and ethical judgement (Chitturi, 2025, p. 3; Chitturi, 2025, p. 6). That hybrid still demands carefully designed interfaces and cognitive-load management for auditors. Without sustained investment in auditor capacity, the formal data-access architecture of Article 40 will remain largely symbolic in practice.

2.7.3 Documentation, Logging and Traceability Expectations

Kostenko's AI Law Model frames traceability as a foundational transparency and explainability requirement: any AI system affecting rights, security or wellbeing must be identifiable, traceable, explainable and accountable as a condition of lawful use, with decision logs, data pipelines and model parameters remaining interpretable enough for independent auditors to identify errors or bias (Kostenko, 2025, p. 35; Kostenko, 2025, p. 36).

Child-affecting systems raise this bar further, requiring a complete technical, methodological, legal and ethical dossier in which decision logs record configuration changes and human interventions; absent safety mechanisms or manipulative interfaces in child-facing contexts count as governance violations justifying suspension (Kostenko, 2025, p. 214). Territorial sovereignty clauses extend traceability into a jurisdictional requirement (Kostenko, 2025, p. 124): only AI systems registered in a National Register of AI Systems may operate within a State's territory, embedding logging and documentation directly into cross-border control. An unregistered system has no lawful path to deployment, whatever its technical merits.

Practice already shows viable implementations. Financial-AI systems stream SHAP-based feature attri-

butions to a write-once immutable ledger, with "circuit breakers" switching models to human-supervised mode when drift crosses a threshold (S, 2026, p. 7). HITL grading in education keeps audit trails of AI outputs and human overrides available for accrediting bodies to inspect (Selvam & González Vallejo, 2025, p. 15), with the evidentiary value of those logs dependent on their tamper-evidence and retention period.

Meßmer and Degeling caution that without firmer implementing acts (Meßmer & Degeling, 2023, p. 6; Meßmer & Degeling, 2023, p. 19), platforms may default to minimal documentation that is formally compliant but not meaningfully accountable. For an Ethical AI Audit Toolkit under an anticipated DFA, these strands converge on the same requirement set: structured technical dossiers and algorithmic passports, tamper-evident logs, documented human-oversight and appeal channels, and independent access for regulators and vetted researchers (Bahangulu & Owusu-Berko, 2025, p. 9; T et al., 2026, p. 12). Continuous audit architectures that stream live, queryable evidence to oversight bodies can double as an operational dataset for improving governance itself (Serhii, 2025, p. 2; Biru & Pallaprolu, 2025, p. 10). That dual function raises confidentiality and false-positive concerns regulators have not yet resolved.

2.8 Child and Youth Protection in Algorithmic Environments

2.8.1 Age-Sensitive Risk Assessment and Design Standards

Child-directed AI requires a qualitatively different assessment standard. Kostenko's AI Law Model mandates "secure by default" architecture for such systems: algorithmic logic, data categories and training processes must be understandable, verifiable and controllable across the system's entire lifecycle, not retrofitted as a cosmetic setting. Operationalisation runs through a multi-stakeholder documentation set (Child Impact Assessments, Child Safety Data Sheets, audit reports, and a minimum public disclosure baseline covering data sources, operator identity, retention policy, and the degree of AI participation in decisions affecting the child) (Kostenko, 2025, p. 214). Explainability is age-calibrated, requiring plain-language explanations at technical, functional and ethical levels. Children's rights tools, including refusal of profiling, data correction and deletion, and human review, must be implemented and surfaced rather than merely promised.

Deficient design carries direct enforcement consequences. Absent safety mechanisms or manipulative interfaces in child-affecting contexts are classified as governance violations (Kostenko, 2025, p. 223), empowering regulators to suspend systems and revoke certification. Age-appropriate design principles add working UX requirements: profiling, geolocation and biometrics disabled by default; explicit bans on endless scrolling, autoplay, streaks and exit penalties for minors; capped notifications; and symmetrical choices such as a visible "no personalisation" switch and one-click "why am I seeing this" explanations. Table 4 converts these requirements into this report's working audit grid.

2.8.2 Restrictions on Profiling, Targeting and Monetization

Certain restrictions on minor-directed profiling admit no exceptions. Kostenko's AI Law Model categorically bans targeted advertising personalised on a child's psychotype, neurobehavioural signs or predicted emotional state (Kostenko, 2025, p. 20; Kostenko, 2025, p. 224). Automated emotion and biometric tracking requires dual (child plus guardian) informed consent; absent that consent, it too is prohibited. Predictive profiling of a child's "future trajectories" is barred without qualification. Operators must instead offer "child information invisibility" modes, meaning service use without profiling, backed by minimal, non-identifying logs, because consent from children is treated as fragile regardless of its formal collection method (Kostenko, 2025, p. 223; Kostenko, 2025, p. 44).

Table 4: Working audit grid: age sensitive risk assessment and design standards

Dimension	Required evidence	Illustrative audit question
Child Impact Assessment	Documented risk analysis, threat models, Child Safety Data Sheet	Is there a current Child Impact Assessment covering profiling, attention capture, and recommendation logic (Kostenko, 2025, p. 214)?
Secure by default	Architecture, data maps, disabled profiling/geolocation at default	Do defaults disable profiling, geolocation, biometrics, and addictive mechanics for minors (Kostenko, 2025, p. 223)?
Explainability	Multi level, age appropriate explanations	Can a child access "brief/detailed/technical" explanations of "why am I seeing this" in simple language (Kostenko, 2025, p. 214; Kostenko, 2025, p. 223)?
Controls	No personalisation switch, parental controls, time limits	Are "lesson/sleep" modes and refusal of profiling available without service degradation (Kostenko, 2025, p. 223)?
Logging	Decision and event logs with minimum retention	Are tamper evident logs available to regulators and families in case of disputed decisions (Kostenko, 2025, p. 214; Kostenko, 2025, p. 191)?
Governance	Internal and external child safety audits, regulator dossier	Has an external audit of child safety been completed and provided to authorities (Kostenko, 2025, p. 214)?

EU platform law reinforces these constraints. DSA Article 28 prohibits profile-based behavioural advertising targeting at minors once a platform can reasonably infer a recipient is a child (Quinelato, 2024, p. 19), with YouTube Kids cited as a working example of ad delivery without personal-data collection. AI-powered health services for teenagers are barred from cross-platform tracking and third-party SDKs entirely (Kostenko, 2025, p. 223). Monetised "digital twin" simulacra require revocable, double consent, and political or commercial use of minors' simulacra is prohibited outright (Kostenko, 2025, p. 140). Breach is not treated as a minor compliance issue; regulators can suspend operation and impose liability on operators, suppliers and manufacturers simultaneously (Kostenko, 2025, p. 223; Kostenko, 2025, p. 44).

For this report's working taxonomy, any child-facing module combining profiling, targeting and monetisation is presumptively classified at "prohibited or critical risk" (Kostenko, 2025, p. 224; Quinelato, 2024, p. 19) unless the operator can demonstrate information-invisibility modes and strict separation of minor data from advertising and recommendation pipelines. This classification places the burden of proof squarely on the operator, which is consistent with how the Kostenko model handles fragile-consent populations generally.

2.8.3 Safeguards for Youth-Focused Platforms

Design choices on youth platforms shape developmental trajectories, a fact that distinguishes them from ordinary consumer products and justifies a qualitatively different duty of care. Kostenko's model operationalises this through age-segmented standards. Early childhood (0–6) requires zero data collection by default, no personalisation or advertising, and local-only processing. Junior schoolchildren (7–12) face bans on loot-box monetisation and manipulative retention mechanics (Kostenko, 2025, p. 217), including endless scroll, streaks and forced quests; AI decisions that affect this age group must be explained in child-appropriate language. The underlying rationale is developmental: retention mechanics that cause acceptable friction for adults can produce disproportionate distress or compulsive avoidance in 7-to-12-year-olds.

For ages 13–15 on teen social networks directly relevant to this audit, feeds must default to "detox mode": no endless scroll, no autoplay, no aggressive return triggers. Notifications are capped at one per day; profiles are closed by default; and a public Child Safety Data Sheet, quarterly Child Impact Assessments and annual independent audits are mandatory (Kostenko, 2025, p. 221). Older teens (16–17) gain expanded self-determination, but predictive profiling and data monetisation remain prohibited. An additional layer extends minimal-data-by-default and accessibility requirements to children with disabilities, displacement or economic hardship (Kostenko, 2025, p. 219), recognising that these groups face compounded exposure risk when algorithmic engagement is combined with social or economic vulnerability. These tiers convert developmental stage into a legally enforceable design specification.

For this report's working taxonomy, any youth platform combining profiling, behavioural advertising and addictive mechanics such as endless scroll is classified "high or unacceptable risk" regardless of engagement performance figures. A nationwide Register of Responsible Subjects of AI Systems backs this with accountability infrastructure, recording audit completions and sanctions, and placing non-compliant operators on a Register of Violators (Kostenko, 2025, p. 82). Applied evidence adds a necessary caution: mechanics such as streaks and gamified reminders that appear benign for adult users can edge into coercive nudging when aimed at minors (Maisarah et al., 2025, p. 5). Clear anonymity and privacy controls measurably increase users' willingness to disclose sensitive information (Maisarah et al., 2025, p. 5), which means design choices affecting privacy perception are functionally also choices about whether users will engage honestly with safeguards or route around them.

2.9 Human-Centered Design and Autonomy Preservation

2.9.1 Respect for Agency and Meaningful Choice

Kostenko's ethical-responsibility principle defines the operative standard: design must keep "autonomy... [and] the right to an informed decision" intact, explicitly prohibiting systems that manipulate behaviour or suppress autonomy (Kostenko, 2025, pp. 61–62). A complementary digital-ecology right requires a "fair, transparent, humane" environment free from intrusive information flows (Kostenko, 2025, p. 119). These two principles define autonomy not as an absence of influence but as a positive design obligation: every system feature must be testable against whether it expands or constrains the user's capacity for uncoerced decision-making.

Liu and Keshavarz define dark commercial patterns as designs that "steer, deceive, pressure, or obstruct consumers" in ways that impair "informed, preference consistent choice" (Liu & Keshavarz, 2026, p. 3; Liu & Keshavarz, 2026, p. 39). This thesis adopts that functional test as its agency-risk baseline. The DSA

operationalises the same boundary by requiring a non-profiled feed option and prohibiting interfaces that "substantially impair" free decisions (Kustura, 2025, p. 7; Quinelato, 2024, p. 18). The anticipated Digital Fairness Act extends this toward addictive design and unfair personalisation (Sanikidze, 2026, p. 390), with OECD/ICPEN analysis framing manipulative design as a systemic autonomy threat (Sanikidze, 2026, p. 391) that cannot be adequately addressed by individual consent mechanisms alone.

Literacy surveys complicate enforcement. Respondents show high self-assessed digital competence alongside a mean manipulation-detection score of only 2.31/5, despite moderate trust in EU regulation (Chukhray & Koval, 2026, p. 31). Liu and Keshavarz's "conversion-trust" framework translates this gap into three audit checks: information clarity, choice symmetry, and proportional reversibility (Liu & Keshavarz, 2026, p. 39). A European Commission study found 97% of 45 websites and 30 apps using deceptive hierarchy or concealment practices, social media platforms included (Quinelato, 2024, p. 13). Large-scale crawls independently confirm that cookie banners are asymmetrically designed: accepting tracking is systematically faster and easier than refusing it (Liu & Keshavarz, 2026, p. 33). The persistence of these patterns across independently sampled datasets suggests they reflect deliberate architectural choices rather than incidental design inconsistency.

The AI Law Model closes what interface design leaves open, mandating independent ethical reviews, internal audit mechanisms and annual AI Ethics Impact Reports (Kostenko, 2025, pp. 61–62), plus bans on self-replicating addictive environments and suspension powers over non-auditable systems (Kostenko, 2025, p. 119; Kostenko, 2025, p. 108). Agency is treated as respected in this thesis only where non-profiled feeds are genuinely accessible (Sanikidze, 2026, p. 390), consent flows meet symmetry and reversibility criteria (Liu & Keshavarz, 2026, p. 33), and documented ethical audits address addiction risk (Kostenko, 2025, pp. 61–62; Kostenko, 2025, p. 119). Absent all three conditions, the platform bears the regulatory burden of justifying what remained.

2.9.2 Explainability and Comprehensibility for Lay Users

Kostenko's transparency principle requires architecture, data sources, and implications to be "understood, accessible, verifiable, and explanatory" throughout the lifecycle, with covert AI use in socially significant contexts explicitly prohibited (Kostenko, 2025, p. 67). The standard is inherently stakeholder-specific: developers document internally, users need literacy-adapted interfaces, and regulators require a complete machine-readable dossier (Kostenko, 2025, p. 68). One-size-fits-all disclosure satisfies none of these audiences adequately, which is why the AI Law Model explicitly differentiates them rather than treating transparency as a single artefact.

Kostenko's digital safety-of-the-child provisions make the requirement direct for minors: systems must deliver "meaningful, structured and age-appropriate" interpretations free of jargon, backed by decision logs and independent child-safety audits (Kostenko, 2025, p. 214). Age-appropriate design adds mandatory "why am I seeing this" panels (Kostenko, 2025, p. 223). HITL grading demonstrates the toolkit at scale: SHAP explanations highlighted score-contributing features, instructors confirmed or overrode with dual-logged actions, and the result was measurably increased consistency and student trust (Selvam & González Vallejo, 2025, p. 16; Selvam & González Vallejo, 2025, p. 21).

Business-analytics research generalises LIME, SHAP, and counterfactual analysis as improving fairness and compliance when combined with human oversight (Bahangulu & Owusu-Berko, 2025, p. 11). Sebastian's financial-fairness research extends this to SHAP-based adverse-action notices detailing decision factors, backed by ongoing algorithmic impact assessments (Sebastian, 2025, p. 6; Sebastian, 2025,

p. 15). Gerr's recommender-specific analysis proposes XAI "why this content" models supporting both user understanding and regulator-side distributional fairness checks (Mariia, 2025, p. 8; Mariia, 2025, p. 30).

Explanation quality matters independently of availability. Opaque logic triggers guilt, anger, or "ethical disengagement" (Raaj S. et al., 2026, p. 8; Raaj S. et al., 2026, p. 191): users stop attending to warnings altogether when they cannot interpret the reasoning behind them. Four audit expectations follow: stakeholder-differentiated explanation layers (Kostenko, 2025, p. 68; Kostenko, 2025, p. 214); multi-level explanations combining local reasons with global ranking descriptions (Kostenko, 2025, p. 68; Mariia, 2025, p. 8); tamper-evident logging enabling contestation (Kostenko, 2025, p. 214; Kostenko, 2025, p. 241); and HITL mechanisms intelligible to both professionals and affected users (Selvam & González Vallejo, 2025, p. 16; Selvam & González Vallejo, 2025, p. 21). Journalism-ethics research frames this as institutional duty: verifiable AI-content labelling and audit logs are prerequisites for public confidence, not optional enhancements (Kostenko, 2025, p. 94; Olanipekun, 2025, p. 4). Where audit logs are absent, a user's right of contestation is formally intact but practically unreachable.

2.9.3 Meaningful Human Oversight of Automated Decisions

Kostenko's AI Law Model establishes the baseline: all lifecycle actors must implement internal ethical audit mechanisms and human-rights impact monitoring, with regulators empowered to suspend operation and annual AI Ethics Impact Reports mandated (Kostenko, 2025, pp. 61–62). Meaningful oversight means identifiable humans and institutions can be held accountable and can intervene. HITL grading supplies a concrete implementation template: a four-layer framework lets instructors approve, adjust, or override AI-generated scores, with all changes dual-logged against the model's initial assessment (Selvam & González Vallejo, 2025, p. 16). That traceability measurably strengthened trust, with 61% of surveyed students reporting higher confidence because the decision path was transparent and contestable (Selvam & González Vallejo, 2025, p. 8; Selvam & González Vallejo, 2025, p. 20).

Business-analytics research reinforces the same structure. Human-in-the-loop, on-the-loop, and over-the-loop arrangements covering pre-implementation review, monitoring with intervention authority, and ex-post audits are prerequisites for AI Act/GDPR compliance against discriminatory lending or biased recruitment (Bahangulu & Owusu-Berko, 2025, p. 7; Bahangulu & Owusu-Berko, 2025, p. 9), echoed in technical governance frameworks requiring explicit role descriptions for each oversight tier (Mohammed, 2025, p. 5). Hybrid public-sector governance case studies show (Alrwili & Kantharia, 2025, p. 2) that institutionalised oversight outperforms pure self-regulation: combined internal ethics boards and external regulators produced measurable reductions in bias complaints and faster appeal resolution. The appeal-resolution metric is particularly telling because it measures not only whether appeals existed but whether they were processed quickly enough to matter to affected users.

Table 5 translates these patterns into audit criteria for social recommenders.

SHAP-based attributions in HITL grading (Selvam & González Vallejo, 2025, p. 16; Selvam & González Vallejo, 2025, p. 21) enabled targeted overrides where model interpretation failed; the authors caution that AI alone cannot evaluate fairness or respond compassionately. Hoax-detection research reaches a parallel conclusion: low explainability undermines the legal usability of outputs (Syari et al., 2025, p. 3). Enforcement remains harder. Regulators need independent AI audits and external oversight committees to handle black-box opacity (Bahangulu & Owusu-Berko, 2025, p. 9). This thesis accordingly assesses social recommender oversight through observable workflows, logging, and governance structures documented

Table 5: Illustrative oversight patterns and audit questions

Pattern	Key features	Audit question
HITL grading (Selvam & González Vallejo, 2025, p. 16)	Human validation layer, dual logging, SHAP explanations	Are recommendation overrides and rationales logged and attributable to responsible staff?
Hybrid DSS governance (Alrwili & Kantharia, 2025, p. 2)	Internal ethics board, external regulator, appeal portal	Does the platform support user appeals and independent audits of algorithmic outcomes?
AI Law ethical responsibility (Kostenko, 2025, pp. 61–62)	Ban on manipulative and addictive systems, ethics impact reports	Are engagement optimising features assessed for digital addiction and reported in ethics disclosures?
HITL roles in high risk AI (Mohammed, 2025, p. 5; Bahangulu & Owusu-Berko, 2025, p. 7)	Defined human in/on/over the loop responsibilities	Are human oversight roles for recommender decisions explicitly documented and resourced?

in secondary sources. Meaningful oversight requires verifiable human authority and institutional capacity to correct or shut down engagement strategies that breach ethical-responsibility norms (Kostenko, 2025, pp. 61–62; Alrwili & Kantharia, 2025, p. 2). Without that capacity, accountability exists on paper only.

2.10 Justice, Non-Discrimination and Inclusion

2.10.1 Bias, Profiling and Indirect Discrimination Risks

Adaptive-education case evidence illustrates the structural pattern. Minority and non-English-speaking students are disproportionately flagged for remedial instruction because their response patterns are underrepresented in training data (Chakraborty et al., 2025, p. 5); rural low-connectivity students face a compounding infrastructural exclusion axis. An AI Now Institute analysis found 65% of examined educational AI studies showed bias against marginalised groups (Chakraborty et al., 2025, p. 7), suggesting the pattern is not an edge case but a predictable output of training pipelines that under-sample minority populations.

Sebastian's sociotechnical critique of credit-scoring warns that single-metric optimisation using demographic parity, equalised odds, or counterfactual fairness can conceal intersectional discrimination even in apparently balanced models (Sebastian, 2025, p. 14). That risk points recommender audits toward intersectional exposure checks rather than aggregate parity alone (Sebastian, 2025, p. 14; Chakraborty et al., 2025, p. 7). Kostenko's fairness principle operationalises this, prohibiting models built on non-representative data or lacking bias-correction mechanisms, requiring "equity by default," and mandating vulnerable-group participation in development (Kostenko, 2025, p. 73; Kostenko, 2025, p. 41). Obligations are distributed across the lifecycle: preliminary bias audits, rebalancing techniques, and user rights to review and appeal, overseen by an Authorized Body for Ethics of AI maintaining a public discrimination-complaint register (Kostenko, 2025, p. 41; Kostenko, 2025, p. 42).

Research on ethical e-commerce shows engagement-oriented algorithms deprioritising sustainability attributes (Raaj S. et al., 2026, p. 4; Raaj S. et al., 2026, p. 10), with unverifiable green claims widening

the intention-behaviour gap. The EU AI Act's unacceptable-risk tier covers psychological-vulnerability exploitation; proposed algorithmic impact assessments and demographic-performance transparency reports sharpen the regulatory stakes further (T et al., 2026, p. 12; Sebastian, 2025, p. 14; Sebastian, 2025, p. 15). ISO/IEC bias-mitigation reference standards extend these expectations to technical design (Sankaran, 2025, p. 3). Sebastian's finding (Sebastian, 2025, p. 14) that single-metric fixes can conceal intersectional harm is the empirical basis for treating bias, profiling, and indirect discrimination as regulated risk categories requiring documented metrics and external oversight (Kostenko, 2025, p. 73; Kostenko, 2025, p. 41). Aggregate parity alone does not constitute compliance.

2.10.2 Protection against Social Rating and Total Scoring

Social recommenders can drift toward quasi-universal value assessments when engagement or trust metrics are aggregated and reused across contexts. Kostenko's non-simulacry principle explicitly prohibits this: digital twins and shadow profiles must not replace a person's autonomy or self-determination (Kostenko, 2025, p. 128). The operationalised ban covers "total scoring" that influences access to labour, credit, insurance, or justice through multi-domain data aggregation, cross-platform ID stitching, function creep, or proxy-variable discrimination.

Only narrowly targeted scoring remains permissible under five cumulative conditions: a specific lawful purpose confined to one domain, non-transferability, exclusion of protected-feature proxies, auditable transparency, and mandatory human-adjustment review (Kostenko, 2025, p. 130). Sebastian's domain-specific fairness constraints in financial AI echo the same logic (Sebastian, 2025, p. 3). The implication for social recommenders is a clear boundary: engagement or trust scores generated within a platform cannot legitimately migrate into gatekeeping for unrelated rights or services.

Meaningful human oversight is the structural antidote. Kostenko's AI Law Model bans configurations that neutralise supervision, requiring mandatory human participation and external audits for justice, health, labour and finance decisions, with automatic denial of basic services explicitly disallowed (Kostenko, 2025, p. 127). Event logs and appeal-rate monitoring detect "rubberstamping" (Kostenko, 2025, p. 129). Cross-border provisions extend extraterritorial jurisdiction where AI decisions significantly affect a State's rights, security, or economy, triggering joint liability for providers, integrators, and data brokers under "flow down" due-diligence clauses (Kostenko, 2025, p. 250). This thesis defines Code Category C: "Prohibited Total Scoring" as covering any practice that aggregates multi-domain data into a universal trust index (Kostenko, 2025, p. 128), extends engagement scores via ID stitching into credit or hiring decisions, or maintains unreviewed blacklists derived from recommender traces (Kostenko, 2025, p. 130). Systems operating within domain-specific, proxy-excluding, human-reviewed scoring qualify as "Targeted Scoring under Oversight" instead, though still requiring documented fairness audits (Bahangulu & Owusu-Berko, 2025, p. 6; Bayona, 2025, p. 6; Sebastian, 2025, p. 3). Kostenko's five-condition permissibility test (Kostenko, 2025, p. 130) is the operational dividing line between the two categories.

2.10.3 Cultural and Linguistic Diversity in Content Curation

When algorithms consistently surface dominant-language content at the expense of minority voices, the effect is structural exclusion. Kostenko's dedicated cultural-diversity principle prohibits algorithms that impose cultural stereotypes or contradict national, ethnic, or linguistic identity, requiring cultural-sensitivity mechanisms, bias testing, and complaint channels. Language support is treated as structural non-discrimination: AI systems must support the state language plus, "within technical feasibility," minority languages without

degrading translation quality or response speed (Kostenko, 2025, p. 182). For recommenders ranking news or educational content, multilingual parity is not discretionary.

Kostenko's broader fairness provisions reinforce the obligation: systems trained on data with systematic cultural bias require multidimensional fairness testing (Kostenko, 2025, p. 73), along with ongoing fairness-metric audits and participation of diverse groups under a "fairness by design" approach (Kostenko, 2025, p. 75). Without those mechanisms in place, recommenders built on majority-language corpora will replicate structural exclusion as a byproduct of their training data.

An AI-supported mental health audit in Nigeria illustrates the tested stakes. Seven pilots used training corpora containing less than 3% Fulfulde, Tiv, and Eggon content; Fulfulde proverbs were misflagged as "hostile," and Hausa female weeping was misclassified as "anger," causing PTSD under-detection. Algorithmic sensitivity was measurably higher for urban literate users than rural low-literacy groups (Grace Malachi et al., 2025, p. 10). Community Health Extension Workers acting as "algorithmic interpreters" partially mitigated the gap, translating outputs into culturally meaningful narratives and reducing intergroup prejudice through mixed audit-circle governance (Grace Malachi et al., 2025, p. 11; Grace Malachi et al., 2025, p. 2). AI-enabled career-counselling research reaches a parallel conclusion: since large language models are trained primarily on English-language corpora, responsible implementation requires localisation and cultural adaptation (Malik, 2024, p. 9762591). Kostenko's algorithmic-education principle adds that AI-literacy materials must be available in multiple languages adapted to age and social group (Kostenko, 2025, p. 98). PTSD under-detection documented in the Nigerian pilots (Grace Malachi et al., 2025, p. 10) illustrates the clinical stakes of this structural root: majority-group training data produces minority-group error rates measurably above baseline, and no amount of post-deployment correction substitutes for representative corpora at the design stage.

Gerr's diversity-index methodology using Shannon entropy and Simpson's index on Facebook, YouTube, and TikTok (Mariia, 2025, p. 3; Mariia, 2025, p. 5) provides a measurement template extendable to cultural and linguistic exposure fairness. HITL grading research confirms the corrective value of human oversight: in an 800-essay pilot, 13% of AI scores were fully overridden because the model misread culturally embedded language from second-language writers (Mariia, 2025, p. 8; Selvam & González Vallejo, 2025, p. 19). The SHAP output identified which features drove the anomalous scores; the educators then determined whether those features reflected genuine writing weakness or cultural encoding the model had not been trained to handle.

2.11 Wellbeing, Safety and Prevention of Harm

2.11.1 Psychological and Social Harm from Addictive Design

Among 1.074 billion Chinese short-video users, college students show diminished academic concentration, weakened social skills, and heightened anxiety and depression under excessive use (Li, 2026, p. 2). Li identifies the mechanism: neural-network recommendation combined with micro-duration clips, infinite scrolling, and push notifications forms an "addiction loop." Hu's synthesis links these mechanics to "dopamine driven feedback loops," with 22% higher compulsive-checking rates in algorithmically curated feeds (Hu, 2025, p. 2). Prolonged personalised-feed use further correlates with reduced grey-matter density and prefrontal activity, structurally comparable to substance dependence, and a 23% higher incidence of depressive symptoms than chronological-feed controls (Hu, 2025, p. 20; Li, 2026, p. 3; Li, 2026, p. 4).

Harm extends beyond the individual. Engagement-driven curation (Hu, 2025, p. 19) amplifies misinfor-

mation and polarisation while users remain unaware of the underlying mechanism. Gerr's European platform analysis shows personalisation systematically diminishing heterogeneous viewpoints (Mariia, 2025, p. 29). Instagram's Eurocentric beauty-standard prioritisation is linked to increased body dissatisfaction among adolescent women of colour (Hu, 2025, p. 3). The DSA's TikTok investigation singles out infinite scroll and autoplay as potential manipulative design (Li, 2026, p. 2), and the AI Act's unacceptable-risk tier (T et al., 2026, p. 2248) bans systems exploiting psychological vulnerabilities. The anticipated Digital Fairness Act's (Sanikidze, 2026, p. 390) prioritisation of addictive product design indicates this is moving from soft guidance toward enforceable constraint.

Kostenko's ethical-responsibility principle ties this directly to prohibited conduct, banning systems causing digital addiction or suppressing autonomy (Kostenko, 2025, pp. 61–62), with confidentiality provisions barring profiling-driven advertising that restricts free choice without explicit consent (Kostenko, 2025, p. 45). Dark-pattern research on Indian e-commerce (Khare et al., 2026, p. 8; Khare et al., 2026, p. 7) confirms that the same cognitive mechanisms (default bias, social pressure, and friction-loaded exits) operate through interface structure alone, without requiring algorithmic intent. The Digital Fairness Act (Sanikidze, 2026, p. 390) is expected to codify this recognition: structural harm through design is not a lesser category of harm than programmatic manipulation.

Hu's Algorithmic Impact Assessment Scale (AIAS-5) quantifies emotional volatility, sleep disruption, and compulsive usage, feeding a proposed "digital nutrition label" disclosure regime and dynamic transparency index (Hu, 2025, p. 20; Hu, 2025, p. 19). Corporate-governance research adds independent audit trails as a mechanism for surfacing ethical liabilities (VERMA, 2025, p. 12). Engagement strategies materially elevating AIAS-style dimensions, especially combined with opaque profiling and no meaningful exit option, fall into high-risk or prohibited zones in this thesis (T et al., 2026, p. 2248; Sanikidze, 2026, p. 390). HITL grading research (Selvam & González Vallejo, 2025, p. 21; Kostenko, 2025, pp. 61–62) offers a counter-example: frameworks maintaining educator authority and transparent dual-sourced feedback reported improved student trust, demonstrating that comparable profiling technology can produce better outcomes when governed accountably rather than optimised for retention alone.

2.11.2 Risk Thresholds and Unacceptable Practices

The EU AI Act's "unacceptable risk" tier explicitly names systems exploiting psychological vulnerabilities alongside social scoring (T et al., 2026, p. 2248). Kostenko's AI Law Model converges on the same boundary, prohibiting behaviour manipulation and digital addiction as a mandatory certification criterion (Kostenko, 2025, pp. 61–62). Together these define Code Category C, "Unacceptable Manipulative Practices": designs intentionally exploiting psychological vulnerabilities (T et al., 2026, p. 2248; Kostenko, 2025, pp. 61–62), addictive infinite-scroll and variable-reward mechanics without meaningful exit, and architectures blocking the DSA-mandated non-profiled feed option. The anticipated Digital Fairness Act's targeting of addictive design suggests structural constraint rather than mere discouragement (Sanikidze, 2026, p. 390). That convergence also has practical enforcement consequences: Code Category C violations now expose platforms to overlapping liability under at least three distinct enforcement regimes simultaneously.

A second ceiling covers discrimination and social rating. Sebastian's pre-deployment fairness verification research warns that single-metric fixes mask intersectional harm (Sebastian, 2025, p. 14). Kostenko's fairness principle bans social-scoring "integral assessments" affecting service access (Kostenko, 2025, p. 41). Cross-domain trust indices derived from engagement traces or unmitigated disparate content

exposure fall under Code Category C: "Unacceptable Scoring and Discrimination" (Kostenko, 2025, p. 41; Sebastian, 2025, p. 14).

Between prohibited and compliant practice, operational thresholds are achievable. The Algorithmic Impact Navigator (Leon, 2026, p. 2742041) documents this specifically: a 30% lending-processing-time reduction remained fair where denial gaps were statistically insignificant, given routine audits and continuous monitoring. Sebastian's domain-specific fairness constraints (Sebastian, 2025, p. 14) supply the design logic: hybrid human-AI governance with compulsory authorisation for sensitive decisions built into workflows is the configuration this thesis classifies as Code Category B, "High Risk but Potentially Mitigable."

Unacceptable practice is as much a governance failure as a design one. A Nigerian mental-health AI trial improved PTSD remission under Pastoral Data Trust governance (Grace Malachi et al., 2025, p. 5419804), yet also documented a false-positive suicide alert triggering a violent vigilante patrol and a privacy breach from bypassed controls. Conservative thresholds and enforceable misuse sanctions are non-negotiable. Findings combining high engagement optimisation with weak governance are classified Code Category C in this thesis regardless of nominal metric compliance (Kostenko, 2025, pp. 61–62). ISO/IEC risk-management standards and AI Act/GDPR compliance research (Sankaran, 2025, p. 8589961; Bahangulu & Owusu-Berko, 2025, p. 7461574; T et al., 2026, p. 2248; Sanikidze, 2026, p. 390) extend these expectations globally. The Nigerian trial's false-positive suicide alert and the resulting vigilante patrol (Grace Malachi et al., 2025, p. 5419804) are the clearest illustration of what metric compliance without enforceable governance produces in practice: documented harm that nominal compliance scores cannot retroactively prevent.

2.11.3 Duty of Care for Vulnerable and Minor Users

Kostenko's digital-safety-of-the-child principle requires systems processing children's data to be "secure by default" across the lifecycle, with developers maintaining Child Impact Assessments and Child Safety Data Sheets, and regulators entitled to full technical dossiers (Kostenko, 2025, p. 214). Explainability standards (Kostenko, 2025, p. 214) are explicitly age-sensitive: plain-language decision-logic interpretations and children's-rights tools such as profiling refusal are mandatory, and manipulative or obscure interfaces are classified as violating the child's best interests. This classification carries regulatory teeth: it authorises suspension, not merely a reprimand.

Age-appropriate design provisions translate these principles into observable interface constraints. Minors' defaults must minimise data collection and disable profiling. Endless scrolling, autoplay, and forced-exit penalties are prohibited. "No personalisation" switches and "why am I seeing this" explanations are mandatory (Kostenko, 2025, p. 223). These requirements map directly onto the short-video mechanics tied to dopamine-driven compulsive checking and depressive symptoms among adolescents (Li, 2026, p. 2; Hu, 2025, p. 2). Hu's 23% higher incidence of depressive symptoms in algorithmically curated versus chronological-feed controls (Hu, 2025, p. 20) is a particular measure of that exposure for a group that lacks fully developed inhibitory control.

Protection extends to profiling and monetisation. Psychotype- or emotion-based targeted advertising to minors is banned (Kostenko, 2025, p. 20; Kostenko, 2025, p. 224). Predictive "life path" profiling is outlawed, and data resale or third-party model training requires a separate best-interests legal basis (Kostenko, 2025, p. 223). Business models premised on behavioural advertising to minors conflict with this normative baseline outright. The DSA's systemic-risk framework for very large platforms and the UK Online Safety Act's parallel duty-of-care regime both impose stricter requirements for child-harmful content (T et al., 2026,

p. 2248; T et al., 2026, p. 2249). UNESCO's AI Ethics Recommendation adds a global reference point calling for algorithmic diversity to ensure pluralism, positioning platforms as custodians of psychological welfare (T et al., 2026, p. 2249). A platform whose engagement metrics confirm high adolescent usage has the data to know the audience; the question is whether governance structures require acting on that knowledge.

Operationalising this requires governance architecture beyond UX adjustments. Kostenko's ethical-responsibility principle mandates independent ethics commissions with suspension authority and annual AI Ethics Impact Reports (Kostenko, 2025, pp. 61–62). HITL grading demonstrates such structures in practice: mandatory educator confirmation and dual logging produced measurably higher student trust (Selvam & González Vallejo, 2025, p. 20). Corporate human-AI audit frameworks similarly show clearly defined roles improving detection power without sacrificing professional judgment (Chitturi, 2025, p. 2; Chitturi, 2025, p. 3). Platforms serving minors that combine intensive profiling, addictive mechanics, and undocumented child-safety dossiers fall into "Code Category C: Prohibited or Critical Risk"; secure-default, age-appropriate, human-audited configurations exemplify duty-of-care-compliant design (Kostenko, 2025, p. 214; Kostenko, 2025, p. 223; Asante & Adeoba, 2026, p. 16). Regulators should additionally scrutinise how facial and interactional data are repurposed for psychometric targeting of children, since innocuous signals can be recombined to reveal sensitive traits (Bojić et al., 2021, p. 4). Filling that gap requires documented accountability chains at each handoff point: from first-party collection through third-party resale and downstream model training, with a named responsible party at every stage.

3 Qualitative Methodology and Audit Framework

3.1 Overall Qualitative Research Design

3.1.1 Choice of Secondary Qualitative Synthesis

Proprietary recommender logs are unavailable. That structural fact, not methodological preference, drives the choice of secondary qualitative synthesis here. Rich documentary material exists across regulatory, managerial and ethical domains. Dark-pattern research on Indian e-commerce justifies exactly this design where "primary experimentation or access to proprietary platform data is limited" (Khare et al., 2026, p. 6). Asante and Adeoba's Reflexive Thematic Analysis spanning marketing, information systems and CSR literature (Asante & Adeoba, 2026, p. 8; Asante & Adeoba, 2026, p. 10) confirms that the same analytic backbone transfers across domains, as does Malik's thematic synthesis of policy and practice sources for AI-enabled career counselling (Malik, 2024, p. 9762591). Systematic coding of secondary material generates analytical themes around addiction, manipulation and auditability, with every coding decision explicitly documented. Line-by-line coding and multi-coder CASP appraisal used in clinical-ethics thematic synthesis (Unknown, n.d., p. 7) supply the procedural template.

Understanding how the AI Law Model and DSA-style regimes articulate risk-management duties is a conceptual question, not an estimation task. The doctrinal-empirical hybrid Khare et al. (Khare et al., 2026, p. 6) use to evaluate regulatory adequacy under proprietary secrecy is the closer methodological analogue. Quality safeguards follow Lincoln and Guba's trustworthiness criteria as operationalised by Asante and Adeoba, specifically methodological description, audit trails and confirmability (Asante & Adeoba, 2026, p. 10). This design avoids direct human-subjects engagement, relying instead on regulatory reports, judicial decisions and complaint statistics. Kostenko's AI Law Model (Kostenko, 2025, p. 54) calls for

Human Rights Impact Assessments and incident logs grounded in ISO/IEC 23894 and the NIST AI RMF, making thematic synthesis of textual artefacts both a methodological necessity and thematically consistent with the audit-toolkit target (Kostenko, 2025, p. 12; Nesterov, 2024, p. 23).

3.1.2 Alignment between Research Questions and Methods

Three research questions, fixed since the approved Capstone exposé (see Appendix A), drive this project. Each addresses conceptual relationships rather than statistical effects: how engagement-optimised recommenders instantiate manipulative design and algorithmic addiction; how EU-style regulation frames duties and prohibitions; and how an Ethical AI Audit Toolkit can be constructed using CSR-based organisational governance. Secondary qualitative synthesis fits this configuration because the aim is to map patterns and interpret mechanisms, not estimate causal magnitudes. Khare et al. describe the same stance: their goal is to "map patterns, interpret behavioural mechanisms, and evaluate regulatory adequacy" rather than estimate causal effects (Khare et al., 2026, p. 6). Asante and Adeoba similarly use secondary synthesis (Asante & Adeoba, 2026, p. 8; Asante & Adeoba, 2026, p. 10) to theorise how Communities of Practice and CSR mediate ethical AI implementation.

Gerr's convergent mixed-method design pairing DSA/AI Act policy-document analysis with bias and diversity metrics on Facebook, YouTube and TikTok (Mariia, 2025, p. 3) provides a methodological anchor for the policy-document coding used here, even though this thesis does not reproduce the quantitative component. Coding transparency draws further on the MMR-RHS checklist's emphasis on documenting sampling, integration logic and credibility strategies (Tovin & Wormley, 2023, p. 2), and on the grounded-theory triangulation practices Kishore et al. (Raaj S. et al., 2026, p. 6) use for AI-mediated finance. Absent new interviews, the same coding logic is applied to documentary material instead, with that substitution declared as an explicit limitation. Research questions are formulated at the level of observable behaviour and documented harm, explicitly excluding reverse engineering or telemetry collection. Asante and Adeoba and Khare et al. both work only with published accounts rather than operational-system access (Asante & Adeoba, 2026, p. 11; Khare et al., 2026, p. 6). Mixed-methods mental-health research using an ethnographically co-designed "CHEW Competency Wheel" (Grace Malachi et al., 2025, p. 4) further supports framing questions around AI interaction with vulnerable communities through qualitative design, and that precedent applies directly to the audit-regime questions at stake (Khare et al., 2026, p. 6; Asante & Adeoba, 2026, p. 8).

3.2 Data Sources and Selection Strategy

3.2.1 Regulatory and Policy Texts as Data

Regulatory and policy documents are not background. They are primary empirical data. Khare et al. code ASCI guidelines, CCPA advisories and enforcement orders such as BookMyShow (Khare et al., 2026, p. 6) directly as evidence of normative expectations and governance performance. Gerr treats DSA, the German Action Plan and EU AI Act texts the same way, coding them alongside platform policies for transparency and pluralism themes (Mariia, 2025, p. 3). Kostenko's AI Law Model is analytically rich data by the same standard. Its technological-sovereignty principle allows states to prohibit AI systems where independent audit is impossible or critical dependence exists (Kostenko, 2025, p. 107), and its generative-AI regime mandates registration, labelling and independent audits covering manipulative impact and bias (Kostenko, 2025, p. 31). Both clauses are coded directly as candidate audit-toolkit criteria in this thesis.

DSA analyses supply further discrete taxonomy variables. Very large platforms must conduct systemic

risk assessments, offer a non-profiled feed option, and submit to annual audits, with Article 40 granting researcher data access (Kustura, 2025, p. 7; Meßmer & Degeling, 2023, p. 19). The anticipated Digital Fairness Act targets addictive design and unfair personalisation, aiming to close gaps between the DSA, DMA and unfair-commercial-practices law (Sanikidze, 2026, p. 390). "Non-profiled feed requirement" is one such variable extracted directly from these texts. The EU AI Act's four-tier risk scheme, classifying systems exploiting psychological vulnerabilities as unacceptable risk, anchors Code Category C (T et al., 2026, p. 12). Kostenko's explicit bans on manipulative and addictive systems reinforce that anchor (Kostenko, 2025, pp. 61–62; Kostenko, 2025, p. 107). Regulatory texts also encode implementation-capacity limits. Research on Ukraine's "imitative modernization" of GDPR/AI Act/DSA frameworks documents a "double imitation" gap between formal transposition and effective governance (Sikalo, 2026, p. 19). Literacy surveys show high self-assessed digital competence but low ability to recognise manipulation, alongside scepticism about cross-border enforcement (Chukhray & Koval, 2026, p. 10). Both are coded here as contextual constraints on any audit toolkit that assumes strong supervisory capacity.

3.2.2 Academic and Grey Literature on Algorithmic Addiction

Two neuropsychological strands anchor coding of algorithmic addiction. Li's synthesis on short-video platforms identifies a clear mechanism: neural-network recommendation combined with 15–60 second clips, infinite swipe navigation, autoplay and push notifications creates what Li terms an "addiction loop," with Chinese usage data averaging near 60 minutes and 21+ sessions daily, especially among students (Li, 2026, p. 2; Li, 2026, p. 3). Dopamine surges occur before each swipe, structurally akin to slot-machine variable rewards. Chronic stimulation elevates reward thresholds (Li, 2026, p. 3; Li, 2026, p. 4). Hu adds quantitative weight: algorithm-driven feeds produce 22% higher compulsive-checking rates, link to depressive symptoms, and are associated with reduced prefrontal activity and reduced grey-matter density paralleling substance-dependence patterns (Hu, 2025, p. 2; Hu, 2025, p. 20). Kishore et al.'s grounded-theory study of AI-embedded financial tools identifies a second harm pathway: users treat algorithms as advisory rather than authoritative and often override recommendations, flagging automation bias and diminished agency (Raaj S. et al., 2026, p. 7; Raaj S. et al., 2026, p. 6).

Methodological precedents justify synthesising these diverse sources without primary data collection. A PRISMA-based review of AI-powered mobile proctoring screened 180 records and synthesised 20 on surveillance anxiety and algorithmic bias (Mogoi et al., 2025, p. 2; Mogoi et al., 2025, p. 9). A structured review of AI in social innovation (Hassen, 2025, p. 3) used inductive three-stage coding to identify bias and an "accountability void." A conceptual synthesis on AI-enabled career counselling derived governance clusters through secondary sources alone (Malik, 2024, p. 14). Together these precedents justify reliance on structured secondary searches and inductive coding. Grey and model-law texts are treated as data on the same terms: Kostenko's AI Law Model prohibits systems causing digital addiction or suppressing autonomy under the ethical-responsibility principle, articulating a right to a psycho-emotionally safe digital environment (Kostenko, 2025, pp. 61–62; Kostenko, 2025, p. 119), and this is coded here as a normative threshold informing Code Category C. Indian dark-pattern studies show that manipulative interfaces exploiting dual-process cognition can be documented through consumer surveys and enforcement records alone, without proprietary platform access (Khare et al., 2026, p. 6; Khare et al., 2026, p. 7). Coding across all these sources distinguishes direct empirical measures from inferred neurocognitive mechanisms and normative governance responses.

3.2.3 Public UX Analyses and Advocacy Reports

Roughly 87% of approximately 30 high-impact Indian platforms deployed at least one dark pattern. That finding from Khare et al.'s audit (Khare et al., 2026, p. 7) is treated here as primary data, not secondary commentary. Their coding against established HCI taxonomies maps interface interference, obstruction, subscription traps and privacy zuckering onto System 1 biases including default bias, scarcity and social proof, and their "sludge"-based obstruction construct directly informs working codes such as "false urgency widgets" and "roach motel subscription flows" (Khare et al., 2026, p. 7). EU regulatory advocacy supplies a second type of UX evidence. Platform X's €120 million fine for deceptive verification features and Temu's DSA breach investigation over addictive design and false urgency cues (Sanikidze, 2026, p. 390) are documented examples informing "Code Category C: deceptive verification and cancellation flows." The Georgian Competition and Consumer Agency's sanctioning (Sanikidze, 2026, p. 390) of false countdown timers, fake reviews and deceptive brand imagery (the "Maxbosch"/BOSCH case) adds further documented UI artefacts usable in this study's coding matrix, confirming that these patterns are regulator-identified harms rather than merely academic constructs.

A subset of UX literature addresses AI-mediated rather than commercial environments. Mogoi et al.'s review of AI proctoring (Mogoi et al., 2025, p. 2055956) records user perceptions of being "constantly watched," linking facial recognition and gaze tracking to surveillance anxiety, with privacy-by-design and transparent communication treated as UX governance benchmarks. HITL grading research shows that explicit AI/human labelling and accessible appeal procedures increase student trust (Selvam & González Vallejo, 2025, p. 6805789), coded here as positive UX patterns contrasting with e-commerce dark patterns. These sources anchor abstract concepts like "manipulative interface" in observable structures, following Khare et al.'s content analysis of screenshots and flows and Asante and Adeoba's audit-trail approach to heterogeneous secondary data (Khare et al., 2026, p. 7; Asante & Adeoba, 2026, p. 10). Internal algorithm logic is not directly observable; this analysis works only through documented UI behaviour and enforcement decisions.

3.3 Development of a Working Coding Matrix

3.3.1 Code A Interface Mechanics and Engagement Features

Code A covers observable interface mechanics implicated in algorithmic addiction. Every slot is coded at the level of explicit UI elements rather than inferred ranking logic. Proprietary algorithms are assessed via published behaviour descriptions (Khare et al., 2026, p. 6; Li, 2026, p. 2), not code or telemetry. Three clusters structure the scheme.

The first cluster addresses high-frequency low-friction engagement loops. Li's "addiction loop" of infinite swipe feeds, autoplay and push notifications shifts users into "auto pilot" scrolling averaging near 60 minutes and 21+ sessions daily (Li, 2026, p. 2; Li, 2026, p. 3). Hu links those same mechanics to 22% higher compulsive checking and reduced prefrontal activity (Hu, 2025, p. 2; Hu, 2025, p. 20). These findings motivate subcodes A1 (infinite scroll), A2 (autoplay) and A3 (high-frequency push notifications). The second cluster addresses salience and urgency constructs. Countdown timers, "only two left" stock alerts and visually dominant acceptance prompts (Bansal, 2026, p. 136; Khare et al., 2026, p. 7) motivate A4 (countdown/scarcity banners) and A5 (visual hierarchy favouring acceptance over decline). The third cluster captures transparency-supporting mechanics: HITL grading's explicit AI/instructor labelling and override visibility (Selvam & González Vallejo, 2025, p. 20; Selvam & González Vallejo, 2025, p. 21), and the AI Law

Model's requirement (Kostenko, 2025, p. 59.1) that AI outputs be labelled and logged with model identifiers, together motivate A6 (AI labelling), A7 (non-personalised mode toggles) and A8 (human-validation indicators). These codes originate in educational contexts but apply to social recommenders because their presence signals whether engagement optimisation is coupled to meaningful oversight. Research on AI-driven review systems (Khushboo Sharma, 2024, p. 4) shows that explicit labels and audit logs help restore user trust in automated outputs. Transparency affordances are coded separately from inferred algorithmic behaviour, because their presence or absence is directly observable from published interface descriptions.

Table 6 summarises Code A slots.

Table 6: Code A slot definitions for interface mechanics and engagement features

Slot	Description	Key sources
A1	Infinite scroll or continuous feed without natural stop cues	Short video "addiction loop," persistent scrolling (Li, 2026, p. 2; Li, 2026, p. 3); dopamine driven checking (Hu, 2025, p. 2)
A2	Auto play of next item without user click/tap	Auto play in short video architectures (Li, 2026, p. 2)
A3	High frequency, behaviour based push notifications	Push notifications within addiction loop (Li, 2026, p. 2; Li, 2026, p. 3)
A4	Countdown timers or scarcity banners in decision flows	Urgency marketing cues (Bansal, 2026, p. 136); false urgency patterns
A5	Visual or interaction hierarchy favouring acceptance over refusal	Interface interference and obstruction (Khare et al., 2026, p. 7)
A6	Explicit labelling of AI generated outputs	Labelling requirements in education (Kostenko, 2025, p. 59.1); dual sourced feedback UI (Selvam & González Vallejo, 2025, p. 20)
A7	User accessible non personalised or "off" modes	Non profiling recommendation option under DSA (Kustura, 2025, p. 7); age appropriate "no personalisation" switches (Kostenko, 2025, p. 223)
A8	Visible human validation or override indicators	HITL grading override logs and authority retention (Selvam & González Vallejo, 2025, p. 20; Kostenko, 2025, p. 59.1)

Coding follows open-then-closed slot assignment for inter-coder consistency, similar to YAML-based codebooks used in large language model prompting research (Torkestani et al., 2025, p. 7286763; Karim & Hasan, 2026, p. 8584408), validated by cross-checking Li's and Hu's descriptions against DSA enforcement concern over manipulative design impairing free decision-making (Li, 2026, p. 2; T et al., 2026, p. 2248). High-level fairness and transparency principles are a useful normative frame but require actionable operational rules to apply reliably across platform designs (Hussain & Sheemar, 2026, p. 3). One

caveat applies broadly: models jointly considering temporal dynamics and social trust (Tahmasbi et al., 2020, p. 2) suggest that what appears as an addictive A1–A5 pattern in one cohort could reflect shifting preference dynamics in another. Auditors should therefore record cohort characteristics alongside code assignments rather than treating any single observation as generalisable.

3.3.2 Code B Psychological Triggers and Motivational Effects

Code B records psychological and motivational mechanisms exploited by interface patterns. Every slot is coded from triggers explicitly documented in empirical and regulatory sources; inferred design intent alone is not sufficient grounds for coding.

Time pressure and scarcity form the first cluster. Countdown timers, "only two left" notices and expiring discounts reframe routine purchases as time-sensitive decisions targeting budget-constrained consumers (Bansal, 2026, p. 136). Indian dark-pattern audits formalise this as "false urgency/scarcity" (Khare et al., 2026, p. 7), yielding B1 (time-pressure cues) and B2 (scarcity/loss framing). Cognitive load and decision fatigue form the second cluster. "Choice obstruction" describes subscription traps and roach motels that keep users in fast, heuristic-driven decision modes (Khare et al., 2026, p. 6). Urgency-marketing interviews separately document "decision fatigue" and "psychological targeting" (Bansal, 2026, p. 8745065), yielding B3 (friction-based obstruction) and B4 (decision-fatigue amplification). The third cluster covers temporal orientation. Short-video platforms' "just one more swipe" aligns with present-oriented focus and undermines long-term goals (Li, 2026, p. 2; Li, 2026, p. 3). Hu's 22% compulsive-checking and 23% depressive-symptom findings capture the shift toward immediate algorithmic reward (Hu, 2025, p. 2; Hu, 2025, p. 20), yielding B5 (instant reward salience) and B6 (temporal discounting bias). A fourth cluster addresses social and identity triggers. Fabricated social proof, "confirm shaming" (Khare et al., 2026, p. 7) and peer/influencer-driven guilt documented in grounded-theory consumption research (Raaj S. et al., 2026, p. 4245543) yield B7 (social proof/conformity pressure) and B8 (guilt/shame messaging).

Gamified-system research provides a positive-coding counterpoint. Intrinsic-motivation taxonomies applied to avatars, rewards and pacing show genuine autonomy/competence support when aligned with educational aims (Tokarieva & Chyzykova, 2026, p. 5645864). Token-economy research shows that "technostress" from complexity can cancel those gains (Anisa & Maisa, 2025, p. 9283805), yielding B9 (autonomy-supportive feedback loops) and B10 (technostress-induced demotivation), distinguishing designs that reinforce motivation from those that overload users. A final boundary condition comes from affective computing: emotional-AI systems risk "behavioral interaction addiction" and "social relationship alienation" absent transparency and assistive, non-anthropomorphic design (Song, 2026, p. 9738431), yielding B11 (emotional dependence triggers), coded where systems simulate companionship without signalling artificiality. Table 7 summarises Code B slots and anchor sources.

3.3.3 Code C Legal Risk Levels and Compliance Status

Code C assigns legal risk by reference to explicit prohibitions and documented duties. Three binding or model-law sources provide the anchors: the AI Law Model's bans on manipulative and non-auditable systems (Kostenko, 2025, p. 68.5; Kostenko, 2025, p. 70), its TSAI provisions on social rating and covert censorship, and the EU AI Act's unacceptable-risk classification for systems exploiting psychological vulnerabilities (Kostenko, 2025, p. 31.4).

The AI Law Model prohibits systems that cannot be independently audited, lack national control over critical components, or embed hidden behavioural-influence mechanisms, requiring placement in a register of

Table 7: Code B psychological triggers and motivational effects

Slot	Description	Key sources
B1	Time pressure cues (count-down timers, expiring discounts)	Urgency marketing cues (Bansal, 2026, p. 136)
B2	Scarcity and loss framing messages (stock notices exaggerating availability)	False urgency / scarcity patterns (Khare et al., 2026, p. 7)
B3	Friction based obstruction (subscription traps, roach motels, complex cancellations)	Choice obstruction (Khare et al., 2026, p. 6)
B4	Decision fatigue amplification through design induced cognitive exhaustion	Decision fatigue and psychological targeting (Bansal, 2026, p. 8745065)
B5	Instant reward salience from swipe or scroll interactions	Short video addiction loop (Li, 2026, p. 2); compulsive checking (Hu, 2025, p. 2)
B6	Temporal discounting bias favouring present oriented, low cost interactions	Present oriented focus undermining long term goals (Li, 2026, p. 3; Hu, 2025, p. 20)
B7	Social proof and conformity pressure (fabricated or amplified testimonials)	Confirm shaming and generative social proof (Khare et al., 2026, p. 7)
B8	Guilt or shame based messaging exploiting emotional or identity cues	Peer influence and post purchase regret (Raaj S. et al., 2026, p. 4245543)
B9	Autonomy supportive feedback loops in gamified systems	Intrinsic motivation via avatars and pacing (Tokarieva & Chyzhykova, 2026, p. 5645864)
B10	Technostress induced demotivation from complexity or instability	Token economies and technostress (Anisa & Maisa, 2025, p. 9283805)
B11	Emotional dependence triggers from simulated companionship or reciprocity	Behavioural interaction addiction in affective computing (Song, 2026, p. 9738431)

prohibited AI (Kostenko, 2025, p. 31.4). TSAI provisions separately ban social rating, manipulative political targeting and covert censorship (Kostenko, 2025, p. 68.5). Neurorights research documents how covert influence operates through interface and reward-schedule mechanics that bypass conscious deliberation (Papari et al., 2026, p. 11). Configurations matching these patterns are assigned to "Code Category C: Prohibited Systemic Practices," signalling risk not mitigable by incremental adjustment. A second rule cluster defines non-negotiable traceability conditions. High-risk systems must maintain decision-architecture documentation, full audit trails linking inputs to outputs, and training-data bias metrics as part of a technical passport accessible to state bodies (Kostenko, 2025, p. 19.5). TSAI further requires unique identifiers, algorithmic passports and automated suspension capability for cross-border systems (Kostenko, 2025, p. 68.6). Absence of these artefacts in high-impact recommendation contexts is coded "Critical Traceability Failure." Digital-sovereignty provisions add a territorial dimension: localisation requirements prevent "digital colonization" by foreign-controlled critical infrastructure, envisaging restrictions where public registers or decisions rely on government-inaccessible code (Kostenko, 2025, p. 33.2; Kostenko, 2025, p. 33.4).

Recommenders routing governance functions through opaque foreign infrastructure fall into "high or unacceptable" risk bands accordingly.

Table 8 summarises these working legal risk levels.

Table 8: Code C working legal risk levels and compliance cues

Code C category	Indicative practices	Key legal anchors
Prohibited systemic practices	Social rating indices, manipulative political targeting, covert censorship, non auditable AI	TSAI prohibitions on social rating and manipulative targeting; auditability bans (Kostenko, 2025, p. 31.4; Kostenko, 2025, p. 68.5)
Critical traceability failure	No technical passport, missing logs, no event evidence for decisions in high risk settings	Documentation and logging duties for high risk systems (Kostenko, 2025, p. 19.5; Kostenko, 2025, p. 68.6)
High sovereignty risk	Critical decisions based on foreign controlled servers, inaccessible code or data relocation	Digital sovereignty, local storage and anti colonization clauses (Kostenko, 2025, p. 33.2; Kostenko, 2025, p. 33.4)
Statutory compliance oversight	Registered systems, annual audits, public trust labelling, accessible registry APIs	TSAI supervision, auditing and trust marks (Kostenko, 2025, p. 68.7)

Code C can also mark positive compliance. Healthcare and civil-service HR systems under statutory or co-regulatory models show continuous bias audits and appeal mechanisms, producing measurable bias reduction and faster appeal resolution (Alrwili & Kantharia, 2025, p. 4), mapped here to "Statutory Compliant with Effective Oversight." Revenue-process audit research reinforces that automation architectures require explicit automated/human-component documentation and feedback loops for anomaly detection (Chitturi, 2025, p. 6539584). Practical recommender-audit guides stress that platform-profile and stakeholder mapping are essential first steps. Absent such documentation, high-impact social recommenders shift from "moderate" into higher risk categories (Kostenko, 2025, p. 70; Alrwili & Kantharia, 2025, p. 4; Chitturi, 2025, p. 6539584).

3.4 Methodological Limitations and Quality Criteria

3.4.1 Constraints of Observing Black-Box Algorithms Indirectly

Only outward behaviour is observable. That constraint is structural, not a design choice, matching Khare et al.'s reliance "exclusively on secondary data" where "primary experimentation or access to proprietary platform data is limited" (Khare et al., 2026, p. 6). The contrast with internal access is instructive: embedded-audit pilots on a mental-health triage model report subgroup sensitivity metrics down to dialect-level error patterns (Grace Malachi et al., 2025, p. 9), and HITL grading research reports override rates and criterion-level $F1$ scores because the dashboard was under researcher control (Selvam & González Vallejo, 2025, p. 17; Selvam & González Vallejo, 2025, p. 21). Neither type of access is available for commercial social recommenders.

Regulatory texts assume that internal observability exists. The AI Law Model demands technical passports with full input-output tracing available to authorities (Kostenko, 2025, p. 71). ISO/IEC standards for bias as-

assessment aim to reduce black-box opacity (Sankaran, 2025, p. 2). External scholars, though, can typically verify only that a standard is claimed, not how it is implemented in a specific pipeline. Hybrid-governance case studies that report quantitative efficiency and bias-complaint reductions rely on internal organisational documentation unlikely to be available for commercial platforms (Alrwili & Kantharia, 2025, p. 2). Three linked limitations follow. Systematic ranking bias cannot be quantified at model level; the analysis must rely on regulators' findings and UX audits that carry their own sampling and jurisdictional constraints (Khare et al., 2026, p. 6; Mariia, 2025, p. 3). Compliance with documentation duties central to the AI Law Model remains unobservable unless platforms voluntarily publish audit summaries (Kostenko, 2025, p. 71; Kostenko, 2025, p. 51). Causal attribution from interface mechanics to individual harm is mediated through third-party research using context-specific samples (Grace Malachi et al., 2025, p. 9). Khare et al.'s explicit position that such designs "map patterns" rather than "establish causal relationships" (Khare et al., 2026, p. 6) defines the appropriate scope. This thesis analyses algorithmic addiction through documented patterns in external audits and regulatory texts, treating Kostenko's safety-and-reliability provisions on incident recording and independent verification (Kostenko, 2025, p. 38) and related procedural-safeguard proposals for emergency shutdown and human override (Stiefenhofer, 2025, p. 86) as normative benchmarks rather than empirically verified descriptions.

3.4.2 Reliability and Transparency in Qualitative Audit Work

Systematic procedures, not analyst intuition, carry the reliability burden here. Asante and Adeoba's trustworthiness table operationalises credibility through triangulation and confirmability through reflexive interpretation (Asante & Adeoba, 2026, p. 10; Asante & Adeoba, 2026, p. 11). Kishore et al. add open/axial/selective coding with analyst triangulation and saturation checks for AI decision-support research (Raaj S. et al., 2026, p. 6). Debt-behaviour research contributes member checking and coding audit trails (Munyaradzi & Lovichakorntikul, 2026, p. 8). This project's reliability baseline follows these precedents: explicit coding schemas, saturation checks, documented analytic decisions, and verbatim coding procedures with multi-researcher dispute resolution (Unknown, 2023, p. 4; Unknown, 2023, p. 5).

Transparency operates through the same mechanism Khare et al. use, describing their ~30-platform sampling frame and HCI-typology coding explicitly (Khare et al., 2026, p. 7). The working coding matrix (Codes A, B, C) is explicitly defined and never presented as official regulatory classification. An Algorithmic Governance Assessment Matrix for zakat institutions (Mutawa et al., 2025, p. 59) shows how secondary quantitative indicators (eligibility accuracy, transparency scores) integrate into qualitative audit while keeping observed values distinct from interpretive conclusions. That distinction is maintained throughout between reported metrics and categorisations like "Code Category C." Hybrid-governance case studies reinforce that reliable audit work requires triangulation across reports and interviews, contextualising quantified indicators within governance models (Alrwili & Kantharia, 2025, p. 2). Explainable-AI research on hoax detection (Syari et al., 2025, p. 2) argues that black-box classification without explanation undermines public trust and legal usability; audit findings must therefore trace to specific UI features or documented metrics. Multimodal emotion-recognition research sets a realistic bound: high classification performance can introduce steering vectors difficult to trace from public reports, limiting what can be inferred from platform disclosures (Zhou, 2025, p. 3). The AI Law Model embeds transparency structurally, requiring technical documentation and cross-border liability rules guaranteeing supervisory access (Kostenko, 2025, p. 107; Kostenko, 2025, p. 111), and published platform reports are assessed against those criteria even without code inspection.

3.4.3 Reflexivity and Researcher Positionality

Secondary qualitative synthesis does not produce findings from neutral ground. Position shapes interpretation. Asante and Adeoba ground exactly this in constructivism: conceptual understanding emerges through interpretation of existing evidence rather than direct participant engagement (Asante & Adeoba, 2026, p. 8; Asante & Adeoba, 2026, p. 10). The analysis here is unavoidably theory-laden, filtered through prior commitments about AI governance and CSR. Selective reading is a real risk, mitigated by the analyst-triangulation and audit-trail practices used in constructivist grounded theory and reflexive thematic analysis (Raaj S. et al., 2026, p. 6; Asante & Adeoba, 2026, p. 10; Asante & Adeoba, 2026, p. 11).

Interpretive bias is especially sensitive where AI intersects with ethical or religious values. Research on Islamic AI applications (Marwantika & Dauda, 2025, p. 5) shows how researchers explicitly confronted their own positionality, maintained reflexive logs and consulted scholars from diverse orientations to resolve coder disagreements. A comparable risk exists here given this author's professional orientation toward AI ethics, specifically a tendency to treat precautionary regulatory proposals as inherently desirable. Reflexive memoing and inclusion of divergent capacity assessments (Sikalo, 2026, p. 19; Chukhray & Koval, 2026, p. 10) serve as safeguards against that tendency. This thesis privileges regulatory documents and enforcement records for interface-manipulation questions, following Khare et al. (Khare et al., 2026, p. 6), and peer-reviewed or policy literature for governance questions, following Asante and Adeoba (Asante & Adeoba, 2026, p. 8). That choice embeds a judgement that formal law is authoritative for defining acceptable recommender behaviour; it may under-represent governance imaginaries outside the EU model-law tradition (T et al., 2026, p. 12; Kostenko, 2025, p. 22).

How black-box opacity is read also reflects this orientation. Mixed-methods AI-mental-health research stresses human-mediated interpretation to avoid "AI mental health colonialism" (Grace Malachi et al., 2025, p. 11). AI-proctoring research treats institutional readiness as equally critical to algorithmic accuracy (Mogoi et al., 2025, p. 9). Both reinforce this thesis's focus on governance architecture over purely technical fixes. Kostenko's AI Law Model, treated throughout as a normative benchmark prioritising human rights and independent oversight (Kostenko, 2025, p. 22; Kostenko, 2025, p. 37; Kostenko, 2025, p. 44), and Asante and Adeoba's framing of CSR as the natural locus of reform (Asante & Adeoba, 2026, p. 11), together shape the coding matrix's assumptions. An innovation-first or minimal-intervention standpoint would likely weight the same evidence toward different risks and opportunities; readers should bear that interpretive latitude in mind when evaluating the findings (Alrwili & Kantharia, 2025, p. 2; Bahangulu & Owusu-Berko, 2025, p. 7).

4 Comparative Platform Audit

4.1 Selection Rationale and Platform Profiles

4.1.1 Short-Form Video Platforms with Personalized Feeds

With 1.074 billion short-video users in China alone (95.4% of netizens), and neural-network engines processing micro-duration clips at scale, Li's description of TikTok-type feeds as an "addiction loop" (Li, 2026, p. 2) is not metaphor but architecture. Infinite scroll removes natural stopping points; autoplay removes the need to choose; push notifications reactivate users who managed to leave. Hu traces the measurable consequences: "dopamine driven feedback loops," 22% higher compulsive checking versus non-algorithmic feeds, reduced prefrontal activity, and 23% higher depressive-symptom incidence (Hu, 2025, p. 2; Hu,

2025, p. 20). The DSA's 2026 TikTok proceeding names exactly these features as potential "systematic manipulative design" (Li, 2026, p. 2; T et al., 2026, p. 15), and Chitra et al. document the same recommender driving viral misinformation spread with community guidelines as the only visible restraint. For college students specifically, short-video addiction correlates with impaired sleep and "information cocoons" that attenuate critical thinking (Li, 2026, p. 6; T et al., 2026, p. 8), while accounts signalling political interest are "quickly directed toward extremist material." Kostenko's digital-ecology provisions directly ban algorithms that "form addiction or create self-replicating digital environments" eroding autonomous thought (Kostenko, 2025, p. 119). Short-form personalised video sits at the highest-priority intersection of Code A features, Code B triggers and Code C exposure in this report's working taxonomy (Li, 2026, p. 2; Hu, 2025, p. 20; Kostenko, 2025, p. 119; T et al., 2026, p. 15; Hu, 2025, p. 2; Khare et al., 2026, p. 6).

4.1.2 Image and Story-Centric Social Networks

Instagram's "Explore" page amplified body-negative imagery by 23% relative to chronological feeds; algorithmically prioritised content shows 31% higher negative affect than baseline (Hu, 2025, p. 3), with statistically significant self-esteem declines documented among adolescent women of colour. Quinelato classifies these platforms as "social media" within virtual hosting services that algorithmically reference third-party content while collecting granular interaction data for profiling (Quinelato, 2024, p. 4), and Ferraro et al. describe recommendation-driven cultural consumption as "monitoring-based marketing" where each user interaction is simultaneously cultural production and ranking-logic input (Ferraro et al., 2023, p. 5). The coupling runs all the way to revenue: platform type, content format and audience size all moderate how AI engineering affects creator earnings, so engagement metrics bind directly to business value for every actor in the stack (Zeng et al., 2025, p. 3). Regulatory scrutiny focuses on behavioural advertising, treating vulnerable demographics profiled via third-party tracking as the primary risk population, with opaque persuasion rather than technical architecture as the operative frame (Quinelato, 2024, p. 4). Literacy surveys sharpen the exposure risk: users rate themselves confident in routine digital tasks yet score only a mean 2.31/5 when asked to recognise manipulation (Chukhray & Koval, 2026, p. 10), with scepticism about EU enforcement particularly acute outside EU jurisdictions. Fairness in these systems is processual, not a one-off check; documentation practices proven in clinical and criminal-justice AI apply equally to revealing representational gaps in image-based recommendation (Clement et al., 2025, p. 20). The Algorithmic Impact Navigator's fairness and explainability requirements (Leon, 2026, p. 2) and the zakat-distribution governance matrix's transparency and accountability dimensions (Mutawa et al., 2025, p. 4) are both operative wherever AI-intensive architectures produce documented psychosocial impacts at this scale (Hu, 2025, p. 3; Ferraro et al., 2023, p. 5).

4.1.3 Professional Networking Platforms with Algorithmic Feeds

Professional networking platforms with algorithmic feeds warrant inclusion in this audit, but a firm evidentiary limit must be stated first. None of the sources gathered for this report name LinkedIn specifically. Claims in this subsection are extrapolated from adjacent recommender-driven platforms and are marked illustrative rather than platform-verified, in line with this thesis's Integrity Rule 1.

Professional feeds combine engagement mechanics with economic gatekeeping around jobs and reputation, a combination with potentially higher stakes than entertainment feeds. Chitra et al. list Facebook, Instagram, TikTok, X and Google Search as nodes where recommenders act as "invisible gatekeepers" in political information flows (T et al., 2026, p. 2); the engagement-ranking risks documented there transfer by extrapolation to professional contexts. Gerr shows algorithmic personalisation systematically reduces

viewpoint diversity (Mariia, 2025, p. 5; Mariia, 2025, p. 29); a professional feed surfacing "thought leadership" through comparable logic would plausibly import bias into hiring and reputational signals. That remains an illustrative inference, not a documented platform finding. Profiling-based advertising documented for Instagram and TikTok (Quinelato, 2024, p. 4) could analogously support job ads targeted on inferred vulnerability rather than qualification, compounded by literacy-survey evidence that active users overestimate their manipulation-recognition ability (mean 2.31/5) (Chukhray & Koval, 2026, p. 10). Gerr's finding that formal DSA/AI Act transparency obligations have not removed underlying engagement incentives (Mariia, 2025, p. 7; Mariia, 2025, p. 28) and Chitra et al.'s call for mandatory auditing with conditional liability (T et al., 2026, p. 17) would apply with equal force to any professional feed influencing employability, were platform-specific data available. Professional networking is flagged as a priority category for future primary research precisely because such evidence does not yet exist, not because current evidence establishes the same harms (Mariia, 2025, p. 5; T et al., 2026, p. 2244; Kostenko, 2025, p. 25). Recommender-systems scholarship reinforces why the gap matters: human-centred fairness metrics (Castells & Jannach, 2023, p. 53) are needed because offline accuracy measures miss the systemic harms that professional-opportunity mediation could create, making primary empirical work on professional feeds a warranted next step.

4.1.4 Original Heuristic Evaluation: A Single-Evaluator LinkedIn Walkthrough

The extrapolation above can be partially tested rather than left as inference alone. This report's author conducted a structured, single-evaluator heuristic walkthrough of LinkedIn's publicly visible desktop and mobile interface, scoring observed features against this report's own Code A/B/C matrix (Code A interface mechanics, Code B psychological triggers, Code C legal risk tiers). The walkthrough covers only features visible to any logged-in user during ordinary use; it has no access to backend ranking logic, A/B-test variants or internal telemetry, was not conducted as a formal timed usability protocol, and reflects a single evaluator's observations and recollections rather than a longitudinal or multi-rater study. Confidence is therefore lower throughout than for the TikTok and Instagram findings elsewhere in this audit, which rest on a substantially larger published literature base. The exercise is offered as a demonstration that the toolkit generalises to a platform with no dedicated dark-pattern literature, not as a claim to have closed the LinkedIn evidence gap – and, notably, it surfaces a genuinely mixed picture rather than uniformly confirming the more severe patterns documented for other platforms.

These observations do not establish prevalence, user harm or regulatory exposure the way the cited TikTok and Instagram literature does; they establish only that some, but not all, of the interface mechanics and psychological triggers documented elsewhere are also present on a platform this audit's source base does not otherwise cover. The walkthrough deliberately reports the negative and moderate findings alongside the concerning ones: no autoplay was observed, notification badges cleared after a single view rather than escalating, and the cancellation flow this evaluator most recently used did not exhibit the severe obstruction documented in the audited subscription-platform literature. Against that, the non-chronological feed with no opt-out toggle, the multi-surface connection-nudging, and the free/Premium split on profile-view analytics do align with the concerning pattern documented elsewhere. Read alongside the Integrity Rule stated earlier in this subsection, this walkthrough sits between pure extrapolation and verified platform-specific research: it is original observation, single-evaluator and non-longitudinal, and its mixed result should itself be read as evidence that platform-specific verification – rather than assuming a platform matches the worst-documented pattern by default – is the right posture, consistent with the researcher-administered evaluation recommended in Section 6.2.

4.2 Interface Mechanics across Platforms

4.2.1 Feed Structures and Scrolling Patterns

TikTok's default landing page is not "Following" but the personalised "For You" feed (Meßmer & Degeling, 2023, p. 42), meaning every session begins with algorithm-selected content before the user has made any deliberate choice. Micro-duration clips with infinite swipe then remove the need to decide again. Li links this architecture to an "auto-pilot mode" associated with weakened social skills and behavioural addiction particularly among college students (Li, 2026, p. 2; Li, 2026, p. 6). YouTube's default autoplay mirrors this pattern, privileging recommended content over manually selected items (Meßmer & Degeling, 2023, p. 42). Intentional decisions per unit of content consumed fall measurably as a result. Hu connects the resulting feed structure to 22% higher compulsive checking and correlated neurocognitive change (Hu, 2025, p. 2; Hu, 2025, p. 20). The regulatory response follows the evidence: the DSA's 2026 TikTok investigation names these features as potential "systematic manipulative design," and Meßmer and Degeling link default autoplay and personalised-feed landing pages directly to DSA Article 25's dark-pattern prohibition (Li, 2026, p. 6; Meßmer & Degeling, 2023, p. 42). Feed structure also shapes exposure composition, not only session length. Gerr shows engagement-optimised trending surfaces reduce viewpoint diversity (Mariia, 2025, p. 3; Mariia, 2025, p. 29); Instagram's "Explore" ranking increased body-negative imagery exposure by 23% versus chronological feeds (Hu, 2025, p. 3). A practical audit variable follows: how many clicks to reach a non-personalised alternative (Meßmer & Degeling, 2023, p. 42)? HITL grading demonstrates the alternative architecture is viable: AI outputs presented alongside editable, logged human review embed oversight without removing the content stream (Selvam & González Vallejo, 2025, p. 17). Infinite scroll and default autoplay are the primary cross-platform inspection targets for the Ethical AI Audit Toolkit (Khare et al., 2026, p. 6; Li, 2026, p. 2; Hu, 2025, p. 2; Meßmer & Degeling, 2023, p. 42).

4.2.2 Use of Autoplay and Recommendations

On short-video feeds, the next clip begins before the user acts; stopping requires deliberate interruption, and opting out is the non-default (Li, 2026, p. 2). YouTube's default autoplay receives the same regulatory designation, explicitly identified as privileging recommended items over manually selected ones (Meßmer & Degeling, 2023, p. 42). Each served item is a personalised selection continuously refined from prior behaviour (Li, 2026, p. 2). TikTok compounds this by opening on "For You" rather than "Following," making both the entry point and every subsequent item algorithm-determined (Meßmer & Degeling, 2023, p. 42). The measurable toll is well-documented: algorithmically curated feeds produce 22% higher compulsive checking and 23% higher depressive-symptom incidence alongside reduced prefrontal and grey-matter activity (Hu, 2025, p. 2). Li frames the resulting loop directly as an "algorithmic addiction loop." Regulators have treated this as actionable (Li, 2026, p. 2): the DSA's TikTok proceeding targets the combination (Li, 2026, p. 2; Meßmer & Degeling, 2023, p. 42), and Kostenko's AI Law Model bans autoplay outright for minors while mandating "detox mode" defaults with visible "no personalisation" switches for teen social networks (Kostenko, 2025, p. 221; Kostenko, 2025, p. 223). Indian platform audits document pre-selected defaults and interface interference operating by the same logic, tilting attention toward recommendation-driven content before users consciously engage (Khare et al., 2026, p. 4; Meßmer & Degeling, 2023, p. 42). Autoplay and infinite scroll are classified as Code A features, variable rewards and temporal discounting as Code B triggers, and violations of age-sensitive design bans as Code C in this report's working taxonomy (Li, 2026, p. 2; Hu, 2025, p. 2; Kostenko, 2025, p. 221; Khare et al., 2026, p. 6; Hu, 2025, p. 2; Kostenko, 2025, p. 12). Kostenko's ban applies specifically to minors, though the underlying architecture operates

identically for adults without comparable protection.

4.2.3 Notification Design and Return Incentives

Push notifications occupy a structurally distinct position from autoplay or infinite scroll: they reach users after they have already left. 47% of users surveyed by Khare et al. reported difficulty cancelling services, and "nagging" notifications with repeated post-refusal prompts generated a documented "wear down effect" steering users toward platform-preferred choices (Khare et al., 2026, p. 8). Urgency-marketing executives describe late-night and high-frequency push campaigns as commercially advantageous, premised on explicit assumptions about temporal vulnerability and low self-control windows (Bansal, 2026, p. 138); when combined with stored-payment one-click checkout, these prompts compress deliberation almost to zero. India's CCPA advisory frames this as platform accountability rather than consumer weakness, mandating internal dark-pattern audits and explicit consent over pre-checked defaults and shifting from complaint-driven enforcement to preventive regulation targeting sludge architecture before it causes harm (Khare et al., 2026, p. 6; Khare et al., 2026, p. 12). High-frequency, behaviourally-targeted notifications embedded in obstructive flows are classified here as high-risk manipulative return incentives. Table 10 synthesises these patterns. Bansal's synthesis of algorithmic advertising architecture shows notifications function as a system-level binding mechanism (Bansal, 2026, p. 139). A "cognitive trigger layer" deploys scarcity cues; an "algorithmic personalization layer" tailors timing to behavioural histories; a "temporal and friction reduction layer" amplifies urgency while shrinking steps to purchase. Exploitative impulse buying is an architectural outcome of this stack. TikTok's infinite scroll, autoplay and personalised feeds are already under DSA investigation as "systematic manipulative design" (Li, 2026, p. 2); behaviourally tuned notifications re-routing users back into those feeds represent the same pattern in a second channel. Two configurations reach Code C: notifications embedded in obstructive exit flows or subscription traps, and those combining temporal-vulnerability assumptions with high-urgency messaging documented to compress deliberation in adjacent retail contexts (Khare et al., 2026, p. 8; Bansal, 2026, p. 138). The India CCPA's mandate for preventive auditing over complaint-driven response reflects a shared conclusion: current self-declaration is insufficient to stop these mechanics.

4.3 Psychological Trigger Mapping by Platform

4.3.1 Variable Rewards and Intermittent Reinforcement

Short-video platforms engineer intermittent reinforcement into default feed architecture rather than offering it as an opt-in. Li describes neural-network recommenders paired with swipe-based feeds that trigger dopamine release during the expectancy moment before the next clip appears, not only at the point of enjoyment (Li, 2026, p. 2; Li, 2026, p. 3; Li, 2026, p. 6). "Auto-pilot" scrolling follows. The result correlates with diminished academic concentration, heightened anxiety and depression among college students. Hu extends the evidence beyond behaviour to neurocognitive change: 22% higher compulsive-checking behaviour, 23% higher incidence of depressive symptoms versus chronological content, and reduced pre-frontal cortex activity and grey-matter density in impulse-control regions (Hu, 2025, p. 2; Hu, 2025, p. 20). Table 11 maps these triggers to working Code B categories. Countdown timers and "only two left" flags in retail contexts work precisely because their inconsistency is itself a variable reward (Bansal, 2026, p. 136; Khare et al., 2026, p. 7): users learn that delay sometimes costs them, reinforcing impulsive action without stable informational value. Regulatory treatment now reflects this. The EU DSA proceeding against TikTok cites infinite scrolling, autoplay and push notifications as possible "systematic manipulative design"

(Li, 2026, p. 2). Kostenko's AI Law Model goes further, prohibiting systems that cause digital addiction and establishing an inalienable right to an information environment free from excessive flows that generate chronic anxiety without user consent (Kostenko, 2025, pp. 61–62; Kostenko, 2025, p. 119). Hu's interdisciplinary matrix demonstrates audit measurability: neurocognitive metrics map to algorithmic feature vectors via the Algorithmic Impact Assessment Scale (AIAS-5), quantifying emotional volatility, sleep disruption and compulsive usage (Hu, 2025, p. 19; Hu, 2025, p. 20). Leon's AIN framework assigns organisations responsibility for continuous impact evaluation; Chitturi ties oversight to documented splits between automated detection and human review (Leon, 2026, p. 2742041; Chitturi, 2025, p. 6539584). Feeds combining infinite scroll, default autoplay and irregular high-arousal content are classified Code C in this report's working taxonomy, particularly for users confident in digital skills yet poor at recognising manipulation (Chukhray & Koval, 2026, p. 10; Hu, 2025, p. 2; Li, 2026, p. 2). Recommender systems sustain this engagement via collaborative filtering and network-based models (Godoy-Lorite et al., 2020, p. 3) that scale efficiently on large sparse interaction datasets, making the reinforcement loop technically resilient precisely where regulation has not yet reached.

4.3.2 Social Validation, Status Cues and Fear of Missing Out

On Douyin and Xiaohongshu, visible engagement metrics are unreliable signals. Wang reports that likes, follower counts and reviews are routinely distorted by fake accounts and paid engagement (Wang, 2025, p. 13); a 2023 China Consumers Association report found nearly 30% of online shoppers had encountered falsified reviews in the preceding year, converting an aggregated trust signal into a manipulation instrument. Influencer and KOL endorsements compound the distortion: followers read them as socially grounded recommendations even when SAMR sponsorship-disclosure rules apply, because personal credibility and paid promotion are deliberately fused (Wang, 2025, p. 13). Algorithmic curation then amplifies whatever is already popular. "Popularity begets more popularity" under engagement-focused ranking, and Douyin's viral challenges and livestreamed sales create collective-buying moments where social pressure to participate drives impulsive spending and emotional fatigue (Wang, 2025, p. 13). Hu connects this to broader engagement logic: content with strong emotional or identity signalling attracts more reactions, driving further algorithmic promotion, which attracts yet more reactions. Instagram's "Explore" page amplified body-negative imagery by 23% versus chronological feeds, correlating with statistically significant self-esteem declines among adolescent women of color (Hu, 2025, p. 2; Hu, 2025, p. 3). Table 12 summarises these triggers in the working coding matrix. Status-seeking shapes purchasing behaviour across these same platforms. Kishore et al. find university students' marketplace choices dominated by price, urgency and reviews; ethical considerations remain secondary, and unverifiable social proof triggers scepticism that widens the intention-behaviour gap rather than closing it (Raaj S. et al., 2026, p. 2; Raaj S. et al., 2026, p. 4). Yin reports TikTok hashtag challenges make participants 62% more likely to log in daily (Yin, 2024, p. 3), linking routine engagement to visible status contests where missing a challenge risks social invisibility. Distorted social proof, herd-reinforcing algorithms and FOMO mechanics are treated as high-priority Ethical AI Audit Toolkit items on the strength of converging evidence from Chinese e-commerce, social-media research and student-behaviour studies (Wang, 2025, p. 13; Hu, 2025, p. 2; Raaj S. et al., 2026, p. 2; Yin, 2024, p. 3). The mechanism is less visible than autoplay but no less structural.

4.3.3 Emotional High-Arousal and Outrage Content

Chitra et al. synthesise experimental work showing moral-emotional language in tweets raises retweet probability by roughly 20% per additional moral-emotional word (T et al., 2026, p. 7), and content evoking

awe, anger or anxiety is significantly more shareable than content associated with sadness. A ranking stage trained on click-through, watch time, completion and comment signals selects for outrage-maximising content by construction. The loss function contains no term for epistemic quality. Hu documents the measurable result: Instagram's "Explore" page amplified body-negative imagery by 23% relative to chronological feeds, correlating with a statistically significant self-esteem decline among adolescent women of color ($\beta = -0.41, p < 0.05$), and algorithmically prioritised content exhibits 31% higher negative affect than baseline (Hu, 2025, p. 3; Hu, 2025, p. 2). Yin documents both directions of viral mobilisation: Black Lives Matter as catalysed global protest, and COVID-19 misinformation (doctored videos and false cures spreading faster than official retraction, with algorithms promoting "racy" high-interaction posts) alongside creators reporting harassment and burnout and viewers reporting distress from repeated exposure (Yin, 2024, p. 2). Table 13 aligns these with the working coding matrix. Candidate-generation and ranking stages optimise multi-task objectives weighting click-through, completion, watch time, share and comment rate by commercial value, with no explicit wellbeing term in the loss function. YouTube's "radicalization pipeline" is interpreted as a reinforcement-learning agent discovering that escalating content intensity sustains attention, gradually steering users toward more extreme material (T et al., 2026, p. 7). Chitra et al. frame algorithmic manipulation of political information as threatening freedom of expression and electoral rights (T et al., 2026, p. 11); digital-constitutionalist proposals call for ranking transparency, audit trails, independent oversight and non-discrimination in political-information distribution. Emotional high-arousal and outrage content is a predictable outcome of engagement-oriented optimisation (T et al., 2026, p. 7; Hu, 2025, p. 3; Yin, 2024, p. 2), making these patterns central diagnostic signals for any audit toolkit aligned with EU digital constitutionalism (T et al., 2026, p. 7; T et al., 2026, p. 11; Hu, 2025, p. 3). The loss function's structural silence on epistemic quality is not an oversight that can be patched without changing what the system optimises for.

4.4 Audit Table and Findings

4.4.1 Consolidated Cross-Platform Risk Matrix

Sections 4.1 through 4.3 produce four platform-specific tables (Tables 10, 11, 12 and 13). Table 14 consolidates these into a single working matrix, mapping each platform type against its dominant Code A/B feature, Code C risk tier and the SDT indicator most directly degraded. LinkedIn is absent: per the Integrity Rule applied in Section 4.1.3, no source in this audit names the platform specifically, so it is scoped out rather than populated with extrapolated figures. Two findings emerge from reading the matrix as a whole. Every high-tier row combines a Code A interface mechanic with a Code B psychological trigger; none of the documented harms traces to either category in isolation. The "not yet assessable" professional-networking row is not a null result but an indicator that audit coverage tracks the literature's own gaps, not the audit team's priorities.

4.5 Synthesis of Core Vulnerabilities

4.5.1 Cross-Cutting Patterns Behind the Platform Findings

Across every platform examined in Section 4, the same structural weaknesses reappear. Platform architecture changes; the vulnerabilities do not.

Compressed deliberation runs through the findings in every product category. Countdown-style urgency, nagging notifications and obstructive cancellation flows all shorten the window for System 2 reflection,

whether the surface is a short-video feed, a subscription paywall or a notification tray (Khare et al., 2026, p. 8; Bansal, 2026, p. 138; Khare et al., 2026, p. 7). The design compresses rather than respects consideration.

Reward unpredictability is equally portable. Infinite scroll, variable-reward swipe mechanics and algorithmically amplified high-arousal content share identical intermittent-reinforcement logic (Li, 2026, p. 2; Hu, 2025, p. 2): a platform can swap short clips for outrage-bait posts and the addictive mechanism transfers intact.

What ties these together is the asymmetry of awareness. Users consistently rate themselves digitally competent while scoring poorly on recognising manipulation (mean 2.31/5) (Chukhray & Koval, 2026, p. 31; Kustura, 2025, p. 7). Disclosure-only remedies, such as transparency reports, systematically under-protect the same population the DSA's non-profiled feed requirement is meant to reach.

These three vulnerabilities co-occur rather than stack independently. A platform combining compressed deliberation, unpredictable rewards and low user awareness is not accumulating three separate risks; each mechanism reinforces the others. That co-occurrence is the core finding this report's Ethical AI Audit Toolkit (Section 5) is built to operationalise: confirming any one vulnerability should prompt immediate checking of the other two.

5 Practical Recommendations and Pro-Social Pivot

5.1 Conceptual Design of the Audit Toolkit

5.1.1 Objectives, Users and Governance of the Toolkit

Three linked objectives organise audit design here: legal and ethical duties must become operational control points rather than aspirational statements; artefacts generated must be inspectable by external actors, not solely internal teams; and controls must embed into standing governance rather than function as one-off assessments.

Catapang's triple-gate architecture demonstrates what operationalised control looks like (Catapang, 2026, p. 24). Governance, Metric and Eco gates block or throttle releases at each pipeline stage, mapped to AI Act obligations and NIST AI RMF functions, and generate logs as audit artefacts. The toolkit adapts this model to prevent feed-ranking releases that worsen addiction or fairness metrics beyond predefined thresholds, thereby operationalising Kostenko's mandatory ethical-admissibility procedures (Kostenko, 2025, p. 9): notification, explainability, and human control over critical decisions. Gate logic is the mechanism; logged artefacts are the proof.

Envisioned user groups differ in what they require. Product and engineering teams need CI/CD-integrated gate checks aligned to DSA non-profiled-option obligations (T et al., 2026, p. 12). Risk and compliance functions need documentation and scoring outputs comparable to proposed algorithmic impact assessments for credit AI (Sebastian, 2025, p. 15) and governance-matrix scoring sheets (Mutawa et al., 2025, p. 59). External regulators face documented systemic gaps in institutional auditing capacity (Kostenko, 2025, p. 9); Bahangulu and Owusu Berko argue that businesses must adopt external oversight committees precisely because regulators struggle with black-box models (Bahangulu & Owusu-Berko, 2025, p. 8; Bahangulu & Owusu-Berko, 2025, p. 9). The toolkit bridges both groups through observable outputs and logged gate decisions, consistent with the black-box constraint.

Combining internal ethics boards, external oversight and statutory frameworks yields measurably better bias reduction and appeal resolution than self-regulation alone (Alrwili & Kantharia, 2025, p. 2). Pastoral Data Trust governance models prevent "digital colonialism" in fragile mental-health deployments by barring any single party from unilaterally controlling data access (Grace Malachi et al., 2025, p. 11; Grace Malachi et al., 2025, p. 13). This thesis situates toolkit governance co-regulatorily: internal ethics committees manage gate configurations while independent auditors and national registers access gate artefacts and risk assessments. HITL grading's academic oversight committees provide a working precedent for rubric standardisation and recalibration (Selvam & González Vallejo, 2025, p. 21). Most ethical failures are failures of control architecture, not failures of stated principle (Catapang, 2026, p. 24; Bahangulu & Owusu-Berko, 2025, p. 9; T et al., 2026, p. 12; Kostenko, 2025, p. 26). Enforced gates, with logged artefacts as evidence, are what make the stated principle operationally real.

5.1.2 Audit Typologies Relevant for Recommender Systems

Four complementary typologies from adjacent domains inform audit design here.

The first is the compliance-versus-scenario distinction Meßmer and Degeling draw for DSA Article 37 work. Compliance audits verify documented risk assessments. Scenario-based risk audits apply the ISO five-step process to socio-technical testing of ranking behaviour under defined conditions such as self-harm content exposure (Meßmer & Degeling, 2023, p. 21). Empirical archetype-account studies show that such scenario tests can reveal algorithmic saturation toward extreme material within approximately five days of passive viewing (Regehr et al., 2025, p. 4). Short behavioural probes surface harmful dynamics that documentation-only reviews miss.

The second typology is the algorithmic impact assessment proposed by Sebastian for high-risk credit-AI systems: an environmental-impact-review equivalent testing fairness, accuracy and systemic effects both pre- and post-deployment (Sebastian, 2025, p. 15). Engagement-optimising feeds meet this threshold. Periodic assessment is warranted well beyond content legality alone.

Third, sector-specific governance matrices provide a scoring template. Mutawa et al.'s Algorithmic Governance Assessment Matrix for zakat institutions shows that transparency scores correlate with documentation clarity, and AI readiness scores with eligibility accuracy (Mutawa et al., 2025, p. 4; Mutawa et al., 2025, p. 8). The same matrix logic transfers to recommender platforms.

Fourth, Kostenko's AI Law Model separates design-time ethical admissibility tests from ongoing risk-tiered supervisory audits by accredited auditors (Kostenko, 2025, p. 8; Kostenko, 2025, p. 9). Admissibility tests cover notification, explainability and human control over critical decisions. Auditors may reclassify risk based on changing operational metrics; reclassification is not a sanction but the intended supervisory loop functioning correctly.

Financial-AI lifecycle validation pipelines add a further process pattern: adversarial pre-deployment bias testing, a self-audit loop logging SHAP attributions to a write-once ledger, bi-annual third-party review, and automatic circuit-breaker fallback when drift thresholds are breached (S, 2026, p. 7). HITL grading demonstrates the same principle in educational AI. There, operational audits are inseparable from daily workflow, and oversight committees analyse override logs and recalibrate models continuously (Selvam & González Vallejo, 2025, p. 21). This report's toolkit aligns its modules to these four typologies, subject throughout to the black-box constraint introduced in Section 5.1.3.

5.1.3 Alignment with Regulatory and Ethical Requirements

Toolkit choices must trace to hard-law and model-law obligations rather than free-standing aspirations. Any audit item that cannot be grounded in a binding duty or authoritative model standard risks becoming a cosmetic layer dismissed during enforcement review.

The AI Act's unacceptable-risk tier and Kostenko's ban on behaviour-manipulating systems jointly anchor Code Category C flags for addictive feeds and minor profiling (T et al., 2026, p. 2248; Kostenko, 2025, p. 10; Kostenko, 2025, pp. 61–62). Platform-facing DSA duties form a second anchor. The non-profiled-feed requirement, annual independent audits, and Article 25's ban on interfaces that deceive or "substantially impair" free decisions translate directly into observable audit checks on default personalisation and exit-flow friction (T et al., 2026, p. 2248; Quinelato, 2024, p. 20; Khare et al., 2026, p. 6). No internal ranking-code access is required.

Kostenko's model law extends these obligations into finer-grained checklist items. Mandatory notification, explainability and human-control-over-critical-decisions procedures are each individually required (Kostenko, 2025, p. 8). Technical passports and event logging are additionally mandated for high-risk systems overseen by an interdisciplinary AI council (Kostenko, 2025, p. 9; Kostenko, 2025, p. 19.5). Tamper-evident logs and human-override evidence are specified compliance deliverables, not optional extras.

ISO/IEC risk-management, data-quality and bias-assessment standards (Sankaran, 2025, p. 4) support AI Act representativeness and fairness requirements even where technically voluntary outside binding regimes. Adopted as internal toolkit metrics crosswalked to AI Act criteria, they mirror Catapang's triple-gate regulatory mapping (Catapang, 2026, p. 23). Voluntary adoption of a binding-equivalent standard is a defensible audit position.

Hybrid governance combining internal ethics boards, external oversight and statutory frameworks produces measurably better bias reduction and appeal resolution than self-regulation alone (Alrwili & Kantharia, 2025, p. 2). This finding aligns with independent-audit calls given regulators' documented difficulty with black-box models (Bahangulu & Owusu-Berko, 2025, p. 9). HITL grading and AI-proctoring research supply concrete communication design patterns (explicit AI/human tagging, dual logging, privacy-by-design consent (Selvam & González Vallejo, 2025, p. 6805789; Selvam & González Vallejo, 2025, p. 20; Mogoi et al., 2025, p. 9)) informing audit items on user-facing disclosure and appeal accessibility. Within the evidentiary limits of secondary qualitative research, the toolkit remains congruent with AI Act, DSA and AI Law Model expectations (T et al., 2026, p. 2248; Quinelato, 2024, p. 20; Kostenko, 2025, p. 8; Mogoi et al., 2025, p. 9).

5.2 Audit Preparation and Scoping Modules

5.2.1 Defining System Boundaries and Use Cases

Scoping an audit too narrowly misses the manipulation vectors that matter most. This toolkit treats recommenders as black-box services whose internal weights cannot be inspected; assessment relies on observable interface behaviour, regulatory records and secondary studies (Khare et al., 2026, p. 7; Li, 2026, p. 2; Hu, 2025, p. 2).

Boundaries are articulated along three axes. Functional scope draws on Mutawa et al.'s governance matrix to treat the recommender as comprising ranking logic, data capture, user-facing interfaces and

surrounding governance mechanisms (Mutawa et al., 2025, p. 2702055). For a SOCIO SPHERE AI architecture, this must include sentiment analysis and AI-assisted replies alongside ranking proper (Behera, 2026, p. 3413731). Ignoring notification and reply channels misses key manipulation vectors. Institutional scope follows Kostenko's distinction between AI providers and operators, both bearing legal obligations (Kostenko, 2025, p. 20), and extends to include internal ethics boards and external auditors whose presence shapes how identified risks are addressed (Alrwili & Kantharia, 2025, p. 2; Mutawa et al., 2025, p. 2702055). Jurisdictional scope follows Kostenko's technological-sovereignty provisions (Kostenko, 2025, p. 108), alongside AI Act and DSA transparency duties and Chitra et al.'s call for mandatory auditing of recommenders influencing political discourse (T et al., 2026, p. 16). Each audit must specify which stack components fall under EU-style duties and which under model-law expectations.

Table 15 illustrates three archetypal use cases recurring in later chapters. Each archetype defines the "system under audit" differently. The student short-video case centres child profiling and predictive "life path" scoring under enhanced-protection provisions (Kostenko, 2025, p. 20; Kostenko, 2025, p. 62). SOCIO SPHERE AI must include sentiment analysis and AI-assisted replies even where models are shared across surfaces (Behera, 2026, p. 3413731). A platform that shares a recommendation model between its main feed and its notifications layer must be audited across both surfaces, not one in isolation.

Use cases also fix expected artefacts. The zakat governance matrix shows that transparency scores track documentation clarity (Mutawa et al., 2025, p. 2702055); following that logic, each use case here specifies expected passports, audit logs and impact assessments per Kostenko's transparency chapter (Kostenko, 2025, p. 68; Kostenko, 2025, p. 69), while explicitly excluding inaccessible training corpora and live A/B results. The toolkit is a qualitative, documentation-centred instrument (Kostenko, 2025, p. 68; Meßmer & Degeling, 2023, p. 21; Alrwili & Kantharia, 2025, p. 2). It evaluates declared objectives, interface mechanics and published outputs, not internal telemetry.

5.2.2 Stakeholder Mapping and Risk Prioritization

Knowing who carries legal duties is the foundation of risk prioritisation. Kostenko distinguishes developers, operators, regulators and affected individuals, all bound by ethical-responsibility duties (Kostenko, 2025, pp. 61–62). The Algorithmic Impact Navigator similarly positions deploying organisations as "the first line of accountability" (Leon, 2026, p. 2; Leon, 2026, p. 3).

Four stakeholder clusters are central for audit design. Platform organisations include leadership commissioning impact evaluations, product and engineering teams building the ranked feed, and ethics committees responsible for gate configurations (Leon, 2026, p. 2; Kostenko, 2025, pp. 61–62). External oversight actors span EU regulators, national authorities, and the accredited auditors central to co-regulatory models (Kostenko, 2025, p. 10; Alrwili & Kantharia, 2025, p. 2). Users and affected communities include heavy social-media users with low manipulation awareness (Chukhray & Koval, 2026, p. 10), students experiencing short-video addiction (Li, 2026, p. 2), and communities subject to AI-mediated mental-health interventions (Grace Malachi et al., 2025, p. 13). Each cluster carries different legal standing and different information needs. Epistemic stakeholders (researchers and civil society) rely on data access (Meßmer & Degeling, 2023, p. 21) to study systemic risk and may trigger regulatory enforcement. Where their access rights under DSA Article 40 analogues go unused or unenforced, that absence is itself a risk signal worth tracking.

Risk prioritisation follows the AIN's screening logic around sociopolitical context, opacity and critical-infrastructure integration (Leon, 2026, p. 3). Engagement-optimised feeds heavily used by students rank

high: vulnerability is elevated, ranking is opaque, and integration into daily routine is pervasive. Chitra et al.'s argument for mandatory researcher data access to very-large-platform recommenders affecting civic discourse reinforces this prioritisation under the DSA (T et al., 2026, p. 2248). Table 16 outlines a working mapping. Governance model also modulates risk level. Hybrid governance yields greater bias reduction and audit compliance than self-regulation alone (Alrwili & Kantharia, 2025, p. 2); zakat institutions with higher AI readiness and transparency scores achieve better eligibility accuracy (Mutawa et al., 2025, p. 2702055). Platforms under co-regulatory or statutory regimes with documented ethics councils therefore rank lower in structural risk than those relying on voluntary self-assessment.

One methodological limit applies throughout. Dark-pattern prevalence is assessed via secondary institutional reports (Khare et al., 2026, p. 7); engagement-related harms are inferred from controlled experiments rather than telemetry (Hu, 2025, p. 2). This is a deliberate scope boundary, not an oversight. Stakeholder mapping and risk prioritisation here are derived from the same documentary and empirical sources describing behaviours external to proprietary systems (Kostenko, 2025, p. 10; Meßmer & Degeling, 2023, p. 21; Bahangulu & Owusu-Berko, 2025, p. 9). The toolkit is a qualitative documentation-centric instrument, not a substitute for internal model forensics.

5.2.3 Documentation and Data Access Pre-Checks

Before any scenario-based or fairness assessment begins, auditors must establish a simpler baseline: does the required documentation exist? Regulators expect specific artefacts, and their absence signals problems beyond any particular audit finding.

Kostenko's TSAI regime provides a detailed checklist. High-risk systems require a unique identifier and an algorithmic passport specifying owner, risk class, model architecture, data sources, oversight channels and event-logging retention. Absence of an equivalent dossier signals serious traceability gaps before substantive assessment even begins. Data-access duties follow: owners must preserve and provide event logs, with failure triggering fines or access suspension (Kostenko, 2025, p. 244). DSA Article 40 similarly positions vetted researcher data access as central to systemic risk audits (Meßmer & Degeling, 2023, p. 32). Pre-checks must verify that log retention meets minimum periods (at least 12 months under the AI Law Model) and that internal procedures exist for supplying materials on request. Missing documentation is itself a finding.

Logging quality matters as much as logging existence. Kostenko requires logs stored for defined periods and coupled to appeal mechanisms suspending disputed decisions pending review (Kostenko, 2025, p. 191). HITL grading provides a practical compliance model: every decision is time-stamped and linked to reviewer identifiers, forming an inspectable override-pattern audit trail (Selvam & González Vallejo, 2025, p. 16; Selvam & González Vallejo, 2025, p. 17). The zakat governance matrix corroborates that transparency scores correlate with documentation clarity (Mutawa et al., 2025, p. 59). Pre-checks confirm whether platforms maintain model cards and change logs per Kostenko's model risk-management provisions (Kostenko, 2025, p. 191); a gap here indicates governance intention without the evidentiary infrastructure compliance requires.

Documentation and data access are assessed through what is published, consistent with the black-box constraint introduced in Section 5.1.3. A structured pre-check grid recording the existence and quality of passports, event logs, appeal channels and audit histories (Kostenko, 2025, p. 244; Sebastian, 2025, p. 15; Mutawa et al., 2025, p. 59) flags missing or inadequate artefacts as Code Category C: "Critical Traceability Failure." Under AI Law Model expectations, a system lacking these artefacts would not qualify

for high-risk deployment (Kostenko, 2025, p. 244). Qualification is conditional on evidence, not assertion.

5.3 Interface and Choice Architecture Audit Checklists

5.3.1 Assessment of Feed Mechanics and Session Design

Feed mechanics sit at the heart of algorithmic addiction. Infinite scroll, default auto play and personalised landing views collectively determine how easily users slip into extended sessions and how difficult those sessions are to interrupt.

Short-video platforms exemplify this via a closed loop of "personalized → need satisfaction → behavioural data accumulation → re-push" (Li, 2026, p. 2; Li, 2026, p. 3). Hu reports 22% higher compulsive checking on algorithmically curated feeds, correlating with reduced grey-matter density (Hu, 2025, p. 2; Hu, 2025, p. 20). The harm pathway is documented across multiple independent studies.

Four explicit audit checks follow: default entry point, presence of infinite scroll, auto play behaviour, and accessibility of a non-personalised option. TikTok's default "For You" launch and YouTube's default auto play already face DSA scrutiny as potential "systematic manipulative design" (Meßmer & Degeling, 2023, p. 42; Li, 2026, p. 2). Configurations combining always-personalised launch, infinite scroll, default auto play, and no accessible non-profiled option are classified higher risk. Douyin averages 21+ daily sessions, with 56.9% of users exceeding 60 minutes, particularly among adolescents with weaker self-control (Li, 2026, p. 3). The 23% higher depressive-symptom incidence linked to hyper-personalised feeds reinforces the classification (Hu, 2025, p. 20). Such combinations are coded Code Category C: "addiction-enabling feed mechanics."

Mitigating structures must also be documented. HITL grading's four-layer architecture shows that continuous streams can accommodate autonomy: AI scores remain subject to educator validation, and 87% were accepted with only minor edits (Selvam & González Vallejo, 2025, p. 6805789; Selvam & González Vallejo, 2025, p. 19). Auditors should ask equivalently whether feeds expose "why am I seeing this" overlays and whether humans can intervene in ranking with logged effect. Logged intervention is what distinguishes auditable control from stated intention.

5.3.2 Evaluation of Friction, Defaults and Opt-Out Channels

Meaningful control depends not only on whether settings exist, but on whether users can reach and activate them without effort or penalty. Dark-pattern research in India finds interface interference and obstruction on roughly 70%/63% of major platforms. 55% of consumers report unexpected charges; only 39% recognise the term "dark patterns" (Khare et al., 2026, p. 2; Chukhray & Koval, 2026, p. 10). Friction and defaults erode autonomy precisely where digital literacy is weakest.

DMA choice-screen research shows that interoperability mandates reduce switching friction only with vigilant enforcement against manipulative recreation of lock-in (Hermansyah, 2025, p. 10). Calls for bright-line prohibitions signal that complaint-driven enforcement is structurally inadequate against sludge architectures (Khare et al., 2026, p. 13; Khare et al., 2026, p. 14). The audit standard follows that signal. Asymmetric defaults and obstructive flows are coded Category C: "high-risk friction and default configurations," checked across three loci.

On consent: India's CCPA advisory requires "explicit user action rather than pre-checked defaults." Auditors should check whether accept and refuse require equal effort. The gap between 81% of platforms

self-declaring "dark pattern free" and national complaint resolution sitting at 28% warrants Code C flagging regardless of formal notices (Khare et al., 2026, p. 12). Declaration and evidence are different things, and only the latter counts in enforcement.

On subscriptions: roach-motel patterns combining easy entry with obstructive exit (Khare et al., 2026, p. 4) are assessed against the AI Law Model's ban on interfaces that make refusal difficult (Kostenko, 2025, p. 207). Account deletion or feed deactivation must match sign-up in step count.

On notifications: Khare et al.'s "nagging" pattern (Khare et al., 2026, p. 14) is assessed against Kostenko's age-appropriate-design limits on push notifications for minors (Kostenko, 2025, p. 223). Toggles must exist. Refusing notifications must not degrade core service access.

Asymmetric defaults and obstructed notification controls are high-salience manipulation indicators (Khare et al., 2026, p. 2; Chukhray & Koval, 2026, p. 31; Kostenko, 2025, p. 223). They warrant prioritisation alongside feed mechanics.

5.3.3 Classification of Nudging Practices along an Ethics Spectrum

Not every nudge is a dark pattern. Interface elements that guide users become problematic only when guidance secures compliance by degrading comprehension, autonomy or reversibility.

Liu and Keshavarz distinguish online choice architecture generally from dark commercial patterns, which "steer, deceive, pressure, or obstruct consumers" (Liu & Keshavarz, 2026, p. 3). This taxonomy's ethics spectrum has two dimensions: the extent to which a design supports preference-consistent choice, and the presence or absence of manipulation indicators. Both dimensions must be scored, not just one.

At the acceptable end, "assistive nudges" (neutral rankings, symmetric accept/reject consent dialogs) preserve informed choice while still persuading (Liu & Keshavarz, 2026, p. 33). They are coded low risk when paired with transparent CSR governance and regular audits (Asante & Adeoba, 2026, p. 16). Mid-spectrum "borderline nudges" raise short-term conversion but risk eroding trust. Cookie notices making acceptance easier than refusal while still offering a decline path, found across large-scale crawls of roughly 11,000 shopping sites, exemplify this band (Liu & Keshavarz, 2026, p. 33). They require remediation and complaint/churn monitoring rather than immediate prohibition, coded medium risk accordingly.

At the unacceptable end sit "coercive sludge patterns," combining exploitative nudging with artificial friction. Khare et al. document roughly 87% of major Indian platforms deploying at least one such pattern: drip pricing, false urgency, roach motels, forced action. Each exploits System 1 heuristics while blocking System 2 reflection through complexity (Khare et al., 2026, p. 7; Khare et al., 2026, p. 4). Auditors should flag any configuration meeting this description as Code Category C: "Prohibited or Critical Risk Nudges." The classification aligns with EU Digital Fairness discussions and AI law models seeking to eliminate manipulative design (Sanikidze, 2026, p. 390; Kostenko, 2025, pp. 61–62).

Table 17 operationalises this ethics spectrum inside audits: Regulatory analysis reinforces this spectrum. The anticipated Digital Fairness Act targets "addictive product designs and unfair personalisation," complementing DSA and DMA bans on interface pressure and the AI Act's prohibition of subliminal manipulation (Sanikidze, 2026, p. 390; Quinelato, 2024, p. 21). Kostenko's confidentiality principle additionally forbids profiling-driven influence without explicit consent (Kostenko, 2025, p. 44). Together, these instruments provide legal backing for treating Code C patterns as incompatible with emerging EU digital fairness norms. The regulatory direction is consistent across jurisdictions, not fragmented.

Triangulating documentary sources with large-scale user-generated evidence and a transparent scoring rubric (TRAN & PHAN, 2026, p. 5) helps auditors distinguish intentional manipulation from incidental friction. A reproducible, documented scoring process is the only basis on which a finding can withstand challenge.

5.4 Pro-Social Choice Architecture Patterns

5.4.1 Finite Session Design and Break Prompts

Personalised streams produce (Hu, 2025, p. 2; Hu, 2025, p. 20) 22% higher compulsive-checking rates and 23% higher depressive-symptom incidence than chronological alternatives, with neuroimaging showing reduced prefrontal activity in heavy users. The European Commission's DSA proceeding against TikTok names (Li, 2026, p. 2) infinite scrolling, autoplay and push notifications as potential "systematic manipulative design". Kostenko's AI Law Model responds by banning (Kostenko, 2025, p. 223; Kostenko, 2025, p. 221) these mechanics for minors outright and mandating a "detox mode" default with visible "no personalisation" switches. Compliance and evidence here point in the same direction.

A compliant finite-session design rests on three elements. Soft time-based prompts triggered by cumulative usage address a documented cognition gap: users score only 2.31/5 on recognising manipulation (Chukhray & Koval, 2026, p. 31) despite high self-rated confidence, so prompts must be salient rather than buried in settings. Structural "natural exit" points break autoplay's gravitational pull, and staged-progression gamification (Tokarieva & Chyzhykova, 2026, p. 9) builds intrinsic motivation better than open-ended streams. Governance integration requires session triggers to be logged as auditable events, mirroring human-in-the-loop grading's dual-logging of every AI decision (Selvam & González Vallejo, 2025, p. 20). AI Law Model event-logging duties (Kostenko, 2025, p. 19.5; Kostenko, 2025, p. 228) and the DFA's anticipated targeting of addictive product design (Sanikidze, 2026, p. 390; T et al., 2026, p. 2248) reinforce this requirement. Recommenders trained on sequential context produce increasingly extreme confidence scores (Cheng et al., 2022, p. 4) that steer vulnerable users toward harmful communities, making hard session boundaries a technical safety measure rather than optional UX polish.

5.4.2 Batch Feeds, Digest Modes and Time-Boxed Usage

Douyin users average more than 21 sessions daily (Li, 2026, p. 2; Li, 2026, p. 3); over half exceed 60 minutes of watch time, concentrated disproportionately among adolescents with weaker self-control. Hyper-personalised feeds correlate (Hu, 2025, p. 2; Hu, 2025, p. 20) with 23% higher depressive-symptom incidence. Kostenko's model treats (Kostenko, 2025, p. 119; Kostenko, 2025, p. 223; Kostenko, 2025, p. 221) access to a "psycho-emotionally safe digital environment" as an inalienable right, prohibiting algorithms that impose behavioural patterns or form addiction, and mandating that feeds for minors default to "detox mode". That mandated default sets the minimum viable standard.

Compression alone is not enough. A batch or digest mode that compresses sessions without removing manipulative features will not close this gap. Literacy surveys document a "second-level divide": users score between 4.40 and 4.62 on basic-task confidence (Chukhray & Koval, 2026, p. 10) yet only 2.31 out of 5 on recognising manipulation. Batching needs to pair with transparent session-boundary signals to close that gap, consistent with AI Law Model explainability requirements (Kostenko, 2025, p. 68; Kostenko, 2025, p. 69). Systems with hidden behavioural-influence mechanisms risk placement in a register of unwanted AI under the technological-sovereignty principle (Kostenko, 2025, p. 108). Batch and digest configurations backed by event logs (Kostenko, 2025, p. 242) provide the evidential record compliance requires. This

report classifies them as “time-bounded, audit-compatible sessions.”

Audit checks derived from governance-matrix methodology (Mutawa et al., 2025, p. 59) include whether maximum session lengths are documented, whether digest alternatives are surfaced as first-class options, and whether choices are logged for regulator access under DSA Article 40 (Meßmer & Degeling, 2023, p. 19). DMA interoperability research shows behaviourally-informed regulation can reduce lock-in while preserving competition (Hermansyah, 2025, p. 10), suggesting batch reforms can hold engagement within AI Law Model-compatible sessions (Kostenko, 2025, p. 148; Sanikidze, 2026, p. 390). Immersion research (Kutsal & Turan, 2025, p. 8) adds a further caution: strong platform immersion can prolong disengagement from offline context, so explicit session boundaries protect better than informative nudges alone. Legibility, default activation, and auditability together determine whether a batch mode actually satisfies regulatory expectations rather than only appearing to.

5.4.3 User Control Dashboards for Algorithmic Preferences

Kostenko’s model specifies (Kostenko, 2025, p. 22) three obligations: users must be able to know AI’s impact on them, obtain explanations, and retain control; every system must disclose its algorithmic nature and support auditing. The Algorithmic Impact Navigator frames (Leon, 2026, p. 2) dashboards as the operational surface where profiling and recommendation-parameter choices, otherwise buried in internal documentation, become visible. Without a compliant dashboard surface, those governance obligations remain administrative rather than operational.

Regulation sharpens this. The AI Act requires transparent, auditable and bias-free algorithms with user opt-out of personalisation; the DSA obliges (Mariia, 2025, p. 8; T et al., 2026, p. 12) very large platforms to offer at least one non-profiled feed and publish transparency reports. A compliant dashboard must surface the non-profiled mode as a first-class option with selections logged for audit (T et al., 2026, p. 12; Meßmer & Degeling, 2023, p. 21). XAI-based explanation panels stating which factors drove a recommendation, paired with diversity-weighting controls (Hu, 2025, p. 3; Mariia, 2025, p. 8), translate bias metrics into user-facing levers. Chitra et al. go further, calling (T et al., 2026, p. 16) for user-accessible explanations of primary ranking factors backed by mandatory audits and research data access. The non-profiled mode must be the default for minors.

A dashboard without an auditable backend is theatre. ISO/IEC AI risk-management standards and the AI Law Model’s event-logging duty both require (Sankaran, 2025, p. 2; Kostenko, 2025, p. 44; Kostenko, 2025, p. 135) that inputs, calculations and outputs remain traceable and appeal-accessible. HITL grading offers a practical model: instructors see SHAP-based explanations, can override outputs, and every action is dual-logged (Selvam & González Vallejo, 2025, p. 20; Mogoi et al., 2025, p. 2055956), measurably increasing student trust. Recommender dashboards that separate automated from human-curated content and log overrides achieve the same traceability, consistent with the AI Law Model’s human-centred principle (Kostenko, 2025, p. 22; Kostenko, 2025, pp. 61–62). Where that traceability is absent, user-facing control has no verifiable effect.

5.5 Recommender System Redesign for Wellbeing

5.5.1 Diversity-Oriented Ranking and Content Pluralism

YouTube’s watch-time optimisation amplified sensationalist vaccine content; Facebook disproportionately surfaced far-right political material (Mariia, 2025, p. 5; Mariia, 2025, p. 29; Mariia, 2025, p. 7). These are

documented case findings about how engagement-maximising personalisation operates in practice, not hypothetical risk scenarios. Peer-reviewed UX and neurocognitive research, not anecdote, establishes this connection.

Chitra et al. propose (T et al., 2026, p. 16) “value-aligned objective functions” as a technical response: accuracy weights for fact-checked sources, intra-session diversity requirements, and engagement ceilings that cap amplification of any single item. Gerr argues (Mariia, 2025, p. 8; Mariia, 2025, p. 28) that benchmarks such as Shannon entropy and Simpson’s diversity index should be binding design requirements rather than transparency-only obligations, because DSA reporting compliance alone has not reduced engagement-driven bias on YouTube or TikTok. Deldjoo et al. add precision (Deldjoo et al., 2023b, p. 28; Deldjoo et al., 2023b, p. 30) by distinguishing exposure (visibility) from effectiveness (relevance-match) and proposing the Gini index for catalogue concentration as a pluralism indicator suitable for an audit toolkit. Voluntary disclosure of diversity metrics without binding targets has not moved platform behaviour in the literature reviewed.

The trade-offs are real and should not be minimised. Promoting long-tail items without quality controls risks irrelevant recommendations (Deldjoo et al., 2023b, p. 28). Capping virality will reduce short-term engagement revenue, placing diversity-oriented ranking in structural conflict with current platform incentive systems (T et al., 2026, p. 14). This report treats that as a deliberate rebalancing: engagement metrics may worsen while wellbeing outcomes improve, including reduced depressive-symptom incidence associated with hyper-personalised feeds (Hu, 2025, p. 20). Annual independent audits assessing political amplification asymmetries (T et al., 2026, p. 16), together with AI Law Model requirements for technical passports and public registers for high-risk systems (Kostenko, 2025, p. 19.5), provide the enforcement architecture. Entropy, Simpson-index and group-utility measures are the primary audit evidence for whether recommenders shift toward pluralistic curation under the anticipated Digital Fairness regime.

5.5.2 Downranking of Harmful and Sensational Content

Internal Facebook documents assessed (T et al., 2026, p. 2244) the platform’s engagement algorithm as a “significant contributor” to political polarisation, with 64% of extremist group joins traced to recommendation prompts; TikTok accounts signalling interest in weight loss or politics are rapidly steered toward eating-disorder and extremist content. Instagram’s “Explore” page increased (Hu, 2025, p. 3; Hu, 2025, p. 20) exposure to body-negative imagery by 23% relative to chronological feeds, and algorithmically prioritised content shows 31% higher negative affect than baseline. This report’s taxonomy treats content combining high negative affect, polarising framing and documented mental-health harm as a prime downranking target. These findings reflect optimisation outcomes in a system designed to maximise engagement, not edge-case failures.

Three technical levers from Chitra et al.’s value-aligned objective functions address this directly. Accuracy weighting boosts fact-checked sources and penalises repeat misinformation patterns. Diversity incentives penalise ideologically narrow sequences. Engagement ceilings cap virality premiums that reward emotional escalation (T et al., 2026, p. 16). Table 18 maps these levers to audit focus. Chitra et al. accept that downranking inflammatory content will reduce engagement and advertising precision, but treat this trade as necessary given structural misalignment between platform revenue models and democratic values; they call for mandatory annual independent audits with published methodologies and findings (T et al., 2026, p. 2250; T et al., 2026, p. 2252). Platforms failing required auditing risk losing liability protection under conditional safe-harbour regimes (T et al., 2026, p. 2253). The AI Law Model prohibits systems that ma-

nipulate behaviour or cause digital addiction outright and recognises a right to a psycho-emotionally safe digital environment (Kostenko, 2025, pp. 61–62; Kostenko, 2025, p. 119).

Scenario-based probes using DSA Article 40 data-access rights (Meßmer & Degeling, 2023, p. 21) can test recommender behaviour under specific risk narratives. Diversity and distributional fairness metrics (Mariia, 2025, p. 5; Mariia, 2025, p. 27) applied to curated datasets offer a second evidential channel. A pro-social pivot must demonstrate value-aligned objectives (T et al., 2026, p. 2252; Hu, 2025, p. 3), measurable affective-balance improvement, and independent audit trails showing sustained downranking of mental-health-harm content. Without a published audit trail, downranking claims are not independently verifiable.

5.5.3 Explainable Recommendations and User Feedback Loops

Explainability becomes tangible when interfaces show users why items appear and how human oversight operates on top. HITL grading offers a tested template: syllabi disclose which tasks are automated; feedback is tagged (Selvam & González Vallejo, 2025, p. 15; Selvam & González Vallejo, 2025, p. 20) “AI generated,” “human reviewed,” or “human added,” with colour coding that measurably increased student trust when instructors visibly modified machine scores. Recommender feeds can apply the same three-way labelling (“algorithmically suggested,” “editorially curated,” “friend recommended”) with moderator overrides (Selvam & González Vallejo, 2025, p. 11; Selvam & González Vallejo, 2025, p. 17) dual-logged by timestamp and identifier. Without that override trail, the label is disclosure only, not accountability.

Regulation anchors this as a design requirement. Kostenko’s transparency principle requires (Kostenko, 2025, p. 67; Kostenko, 2025, p. 68) that architecture, logic and purpose be “understood, accessible, verifiable, and explanatory” at every lifecycle stage, with covert or manipulative use in socially significant contexts prohibited. A “why this post” panel linked to an appeal channel implements this directly. The Algorithmic Impact Navigator calls (Leon, 2026, p. 2) for continuous stakeholder engagement and transparent records across the full lifecycle, mirrored in HITL grading’s guarantee that appeals are reviewed solely by human faculty with full audit-trail disclosure (Selvam & González Vallejo, 2025, p. 15). Where stakeholder visibility is absent, the transparency obligations in Article 14 DSA and Kostenko’s explainability principle have no operational bite.

SHAP and LIME techniques make black-box models inspectable and support AI Act/GDPR compliance, but regulators still require (Bahangulu & Owusu-Berko, 2025, p. 11; Bahangulu & Owusu-Berko, 2025, p. 9) independent audits and external oversight committees to reach the model itself. An audit toolkit can require XAI pipelines and aggregated explanation summaries for vetted researchers without internal access, consistent with DSA data-access provisions (Meßmer & Degeling, 2023, p. 19; Kostenko, 2025, p. 67). Hu’s Algorithmic Impact Assessment Scale (AIAS-5) quantifies (Hu, 2025, p. 20; Hu, 2025, p. 19) mental-health risk through emotional volatility and compulsive-usage dimensions, proposing a dynamic transparency index and early-warning system. Combined with explainable interfaces and human-validated overrides, these define a concrete pro-social pathway: dashboards that explain and track impact, feeding regulator channels capable of triggering model throttling when harm indicators rise.

5.6 Alternative Engagement and Monetization Models

5.6.1 Subscription and Freemium Revenue Streams

Subscription and freemium designs become ethically and legally charged when dark commercial patterns enter the picture. An audit of Indian digital platforms found (Khare et al., 2026, p. 2) roughly 87 percent used at least one dark pattern; interface interference appeared in 70 percent of cases and obstruction in 63 percent, manifesting as difficult cancellation flows and unexpected charges; of surveyed consumers, 55 percent reported unexpected charges and only 39 percent recognised the term “dark patterns,” signalling a sharp information asymmetry. The “roach motel” is its clearest expression: sign-up is frictionless (Khare et al., 2026, p. 4) while cancellation is deliberately complex, eroding autonomy and informed consent. The same platforms’ public-facing registration flows are frictionless by design, making the cancellation asymmetry a deliberate architectural choice with measurable economic benefit to the platform.

Liu and Keshavarz label this the “conversion–trust tradeoff.” Dark commercial patterns raise immediate compliance but corrode the fairness, autonomy and trust that drive long-term loyalty. Their own typology covers five categories (Liu & Keshavarz, 2026, p. 33) – information distortion, interface asymmetry, social and temporal pressure, choice obstruction, and forced action – all of which map onto subscription tactics such as hidden fees and multi-step cancellation flows; the roach-motel and unexpected-charges evidence above from the Indian platform audit illustrates the same categories in practice (Khare et al., 2026, p. 4). Aggressive implementations carry a documented relationship risk (Liu & Keshavarz, 2026, p. 35), damaging brand relationships and inviting regulatory scrutiny. Liu and Keshavarz find no evidence that short-term conversion gains from dark-pattern tactics persist once consumer trust declines.

DSA and AI Act regimes are operationalised in part through the European Centre for Algorithmic Transparency (ECAT), whose remit includes promoting fair algorithmic practices across the platforms it monitors (Mariia, 2025, p. 3). Applying Chitra et al.’s engagement-ceiling and diversity-incentive requirements (T et al., 2026, p. 2252) to paid tiers specifically – rather than only to free-tier feeds, as the underlying proposal is stated – is this report’s own extension: relaxing diversity constraints for paying users would undermine the regulatory intent of both DSA and the anticipated DFA, even though the source literature does not address paid tiers directly.

This report classifies subscription and freemium designs relying on information distortion, interface asymmetry or choice obstruction under Code Category C (“Prohibited or Critical Risk Nudges”) (Liu & Keshavarz, 2026, p. 33; Khare et al., 2026, p. 4). Platforms with clear pricing and symmetric sign-up/cancellation flows fall into lower-risk categories, pending DFA and ECAT interpretive guidance (Mariia, 2025, p. 3). Audit dimensions include classification against the five dark-pattern categories, measurement of cancellation friction via observable UX steps, and assessment of whether premium tiers modify recommendation objectives against diversity and engagement-ceiling standards – the last of these extending Chitra et al.’s engagement-ceiling framework (T et al., 2026, p. 2252) to the premium-tier case by this report’s own reasoning, not as a claim already made in the cited literature.

5.6.2 Micro-Learning and Positive Habit Loops

The core design shift is from sessions optimised for time-on-platform to bounded sequences that build competence. Gamified education platforms such as Legends of Learning, Kahoot! and GradeCraft already operationalise this. When gamification is embedded systemically (clear tasks, defined stages, diagnostic feedback), personalised configurations show (Tokarieva & Chyzykova, 2026, p. 9) 35–40 percent

higher engagement and 25–30 percent better learning outcomes than generic designs. The performance difference between bounded and open-ended designs in these studies is large enough to treat as an evidence-backed design principle rather than a tentative finding.

Blooket’s token economy illustrates a workable micro-learning arc: compact question batches, instant feedback, and token allocation toward a narrative goal function as a “currency for creative agency” rather than shallow point accrual. Rewards must sit in goal-oriented, context-sensitive sequences (Tokarieva & Chyzhykova, 2026, p. 7); punitive feedback or vague contingencies generate technostress that undermines intrinsic motivation. HITL grading offers a comparable template: each assignment moves through AI suggestions (“automated”), instructor comments (“final”), and dual score logs (Selvam & González Vallejo, 2025, p. 5; Selvam & González Vallejo, 2025, p. 9; Selvam & González Vallejo, 2025, p. 21), with over 80 percent of students valuing explicit AI-involvement transparency. Hu et al.’s finding that over 80 percent of students valued explicit AI-involvement transparency reinforces the importance of visible progression logic over reward accumulation alone.

Mapped onto recommenders, positive habit loops would define finite content batches around user-relevant goals, with explicit completion signals replacing open-ended scrolling. Kostenko’s AI Law Model requires (Kostenko, 2025, p. 25) that session boundaries and reward structures be auditable, not merely disclosed. An aligned audit toolkit requires evidence of finite modules with natural exit points and event logs documenting session start/end and human-oversight intervention (Selvam & González Vallejo, 2025, p. 21; Kostenko, 2025, p. 25).

Self-regulation alone is insufficient. Even where firms like Google and Meta maintain internal AI-ethics divisions, transparency is limited, reinforcing calls (Bahangulu & Owusu-Berko, 2025, p. 1753) for independent audits aligned with the AI Act and OECD AI Principles. Chinese short-video platforms combine neural-network recommendation with micro-duration stimuli and variable reward schedules (Li, 2026, p. 2) in ways that the EU’s TikTok DSA inquiry treats as potentially manipulative design, with college students disproportionately vulnerable given still-maturing executive function. This report’s own recommendation, extending that concern to micro-learning design, is that structured tasks and predictable routines should remain user-configurable for batch size, pacing and notification frequency rather than imposed as a fixed template; explainable-AI techniques such as SHAP and LIME, used elsewhere in algorithmic-accountability contexts for transparency (Cao, 2026, p. 66), offer one plausible route to periodic fairness audits examining differential impacts of engagement loops across demographic and psychological groups, though no cited source yet applies them to this specific use case.

5.6.3 Advertiser Alignment with Digital Wellbeing

Standard advertising metrics track the same dynamics regulators now flag as harmful. Engagement-driven recommendation environments concentrate exposure on emotionally charged content (Hu, 2025, p. 20), and this same mechanism connects directly to the engagement-maximising objective functions Chitra et al. propose constraining (T et al., 2026, p. 2252). Impressions, click-through rates and dwell time are sensitive to the same filter-bubble and high-affect amplification patterns that generate a 23% higher depressive-symptom incidence in hyper-personalised feeds (Hu, 2025, p. 20). For advertisers evaluating inventory on those metrics alone, brand-safety exposure is already built in.

The regulatory pressure compounds this. Sanikidze’s analysis makes the DFA intent explicit (Sanikidze, 2026, p. 390): “addictive product designs and unfair personalisation” are treated as unfair commercial practices alongside DSA/DMA dark-pattern provisions. Roughly 87 percent of Indian digital platforms deploy at

least one dark pattern (Khare et al., 2026, p. 2), and the same filter-bubble dynamics documented earlier in this report are already shown to depress adolescent self-esteem (Hu, 2025, p. 3) – a wellbeing cost advertisers inherit as brand-safety exposure even without a documented finding on brand co-appearance specifically. Kostenko’s AI Law Model adds categorical prohibitions (Kostenko, 2025, p. 217) on psychotype- or emotion-based targeted advertising to minors. The combination of harm evidence and tightening legal obligations means advertising strategies premised on limitless tracking and opaque profiling carry growing regulatory and reputational risk. Algorithmic systems can also reproduce harmful stereotypes (Johar et al., 2025, p. 4) by skewing outputs toward demeaning representations of marginalised groups.

Value-aligned objective functions offer a specific alternative. Chitra et al. propose weighting fact-checked sources, enforcing intra-session diversity and capping per-item engagement (T et al., 2026, p. 2252); applying that objective function to advertising inventory specifically, so that it trades some short-term reach for association with less polarising, audited inventory, is this report’s own extension rather than a claim made in the source. The same paper calls for mandatory annual third-party audits of recommendation algorithms (T et al., 2026, p. 2252); letting advertisers evaluate inventory against a harm-risk index such as the AIAS-5 scale used elsewhere in this literature (Hu, 2025, p. 20), rather than relying on platform self-declarations, is again this report’s proposed application of an audit mechanism the cited sources describe in general terms, not an advertiser-specific finding either source reports directly.

Practically, this means bidding only on inventory subject to published third-party audits with engagement ceilings and diversity incentives – extending Chitra et al.’s audit and engagement-ceiling proposals (T et al., 2026, p. 2252) to advertiser bidding decisions, which the source itself does not address – and prioritising placements that disclose predicted psychological impact rather than burying it in terms of service, in the spirit of the “digital nutrition label” disclosure model discussed in the algorithmic-harm literature (Hu, 2025, p. 20). Internal incentive structures matter as well: this report recommends that campaign teams be rewarded for reduced association with high-risk content alongside standard click-through metrics, better positioning them to act on these requirements once the DFA enters force – a governance-design recommendation of this report’s own, not a finding reported in the cited audit or dark-pattern literature. Advertiser-side pressure for research data access before asserting campaign safety extends Chitra et al.’s call for researcher and regulator data access (T et al., 2026, p. 2252) to advertisers specifically, creating, on this report’s own reasoning, an additional lever for structural change at the platform level.

5.7 Directional Commercial Viability Analysis

5.7.1 Cited Figures versus Modelled Assumptions

A pro-social pivot survives a finance committee only if the revenue case is stated honestly. Three figures in this analysis are directly cited from published sources, not modelled. Roughly 87% of major platforms audited in India deploy at least one dark pattern (Khare et al., 2026, p. 2; Liu & Keshavarz, 2026, p. 4), and 47% of users report difficulty cancelling subscriptions because of obstructive exit flows. Separately, 81% of platforms self-declare “dark pattern free” while complaint resolution sits at only 28% (Khare et al., 2026, p. 12), a gap that functions as a quantifiable regulatory-exposure indicator. Engagement-optimised feeds correlate (Hu, 2025, p. 20) with a 23% higher incidence of depressive symptoms, a wellbeing cost that appears in churn research as a driver of voluntary platform abandonment. These three figures set the factual baseline; everything that follows is modelled reasoning, not additional evidence.

None of these three figures translates directly into ARPU, LTV or subscription conversion. Doing so re-

quires assumptions, stated explicitly here as assumptions.

Where obstructive cancellation currently retains some share of the 47% of users who report wanting to leave, a redesigned symmetric cancellation flow would be expected to increase voluntary churn in the short term. No cited churn-elasticity figure exists for social recommenders. The honest modelling position is to treat this as a near-term revenue cost weighed against DFA-era fine exposure for continuing roach-motel patterns once the Act is in force (Sanikidze, 2026, p. 390). The freemium redesign in Section 5.6.1 (removing diversity-relaxation for paying tiers, enforcing symmetric sign-up/cancellation) is treated as conversion-neutral: Liu and Keshavarz's conversion–trust tradeoff literature (Liu & Keshavarz, 2026, p. 37) indicates dark-pattern removal costs short-term conversion but protects long-term retention and trust-driven referral, without providing a point estimate this report could cite as fact.

On the advertiser side, bidding only on audited, engagement-ceiling-compliant inventory forgoes some volume-driven reach – a real and near-term cost. In return, it reduces brand-safety exposure to the polarising and body-negative content correlated with the cited 23%/31% harm statistics (Hu, 2025, p. 3). The compliance-driven redesign trades a bounded, front-loaded conversion cost for a reduction in regulatory and reputational exposure that the 87%/81%/28% figures show is already material today, before any new legislation takes effect. That directional conclusion is defensible without a full ARPU/LTV model. A fully quantified model would require primary access to a specific platform's cohort data; this report flags that as future empirical work, with the present analysis serving as a structured starting point for platform-specific modelling.

5.7.2 Illustrative ARPU/LTV Scenario Model

The preceding directional argument can be sharpened into a scenario model, provided every figure below is read as a modelled assumption indexed to the report's own baseline, not a cited fact about any named platform. The model exists to make the reasoning auditable, not to forecast a real platform's financials: each row states its modelling basis so a reader can substitute their own assumptions and recompute the outcome.

The break-even logic is deliberately conservative: even at the pessimistic end of the conversion-loss range and the pessimistic end of the retention-uplift range, a single DFA-era enforcement action large enough to trigger the modelled fine band would offset several months of foregone conversion. That asymmetry, not a precise ARPU figure, is the actionable output of this model. A platform-specific version would replace every modelled row with the organisation's own cohort data; this report's contribution is the structure of the calculation, not a claim to have already run it on real figures.

6 Critical Reflection, Limitations and Conclusion

6.1 Critical Reflection

This report's central limitation is methodological and was disclosed from the outset. Every claim about TikTok, Instagram, LinkedIn and comparable platforms rests on secondary sources: regulatory filings, published UX and neurocognitive research, and advocacy-group audits. No claim rests on proprietary ranking code (T et al., 2026, p. 6; Kostenko, 2025, p. 244), training data or internal telemetry. That constraint is defensible given the qualitative, secondary-data scope defined in Section 1. The audit toolkit can classify externally observable design patterns (autoplay defaults, notification frequency, cancellation

friction) with reasonable confidence, while claims about exact optimisation objectives or the causal weight of any single feature remain inferential, triangulated across studies rather than verified against a single authoritative source. Triangulation is a legitimate method; it is not a substitute for primary access.

Coverage across platforms is uneven, reflecting the underlying evidence base rather than selective attention. TikTok and Instagram dominate the audited literature; LinkedIn is genuinely under-studied in the dark-pattern and algorithmic-addiction literature consulted here. Section 4.1.3 addresses this by flagging LinkedIn claims as illustrative extrapolation from adjacent platforms rather than fabricating platform-specific statistics, consistent with the Integrity Rule that every claim traces to a citation or is explicitly marked illustrative. Any organisation applying this toolkit to a professional-networking product should treat the LinkedIn-specific sections as a starting hypothesis requiring primary validation. That gap is an honest acknowledgment, not a structural weakness.

The working coding taxonomy (Code A interface mechanics, Code B psychological triggers, Code C legal risk tiers) is this report's own synthesis, not an official classification under the DSA, AI Act or the still-unpublished Digital Fairness Act. Risk-tier assignments should be read as defensible interpretations of regulatory intent, not binding legal determinations. Because the DFA remains a policy trajectory rather than adopted law at the time of writing, several forward-looking claims about how addictive design will be regulated are necessarily provisional. Where article or clause numbers eventually diverge from current Commission communications, the normative arguments about autonomy, transparency and duty of care should still hold; they are grounded in the DSA and AI Act as already-adopted anchor instruments.

The commercial-viability reasoning in Section 5 raises a distinct concern. It mixes cited figures with modelled assumptions about how those figures translate into subscription conversion or advertiser revenue. These two categories carry different evidentiary weight and should not be read as equally certain. The audit toolkit is best used as a structured starting point for platform-specific investigation, not a substitute for it.

6.2 Recommended Validation Methodology

The Code A/B/C matrix has been applied throughout this report by a single analyst working from secondary sources and, in Section 4.1.3, a single-evaluator heuristic walkthrough. Neither constitutes independent expert validation, and this report does not claim otherwise. What follows is a deliberately scoped protocol for the primary validation this report's methodology has not yet undergone, offered as a concrete next step rather than a general call for "future research."

A useful validation round would recruit 8 to 12 participants across three practitioner groups with direct exposure to the matrix's three code families: compliance officers or legal counsel familiar with the DSA and AI Act (to stress-test Code C legal risk tiers), UX or product designers with production experience on recommender-driven interfaces (to stress-test Code A interface-mechanics classifications), and researchers or practitioners in behavioural science or AI ethics (to stress-test Code B psychological-trigger classifications). Recruiting across all three groups, rather than one, tests whether each code family is legible to the practitioner population it is actually meant to serve, matching the audit methodology's own framework-triangulation logic (Asante & Adeoba, 2026; Meßmer & Degeling, 2023) rather than treating validation as a single monolithic check.

The validation instrument would be a semi-structured interview: each participant reviews a sample of five to eight already-coded audit excerpts (with the assigned Code A/B/C labels visible) and is asked, per excerpt,

whether they would assign the same code, what they would assign instead if not, and how confident they are in their own judgement. A structured post-interview questionnaire would capture perceived practical applicability of the toolkit to the participant's own compliance or design workflow, using a short Likert-scale instrument rather than open-ended impressions alone, so results are at least partially comparable across participants.

Two concrete criteria would distinguish a successful validation round from an inconclusive one: first, inter-rater agreement between participants and the original coding above a stated threshold (a Cohen's kappa or percentage-agreement figure decided in advance, not selected after seeing the results); second, a majority of participants rating the toolkit as practically applicable to their own workflow, not merely conceptually sound. Falling short of either threshold would be a genuine finding, not a validation failure to be explained away: it would indicate which code family or which platform's coding needs revision before the toolkit is used operationally. This protocol is deliberately specific enough to be executed by a future researcher without further design work, which is the point of naming it as a next step rather than leaving primary validation as an unaddressed gap.

6.3 Concluding Strategic Outlook

Engagement-optimised social recommenders combine interface affordances, profiling practices and business incentives in ways that materially affect user autonomy, attention allocation and distributional outcomes. Endless feeds, autoplay, high-frequency re-engagement prompts, asymmetric defaults and opaque opt-out channels create feedback loops the literature (Hu, 2025, p. 2; Li, 2026, p. 2) associates with compulsive checking, degraded wellbeing and narrowed exposure, with neurocognitive, behavioural and UX evidence consistently linking these design choices to reduced self-regulation and elevated affective volatility for vulnerable groups. European legal and model-law frameworks now treat such configurations as regulatory risk rather than acceptable product differentiation. Obligations for transparency, non-profiling feed options, tamper-evident logs and age-sensitive defaults are specified (T et al., 2026, p. 12; Kostenko, 2025, pp. 61–62); where those artefacts are absent or inadequate, governance failure becomes a signal of unacceptable systemic risk under the DSA and AI Act. The compliance question and the product-design question are now the same question.

Compliance readiness and commercial resilience now point in the same direction. The pro-social pivot strategy in Section 5 shows that finite-session design, user control dashboards, diversity-aware ranking and subscription/freemium models decoupled from dark-pattern retention tactics are plausible routes to sustaining engagement and revenue once the current attention-maximising model faces DFA-level scrutiny (Sanikidze, 2026, p. 390). Operationally, a practicable audit and remediation pathway should focus on observable, auditable artefacts: documented algorithmic passports, event logs with human-override trails, user control dashboards exposing non-profiled modes, finite session mechanisms and batch or digest alternatives, explainable recommendation panels, and diversity-aware ranking objectives with downranking levers for sensational content (see Section 5). Embedding these controls (T et al., 2026, p. 16; Kostenko, 2025, p. 244) into CI/CD gates, independent third-party audits and governance cycles creates traceable evidence for regulators and civil society while enabling product teams to iterate on safer objective functions without waiting for finalised DFA text. Observable artefacts are the operationalisation of intent.

Priority empirical work beyond this report should centre on scenario-based probes, representative trace releases for vetted researchers, and longitudinal evaluation of wellbeing outcomes after deploying time-boxed, diversity-oriented interventions. LinkedIn is the clearest gap: the evidence base there remains

thin, and primary research would close a documented hole. Each of these steps feeds directly into audit practice: scenario probes generate the DSA Article 40 datasets that external researchers need; trace releases let regulatory bodies verify downranking claims independently; and longitudinal data would finally answer whether the 23% depressive-symptom reduction correlated with chronological feeds is sustained when platforms move from unlimited engagement optimisation to the bounded, wellbeing-oriented designs described in Section 5 (Kostenko, [2025](#), p. 119; Sanikidze, [2026](#), p. 390).

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List of Appendices

- Appendix A: Capstone Project Exposé
- Appendix B: AI Use Declaration

7 Appendix A: Capstone Project Exposé

Name: Mohit Nirwan

Matriculation Number: IU14087991

Programme: Master of Business Administration (IU MBA 90 ECTS / LSBU Dual-Award)

1. Preliminary Title

Algorithmic Addiction in Social Recommender Systems - Development of an Ethical AI Audit and Pro-Social Pivot Strategy under the EU Digital Fairness Act

2. Introduction: Problem Description and Objective

Social media platforms like TikTok, Instagram, and LinkedIn optimize their recommendation engines primarily for screen time. To hit those numbers, the design leans on infinite scroll, variable reward schedules, and content loops built to provoke outrage. All three exploit the same cognitive weak spots. For example, TikTok's auto-playing feed and pull-to-refresh mechanics act as textbook variable reward schedules. This ad-driven model is highly monetizable, but it's wearing users down. Younger users in particular are showing signs of digital burnout and cognitive overload. That reliance on engagement-hacking has now hit a regulatory wall. The European Commission's upcoming Digital Fairness Act (targeted for Q4 2026) explicitly aims to ban deceptive design and addictive algorithmic patterns. If platforms do not comply, they face revenue-based fines.

Right now, most executive teams have no real way to audit their own algorithms, let alone fix them without tanking retention. My Capstone Project turns this into a consulting report with a specific output: a roadmap for moving away from ad-supported dark patterns toward what I'm calling Pro-Social AI. That means testing whether subscription models and empathetic interaction design can replace outrage-driven engagement without sinking revenue. The goal is to answer a specific Research Question: How can enterprise social platforms redesign their AI recommender systems to comply with upcoming EU Digital Fairness standards while protecting core user engagement and commercial viability?

3. Brief Theoretical Foundation

The analysis is anchored in Behavioural Economics, specifically Thaler and Sunstein's (2008) choice architecture and the taxonomy of dark patterns mapped by Mathur et al. (2019). Two other strands feed into this. Self-Determination Theory (Deci & Ryan, 2000) explains why users are vulnerable in the first place. More recent work on social-platform dark patterns (Mildner et al., 2023; Wiesböck et al., 2026) shows how that vulnerability gets designed into the product. A clear limit to acknowledge is that behavioural economics often assumes users can be "nudged" rationally, which breaks down in highly addictive, frictionless environments. Still, it provides the best diagnostic baseline.

4. Research Methodology (Method and Analysis)

The recommendations rest on application-specific secondary data. I will gather three kinds of secondary data: the regulatory texts themselves (the DSA and the 2026 DFA drafts), public UI/UX audits of major platforms, and consumer advocacy submissions on digital burnout going back to 2021. I will code these inputs using qualitative thematic analysis to map specific algorithmic design patterns against legal risk

categories. One limitation I will have to manage: secondary data regarding proprietary algorithms relies on reverse-engineered audits rather than open internal codebases. I can bridge this gap by focusing on the observable user interface outputs rather than the underlying black-box code.

5. Planned Structure of the Consulting Report

The following outline aligns directly with the IU consulting report requirements:

- 1. Cover Page
- 2. Summary
- 3. Problem description and environment: Algorithmic engagement optimization, addictive design, digital burnout, and the legislative threat of the EU Digital Fairness Act.
- 4. Consulting function
 - 4.1. Objective and challenges: Harmonizing legal compliance and pro-social features with platform business models.
 - 4.2. Method: Qualitative thematic coding of regulatory standards, consumer burnout data, and UI design patterns.
 - 4.3. Analysis: Comparative audit of current platform recommendation loops against proposed legal boundaries.
 - 4.4. Recommendations: An audit tool for compliance and a strategic pivot roadmap toward Pro-Social AI (e.g., subscription models, micro-learning loops).
- 5. Critical reflection: Concluding discussion of process limitations.
- 6. Literature
- 7. Appendix: (Including this exposé for the LSBU award assessment)

6. Initial Bibliography

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8 Appendix B: AI Use Declaration

This thesis was prepared with the assistance of large language model tools that were used to generate, edit, and refine portions of the text under the direct supervision of the author. Such tools were employed to help draft language, suggest rephrasings, and identify stylistic or organizational improvements; all outputs from these tools were reviewed, modified as necessary, and incorporated at the author's discretion.

The author independently verified and integrated all source material, checked factual statements, and ensured that citations, data, and interpretations are accurate and appropriate for an MBA capstone thesis. Where text proposed by a language model was used, the author confirmed its conformity with academic standards, adapted the content to the thesis argument, and ensured proper attribution of ideas and evidence.

Ultimately, responsibility for the content, analysis, argumentation, and academic integrity of this work rests solely with the author. Any errors, omissions, or interpretations in the final document are the author's own.

Table 9: Heuristic Walkthrough of LinkedIn's Publicly Visible Interface

Feature Observed	Code Classification	Confidence
Continuous feed with no natural end point or session boundary	Code A (interface mechanics)	High – directly observed
Notification and connection-request badges (red indicator) appear on unread items but clear after a single view, with no observed re-escalation or repeat nagging	Code A; low observed Code B salience relative to the variable-reward pattern documented for other platforms	High – directly observed
“Who’s viewed your profile” analytics: some detail available free, fuller detail gated behind a Premium subscription upsell	Code A (economic gatekeeping) / Code C (monetised curiosity, DFA-relevant if repeated at scale)	High – directly observed
Profile-completeness section present; explicit completion nudges were not observed on every session, but 2-3 distinct touchpoints recommended a Premium upgrade to unlock profile-enhancement features	Code A (monetisation-linked) more than a pure Code B goal-gradient effect	Medium – based on this evaluator’s recollection across recent sessions, not a single controlled walk-through
“People you may know” and connection-request prompts recurring across three distinct surfaces in a single session (feed, dedicated area, notifications); no single-step opt-out located	Code A / Code B	Medium
Feed interleaves algorithmically ranked “suggested” content among connection updates, not strictly chronological, with no toggle located to switch to a chronological-only view	Code A	High – directly observed, consistent with the ranking-opacity pattern documented for other platforms
Autoplay of video content in the feed	Not observed – videos did not play automatically while scrolling	High – directly observed; noted for completeness, since not every mechanism documented elsewhere generalises to this platform
Subscription downgrade/cancellation flow includes a reason-selection step, but permitted the change without the more severe obstruction patterns documented in the subscription-platform audit elsewhere in this report	Code C, but lower-severity than the audited roach-motel pattern	Medium – single most-recent use recalled by this evaluator, not a fresh systematic test

Table 10: Notification related patterns functioning as return incentives

Pattern	Mechanism	Documented effects
Nagging notifications	Repeated prompts after refusal, often via app alerts	Wear down effect, decision fatigue, coerced consent (Khare et al., 2026, p. 8)
Late night urgency messages	Time based push for expiring discounts or offers	Executives perceive higher responsiveness in "low control" windows (Bansal, 2026, p. 138)
Subscription renewal prompts inside obstructive flows	Notifications point to renewal while exit paths remain complex	Forced continuation of paid services, recurring unwanted payments (Khare et al., 2026, p. 8)
"Dark pattern free" self declarations	Internal audits and voluntary declarations without effective external verification	81% of platforms self declared dark pattern free despite persistent complaints, 28% resolution rate (Khare et al., 2026, p. 12)

Table 11: Variable reward patterns and intermittent reinforcement across platforms (working Code B mapping)

Platform type	Variable reward mechanic	Documented effects
Short video feeds	Swipe to next clip with unpredictable quality; infinite scroll; default auto play	"Auto pilot" scrolling, behavioural addiction, impaired study performance in students (Li, 2026, p. 2; Li, 2026, p. 6)
Algorithmic social feeds	Engagement ranking that privileges emotionally charged content; irregular delivery of highly salient posts	22% higher compulsive checking; 23% higher depressive symptoms; reduced prefrontal activity (Hu, 2025, p. 2; Hu, 2025, p. 20)
E commerce flows	Countdown timers and "only two left" alerts that sometimes expire and sometimes do not	Conversion uplift reported by executives; decision fatigue and regret in consumers (Bansal, 2026, p. 136; Khare et al., 2026, p. 8)

Table 12: Working Code B mapping: social validation, status cues and FOMO

Pattern	Mechanism	Documented effects
Inflated metrics	Fake likes, followers, paid engagement	Illusion of popularity, weakened trust in reviews and ratings
Undisclosed sponsorship	Influencer posts that hide commercial nature	Exploitation of emotional credibility, confusion between advertising and opinion
Algorithmic herd reinforcement	Preferential promotion of already popular content	Pressure to conform to trends, FOMO driven participation in viral challenges and live sales (Wang, 2025, p. 13)
Body and identity sensitive feeds	Prioritisation of high arousal or body negative imagery	Increased body dissatisfaction and lower self esteem among specific groups (Hu, 2025, p. 3)

Table 13: High arousal and outrage content: working Code B mapping

Coding slot	Mechanism	Empirical or conceptual anchor
B1: Moral emotional amplification	Moral emotional wording boosts sharing, driving outrage selection	Moral emotional words increase retweets by $\approx 20\%$ per word (T et al., 2026, p. 7)
B2: Negative affect bias	Ranking favours high negative affect content	Algorithmic feeds show 31% higher negative affect, 23% more body negative imagery (Hu, 2025, p. 3)
B3: Viral echo chamber reinforcement	Viral, racy posts amplified into cohesive cliques	Viral misinformation and doctored videos deepen echo chambers and division
B4: Emotional fatigue and distress	Repeated exposure to polarising material harms creators and viewers	Harassment, burnout, information fatigue and emotional distress reported (Yin, 2024, p. 2)

Table 14: Cross-platform audit matrix: features, DFA-adjacent risk tier, and SDT degradation indicator

Platform type	Dominant Code A/B feature	Code C tier	SDT indicator degraded
Short-video (TikTok-type)	Infinite scroll, autoplay, variable reward (Li, 2026, p. 2; Hu, 2025, p. 2)	High/critical	Autonomy (auto-pilot scrolling); competence (impaired concentration)
Image/story (Instagram-type)	Explore-style ranking, social proof, body-negative amplification (Hu, 2025, p. 3)	High	Relatedness (distorted social comparison); competence (self-esteem decline)
Professional networking (illustrative only)	Engagement ranking, profiling-based ads (extrapolated, not platform-verified) (T et al., 2026, p. 2; Quinelato, 2024, p. 4)	Not yet assessable	Not yet assessable, flagged for future primary research
Cross-platform notification layer	Nagging, obstructive cancellation, temporal-vulnerability targeting (Khare et al., 2026, p. 8; Bansal, 2026, p. 138)	High	Autonomy (compressed deliberation)
Cross-platform outrage/virality layer	Moral-emotional amplification, negative-affect bias (T et al., 2026, p. 7; Hu, 2025, p. 3)	High	Relatedness (echo-chamber reinforcement); autonomy (opinion manipulation)

Table 15: Illustrative use case archetypes and boundary focus (working taxonomy)

Use case archetype	Boundary focus	Primary sources informing scope
Short form video feed for students	Infinite scroll, auto play, push notifications, personalisation pipeline, child data processing, operator governance	Algorithmic addiction loop, student harms, DSA proceeding against TikTok, age appropriate design and child data processing duties (Li, 2026 , p. 2; Li, 2026 , p. 6; Hu, 2025 , p. 2; Kostenko, 2025 , p. 20)
Image centric social discovery	Explore style ranking, social proof widgets, behavioural advertising modules, bias and body image impacts	Amplification of body negative imagery, negative affect bias, social proof manipulation and its impact on trust and identity (Hu, 2025 , p. 3; Wang, 2025 , p. 13)
AI augmented social platform (SOCIO SPHERE AI)	AI recommendation layer, sentiment analysis, AI assisted replies, moderation AI, admin analytics dashboards	Three tier plus AI/ML architecture, AI generated captions and hashtag suggestions, AI based moderation, engagement analytics (Behera, 2026 , p. 3413731)

Table 16: Stakeholder centric risk priorities for social recommender audits (working taxonomy)

Stakeholder cluster	Primary concern	High priority risk cues
Platform leadership and boards	Legal compliance, reputation, business continuity	Presence of addictive design patterns under DSA scrutiny; absence of internal ethical review and AI Ethics Impact Reports (Li, 2026, p. 2; Kostenko, 2025, pp. 61–62)
Engineering and product teams	Technical performance, shipping velocity	Lack of gating controls in CI/CD for fairness, transparency and sustainability; missing logs needed for conformity assessment (Catapang, 2026, p. 24; Kostenko, 2025, p. 19.5)
Regulators and auditors	Systemic risk, fundamental rights	Inability to obtain technical dossiers and logs; absence of algorithmic impact assessments and standardised fairness metrics (Meßmer & Degeling, 2023, p. 19; Sebastian, 2025, p. 15)
Users and communities	Autonomy, psycho emotional safety, fairness	Infinite scroll and auto play combined with personalised feeds; documented mental health or discrimination harms (Li, 2026, p. 2; Hu, 2025, p. 2; Raaj S. et al., 2026, p. 4245543)
Researchers and civil society	Transparency, contestation	Restricted data access under DSA Article 40 analogues; opaque terms and conditions on content and risk management (Meßmer & Degeling, 2023, p. 19; Meßmer & Degeling, 2023, p. 3195574)

Table 17: Working ethics spectrum for nudging practices in interface and choice architecture

Category (working label)	Typical design attributes	Indicative sources / risk signals
Assistive nudges (low risk)	Symmetric effort for accept / decline, clear total costs, visible lower priced alternatives, easy reversal	Preserves informed, preference consistent choice; aligns with ethical digital marketing and CSR based AI governance (Liu & Keshavarz, 2026, p. 33; Asante & Adeoba, 2026, p. 16)
Borderline nudges (medium risk)	Mild interface asymmetry, moderate highlighting of defaults, incomplete but non deceptive urgency messages	Asymmetric cookie banners where refusal is possible but less salient; potential conversion trust trade off (Liu & Keshavarz, 2026, p. 33)
Coercive sludge patterns (Code C)	Hidden fees, false scarcity, confirm shaming, roach motels, obstructive cancellation, forced action	Documented as structural features on $\approx 87\%$ of major platforms; linked to autonomy erosion and regulatory calls for bright line bans and UX audits (Khare et al., 2026, p. 7; Khare et al., 2026, p. 14; Sanikidze, 2026, p. 390)

Table 18: Working mapping: downranking levers for harmful and sensational content

Lever	Mechanism	Audit focus
Accuracy weighting	Boosts fact checked sources, penalises repeat misinformation	Evidence of source level weights; fact checker integration (T et al., 2026, p. 2252)
Diversity incentives	Penalises ideologically narrow sequences	Diversity metrics (entropy, Simpson index) over recommendation sessions (T et al., 2026, p. 2252; Mariia, 2025, p. 6)
Engagement ceilings	Caps exposure of single items	Detection of hard caps on impressions and watch time per item (T et al., 2026, p. 2252)
Explainable ranking	User facing reasons for each recommendation	Dashboards stating factors such as prior viewing or trending status (T et al., 2026, p. 2252; Mariia, 2025, p. 6)
Mandatory audits	External checks of amplification asymmetries and pluralism	Published audit summaries assessing political amplification and diversity (T et al., 2026, p. 2252)

Table 19: Illustrative ARPU/LTV Scenario Model for the Pro-Social Pivot (Modelled, Not Cited)

Scenario Variable	Modelled Range	Basis
Baseline monthly ARPU (indexed to 100)	100 (index)	Illustrative reference unit; not a claimed figure for any named platform
Short-term conversion loss from symmetric cancellation flow	–5% to –12% of baseline ARPU	Modelled against the 47% stated cancellation-difficulty figure and the conversion-trust tradeoff in Liu and Keshavarz (Liu & Keshavarz, 2026, p. 37)
Medium-term LTV uplift from trust-driven retention (12-18 months)	+8% to +15% LTV	Modelled from the same conversion-trust tradeoff literature (Liu & Keshavarz, 2026, p. 37); no churn-elasticity point estimate exists for social recommenders, so the range is deliberately wide
DFA-era regulatory fine exposure avoided per enforcement cycle	1% to 6% of global turnover	Modelled on DSA/AI-Act-era fine-severity bands, anchored to the DFA discussion in Sanikidze (Sanikidze, 2026, p. 390) and the third-party-audit accountability framework in Chitra et al. (T et al., 2026, p. 2252)
Net directional effect (12-24 month horizon)	Positive once fine-avoidance plus retention uplift exceed the short-term conversion loss	Break-even timing depends on the unmeasured churn-elasticity figure; a testable hypothesis, not a forecast